

Early Life Origins of Hypertension and Chronic Kidney Disease

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KEY POINTS

- Being born too early (preterm) or too small (small for gestational age) are risk factors for longer-term hypertension and kidney disease.
- Preterm birth and being small for gestational age are associated with smaller kidneys with fewer nephrons. This constitutes a first kidney “hit,” which enhances vulnerability to further kidney “hits.”
- A reduced nephron number is associated with increased risk of hypertension and kidney disease in animals and humans.
- Clinical surrogates for low nephron number include preterm birth, low birth weight, female sex, short stature, minoritized ethnicities, older age, and glomerulomegaly on kidney biopsy.
- Long-term follow-up of at-risk infants is advised to permit early diagnosis and intervention to prevent or delay kidney disease.
- Avoidance of overweight/obesity in those born too small or too soon is important to reduce life-time risk of hypertension and kidney disease.

The large variation in individual susceptibility to kidney disease and other chronic diseases is not easily explained.¹ Genetic factors are important determinants of development and function of major organ systems, as well as susceptibility to disease. Rare genetic and congenital abnormalities leading to abnormal kidney development are associated with the occurrence of subsequent kidney dysfunction, often manifesting early in life.^{2,3} Most kidney diseases in the general population, however, are not ascribable to genetic

mutations, with the most common etiologic associations with end-stage kidney failure (ESKF) worldwide being the polygenic disorders of diabetes and hypertension. Of note here is the association between mutations in the apolipoprotein-1 (*APOLI*) gene in people of African descent, with increased predisposition to the development of kidney disease including HIV-associated nephropathy and focal and segmental glomerulosclerosis (FSGS). However, searches for other specific gene polymorphisms or mutations have

not implicated specific genes but instead point to a likely complex interplay between polygenic predisposition and environmental factors in the development of hypertension, diabetes, and kidney disease.⁴⁻¹⁰ Hypertension and kidney disease prevalence vary between populations from different ethnic backgrounds, with high rates observed among Australian Aboriginal and Torres Strait Islander (Indigenous) people, Indigenous Americans, and people of African descent.¹¹⁻¹³ It is well established that lifestyle and socioeconomic factors can pose significant risks for the development and persistence of hypertension and diabetes in the general population, with obesity becoming an increasing concern, especially in lower-resource settings, and most concerning beginning in childhood.^{7,14,15} However, more and more evidence is also pointing to far-reaching effects of the intrauterine environment and early postnatal growth on organ development, organ function, and subsequent susceptibility to adult disease.¹⁶⁻²¹ Stresses experienced during fetal life (for which low birth weight [LBW], being small for gestational age, preterm birth, or high birth weight may be surrogate markers), such as maternal malnutrition, ill health, preeclampsia, or gestational diabetes, or those experienced in early childhood, such as poor early nutrition, infections, and environmental exposures, may “program” long-term organ function and may be the first in a succession of challenges or “hits” that ultimately manifest in overt disease.^{22,23}

This chapter outlines the effects of fetal and early life programming on kidney development (particularly nephrogenesis), nephron endowment, and the risks of hypertension and kidney disease throughout the life course. Major congenital renal anomalies are discussed elsewhere in this book (see [Chapter 71](#)). In addition, it must be borne in mind that adverse early life exposures can also predict later-life diabetes, cardiovascular disease, metabolic syndrome, and preeclampsia; therefore kidney function and disease susceptibility may be additionally impacted through developmental programming of these disorders and, in turn, may impact outcomes of these disorders, the discussion of which is beyond the scope of the current chapter.^{18,20,22,24-27}

DEVELOPMENTAL PROGRAMMING

The association between adverse intrauterine events and subsequent cardiovascular disease has long been recognized.^{17,28-30} In early studies, adults born with a LBW were found to have higher cardiovascular morbidity and mortality than those of normal birth weight.³¹ Subsequently, evidence from diverse populations has confirmed these findings and expanded them to include other conditions, such as hypertension, chronic kidney disease (CKD), impaired glucose tolerance, type 2 diabetes, obesity, and preeclampsia.^{28,32-38}

The process through which an environmental insult experienced early in life can predispose to adult disease is known as “developmental programming,” which refers to the observation that an environmental stimulus experienced during a critical period of development in utero or in early postnatal life can induce long-term structural, physiologic, metabolic, and epigenetic effects in the organism.^{17,28} These changes, which can occur to promote short-term

survival, may ultimately affect long-term health and disease susceptibility. Often termed “developmental origins of health and disease,” this phenomenon can have far-reaching implications in that the effects can be perpetuated across generations.³⁹⁻⁴¹ Other whole-organism programming effects may occur from conception or even earlier with respect to gamete quality and epigenetic programming.^{41,42}

KIDNEY DEVELOPMENT

The most comprehensive descriptions of the process and timing of human kidney development are available from microdissection studies performed in the 1960s.^{43,44} Briefly, the metanephric kidneys begin formation in embryos at approximately 5 weeks of gestation through the process of branching morphogenesis (forming the multipapillary collecting system), with the first nephrons (renal corpuscle and associated tubules) forming in a centrifugal manner from approximately 9 weeks (development of the kidneys is discussed in detail in [Chapter 1](#)).^{45,46} Initially, nephrons are formed individually at the ampullae of ureteric bud branches and later in an arcade pattern with multiple nephrons forming per branch^{45,46}; the majority (~60%) of nephrons develop in the third trimester of pregnancy.⁴⁷ Nephrogenesis is normally completed in late gestation, with the timing of the cessation of nephrogenesis varying between individuals; it has been reported to have been completed as early as 32 weeks’ gestation in one case or still ongoing in other cases at 37 weeks’ gestation.^{48,49} In infants born prematurely at a time when kidney development is ongoing, nephrogenesis may continue postnatally, but this process does not proceed normally (as discussed later).⁴⁹⁻⁵² In both term and preterm infants, processes of glomerular and tubular growth and maturation continue after cessation of nephrogenesis into postnatal life.^{53,54} The critical periods of development during which the immature kidneys are vulnerable to programming therefore extend through the majority of the gestation period for fetuses and into the postnatal period. Thus the time period during which the developmental and maturational processes of fetal and infant kidneys may be influenced by adverse exposures, and thereby program life-long renal health, is substantial.

Adverse exposures that impact kidney development are numerous and diverse and may be of maternal, placental, fetal, or environmental origin.^{1,22} The type, timing, duration, and severity of exposure (or combination of exposures) all influence the way in which the kidneys are affected during development (e.g., with a reduction in nephron formation), as well as the long-term impact on blood pressure and kidney function.

MARKERS OF PROGRAMMING RISK

Initial human studies focused on LBW, <2.5 kg as a marker for risk of developmental programming in the kidney. Over time, it has been recognized that preterm birth, being born small for gestational age (SGA), intrauterine growth restriction (IUGR), having a high birth weight (HBW), or being exposed to preeclampsia, a diabetic pregnancy or maternal obesity, are also important indicators of programmed risk.⁵⁵⁻⁵⁸ [Table 20.1](#) outlines the definitions of birth weight and gestational age categories that are referred

Table 20.1 Definitions of Birth Weight and Prematurity Categories

Category	Definition
Birth Weight Categories	
Normal birth weight (NBW)	>2500 g and <4000 g (usually)
Macrosomic birth weight	>4000 g
High birth weight (HBW)	>4500 g
Low birth weight (LBW)	<2500 g
Very low birth weight (VLBW)	<1500 g
Extremely low birth weight (ELBW)	<1000 g
Appropriate for gestational age (AGA)	10-90th percentile of birth weight for gestational age/within ± 2 standard deviations of the mean birth weight for gestational age
Large for gestational age (LGA)	>90th percentile of birth weight for gestational age/>2 standard deviations above the mean birth weight for gestational age
Small for gestational age (SGA)	<10th percentile of birth weight for gestational age (usually)/>2 standard deviations below the mean birth weight for gestational age
Intrauterine growth restriction (IUGR)/Fetal growth restriction (FGR)	Drop in fetal size percentile across gestation and/or abnormalities in umbilical arterial blood flow. Diagnosis after birth based on infant anthropometry.
Extrauterine growth restriction (EUGR)	Infant length or weight <10th percentile for age at hospital discharge
Small and vulnerable newborn (SVN)	Preterm, SGA, and/or LBW
Gestational Age at Birth Categories	
Extremely preterm	<28 weeks of gestation
Very preterm	<32 and >28 weeks of gestation
Moderately preterm	<34 and >32 weeks of gestation
Late preterm	<37 and >34 weeks of gestation
Term	>37 weeks of gestation

The assignment of infants to birth weight for gestational age categories can be variable as they may be based on region-specific growth charts. Some categories, such as SGA, IUGR, and EUGR, do not yet have an international consensus on definition. Gestational age may be determined by date of mother's last menstrual period and/or a dating ultrasound, or by Dubowitz examination after birth.

to throughout this chapter. The term *Small and Vulnerable Newborns* has been coined in a *Lancet* series to encompass all infants born too soon and/or too small, to improve awareness that these conditions are risk factors for neonatal

death, as well as long-term chronic illness⁵⁹ as illustrated in Fig. 20.1. The term LBW has been deemed no longer granular enough, and the new classification of small and vulnerable newborns includes the three mutually exclusive categories of preterm SGA, preterm, and appropriate weight for gestational age (AGA) and term SGA.²³ Globally, in 2020, 11.9 million live births were preterm AGA, 1.5 million were preterm SGA and 21.8 million were term SGA, comprising 35.2 million babies and 26.2% of all live births (Fig. 20.1 A and B).²³ The global incidence of HBW ranges from 5% to 20% in high-income and 0.5% to 15% in lower-income countries.⁵⁷ Maternal hypertension and diabetes during gestation occur with a frequency of around 8% to 10% and 15% of pregnancies, respectively.^{60,61} By 2025 it is estimated that 21% of women will be obese.⁶² Overall, around one in five infants born each year are therefore at risk of adverse developmental programming and will thereby be at risk of chronic disease later in life.⁶³

Despite the fact that birth weight is a widely accessible and easily obtained marker of infant morbidity and mortality risk, in 2020, global data on birth weight was only available from 81% of countries.^{23,64} Data on preterm births and SGA babies were only reported by 53% and 4% of countries, respectively.²³ Tracking of these markers is therefore incomplete, leading to missed opportunities to identify babies at risk for future kidney and cardiovascular disease. In addition, although LBW, SGA, and preterm birth are currently the best available clinical surrogates for an adverse early life environment, some programming stresses may not manifest as such, and therefore infants at risk may not be identified. For example, an infant affected by IUGR may still be delivered AGA if growth restriction occurred late in gestation (see Fig. 20.1B); a large proportion of babies born SGA have a birth weight above 2.5 kg and would therefore not be classified as LBW. These infants may, however, still be at risk of long-term consequences.⁶⁵ Conversely, some babies may be normally constitutionally small and may therefore not be at increased risk. Ongoing work is required to develop more sensitive measures of developmental stress for the identification and mitigation of risk at birth.

THE NEPHRON NUMBER HYPOTHESIS

In 1988, Brenner and colleagues⁶⁶ proposed that congenital (programmed) low nephron number may explain why some individuals are susceptible to hypertension and kidney injury, whereas others may seem relatively resistant under similar circumstances (e.g., sodium excess and diabetes mellitus). Low whole-kidney glomerular filtration surface area (a product of average glomerular capillary surface area and total nephron number) would result in reduced sodium excretory capacity, enhancing susceptibility to hypertension, and a relatively reduced renal reserve capacity, limiting compensation for renal injury (Fig. 20.2). A kidney starting with a lower nephron number would conceivably reach a critical reduction of nephron mass, either through age or in response to additional renal insults (second hits), earlier than a kidney with a greater nephron complement, thereby predisposing to hypertension and/or kidney dysfunction at an earlier age. The Brenner hypothesis proposed that an association between low nephron number and LBW, for example, could explain high prevalences of hypertension

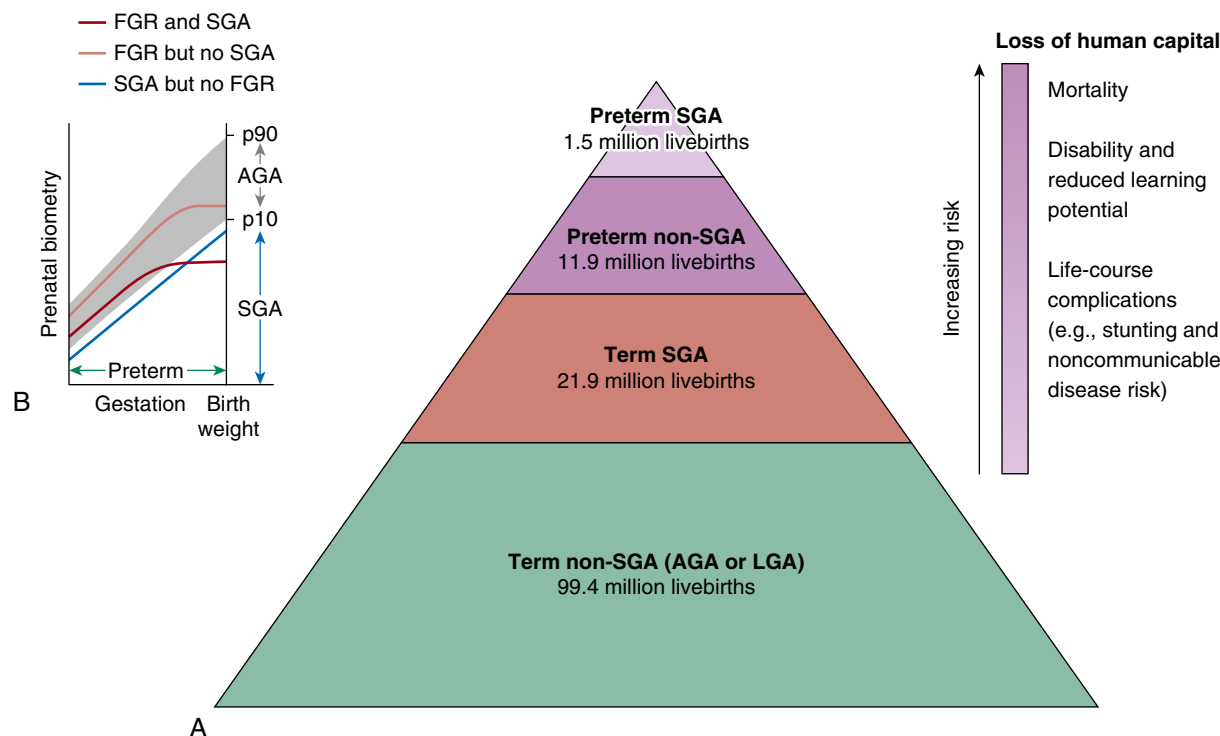


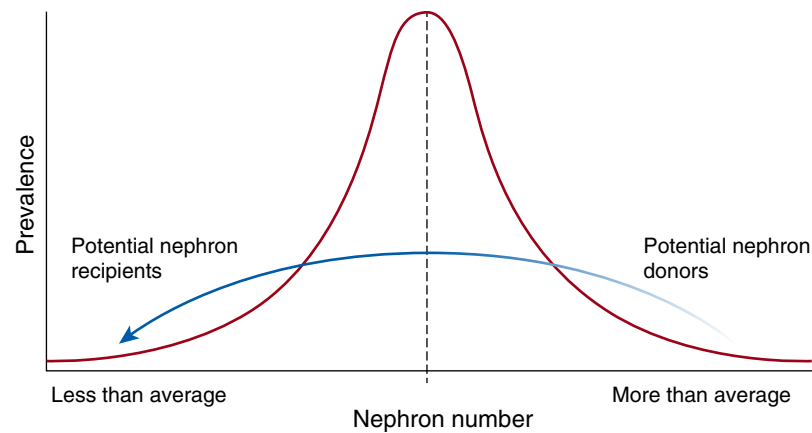
Fig. 20.1 (A) Global Burden of Small and Vulnerable Newborns in 2020. (B) Potential in utero growth trajectories reflecting the presence or absence of growth restriction and timing of birth, which impact weight for gestational age. Low birth weight per se is no longer used as a risk threshold. Instead, appropriateness of weight for gestational age provides more granular identification of programming risk in infants born preterm or at term. AGA, Appropriate for gestational age; FGR, fetal (intrauterine) growth restriction; LGA, large for gestational age; SGA, small for gestational age. (Adapted from Terstappen F, Lely AT. Long-term renal disease after prematurity or fetal growth restriction: who is at risk? *Nephrol Dial Transplant.* 2020;35:1087–1090 and Lawn JE, Ohuma EO, Bradley E, et al. Small babies, big risks: global estimates of prevalence and mortality for vulnerable newborns to accelerate change and improve counting. *Lancet.* 2023;401[10389]:1707–1719.^{23,783})

and CKD observed in populations with coexistence of lower birth weights (e.g., as observed in minoritized populations in high-income settings such as African Americans).^{67–70} Nephron number therefore became a major focus of kidney developmental programming research.

ESTIMATION OF NEPHRON NUMBER

An obstacle to investigation of the role of nephron number in programming of hypertension and kidney disease risk has been the difficulty in accurately counting or estimating the total number of nephrons in a kidney.⁷¹ Acid-maceration techniques, as well as early model-based or design-based stereologic techniques for analysis of histologic samples have been used, but these are prone to bias because of required assumptions, extrapolations, and operator sensitivity.^{71–73} Over the past 30 years, the design-based (often termed “unbiased”) disector/fractionator method has emerged as the gold-standard method for estimating total glomerular (and thereby nephron) number because it generates accurate (no bias) and precise (low-variance) estimates.^{71–73} With the disector/fractionator method, the disector principle⁷⁴ is used to sample glomeruli in representative histologic samples of the three-dimensional renal cortex with equal probability (regardless of their size, shape, and location). Sampled glomeruli are counted in a “known” fraction of kidney tissue, allowing simple algebraic estimation of the total number of glomeruli in the whole kidney.

Most estimates of glomerular number have been obtained from whole kidneys obtained at autopsy. While useful, reliance on autopsy samples precludes studies in living animals and people, as well as longitudinal studies. In recent years, several new approaches for imaging, counting, and sizing glomeruli have emerged, some of which do not require autopsy tissue. For example, the methods of Denic and colleagues^{75–77} and Fulladosa and colleagues,⁷⁸ which use computed tomography (CT) or magnetic resonance imaging (MRI) estimation of cortical volume, and an estimation of glomerular numerical density derived from biopsies, do not require autopsy tissue. The number of functional (nonsclerotic) glomeruli has also been estimated by dividing the whole-kidney ultrafiltration coefficient (K_f ; calculated from renal plasma flow from para-amino-hippurate clearance) by single-nephron K_f estimated from electron microscopic measurements of the glomerular filtration barrier using kidney biopsies.^{79,80} These biopsy-based methodologies have allowed for a number of small-to large-scale studies of nephron number to be conducted in living people including healthy donors and people with kidney disease; although limited to sampling only a small number of glomeruli, the accuracy of these techniques continues to improve (Table 20.2).^{75,76,78–83} An intriguing short report also examined 24-hour urine uromodulin excretion as a potential correlate for nephron number before and after living donor nephrectomy.⁸⁴ Urine uromodulin excretion fell by around half at 3 days after nephrectomy and rose to around two-thirds of the predonation value by



Renal functional reserve	None or poor	Sufficient	Excellent
Preeclampsia risk	High	Low	Low
Hypertension onset	Early in life	Late in life	None or very late in life
Proteinuria level relative to disease stage	High	Adequate	Low
Trigger for sudden SCr↑ or AKI	Mild injury	Modest injury	Severe injury
CKD progression	Fast	Slow	Minimal
Obesity	High risk of adaptive FSGS	Low risk	No adaptive FSGS
Diabetes	Risk of early nephropathy	Late nephropathy	No nephropathy
Glomerulonephritis	Fast CKD progression	Slow progression	Minimal progression
Post-donation kidney failure risk	High	Low	No risk for donor

Fig. 20.2 Nephron endowment follows a normal distribution across the population. Individuals with a low nephron number have a limited capacity to tolerate superimposed kidney insults or to compensate for nephron loss and are therefore at risk of developing (progressive) kidney disease. People with higher nephron numbers are likely more resistant to developing a critical loss of kidney function in the setting of kidney injury because they have an abundant nephron reserve. AKI, Acute kidney injury; CKD, chronic kidney disease; FSGS, focal segmental glomerulosclerosis; SCr, serum creatinine. (From Luyckx VA, Rule AD, Tuttle KR, et al. Nephron overload as a therapeutic target to maximize kidney lifespan. *Nat Rev Nephrol.* 2022;18(3):171–183.⁷⁸⁴)

1 year, possibly reflecting the known adaptive increase in single kidney function.⁸⁴

An MRI technique using cationic ferritin to label glomeruli also shows promise. Cationic ferritin is injected intravenously into animals or into ex vivo human kidneys via the renal artery, and it binds to anionic charges on the glomerular basement membrane. With this approach, all glomeruli in whole rat,^{85–87} mouse,^{88,89} and ex vivo human^{90,91} kidneys have been imaged, counted, and sized, providing, for the first time, the glomerular volume distribution for whole kidneys. In addition, mapping of renal pyramids and the identification of areas of global sclerosis in whole human kidneys have shown this technique to have great potential for improving our understanding of regional renal structure and pathology.⁹¹ New tissue-clearing approaches also facilitate imaging (using light sheet microscopy), counting, and sizing of all glomeruli in ex vivo rodent kidneys,⁹² but whether modifications to this approach can be safely used in vivo remains questionable. Overall, with these advances, accurate quantitation of nephron mass in vivo may become possible in the not-too-distant future.

HUMAN NEPHRON NUMBER VARIABILITY

In early studies, humans were believed to have an average of approximately 1 million nephrons per kidney.⁷² This implied that there was little variability in human nephron number and therefore likely little relationship with adult disease risk. Such studies, however, were performed using techniques that are prone to bias.^{71–73} Moreover, many of the early studies analyzed small numbers of kidneys, and therefore population variability in nephron number was not observed.

The disector/fractionator method requires sampling from a whole kidney; therefore all studies using this gold-standard technique have been performed on autopsy samples. An early study in 1992 among 37 normal Danish adults reported an average glomerular (nephron) number of 617,000 per kidney (range: 331,000–1,424,000).⁷² A positive correlation was noted between nephron number and kidney weight, but this has not been the case in all studies.⁹³ Nephron number tends to be lower in women than men and lower in people with shorter adult height.^{75,77,94} A large variation in nephron number between individuals has subsequently

Table 20.2 Variability of Nephron Number Per Kidney in Humans

Population	Sample Size	Mean	Range	Fold ^a	Reference
British ^b	32	1,309,280			696
British ^b	18		1,200,000-2,300,000	1.9	697
French ^c	28	1,107,000	655,000-1,554,000	2.4	698
Danish	37	617,000	331,000-1,424,000	4.3	72
German (hypertensive + normotensive)	20	1,074,414	531,140-1,959,914	3.7	106
German hypertensive	10	702,379	531,140-954,893	1.8	
German normotensive	10	1,429,200	884,458-1,959,914	2.2	
African American	105	884,938	210,332-2,026,541	9.6	547
White American	84	843,106	227,327-1,660,232	7.3	
White Canadian (infants)	15		246,181-1,106,062	4.5	96
Non-Indigenous Australian	24	861,541	380,517-1,493,665	3.9	439
Indigenous Australian	19	713,209	364,161-1,129,223	3.1	
Senegalese males	47	992,353	536,171-1,764,421	3.3	546, 699
White and African American, Indigenous and non-Indigenous Australian, and Senegalese	420	901,902	210,332-2,702,079	12.8	511
Japanese males (hypertensive + normotensive)	18	544,819	306,092-960,756	3.1	443
Japanese hypertensive (males)	9	423,498	306,092-550,222	1.8	
Japanese normotensive (males)	9	666,140	419,282-960,756	2.3	
MCD patients (Japanese) ^d	75	931,000	107,000-1,877,000	17.5	82
Kidney donors (Japanese) ^d	44	710,000	290,000-1,340,000	4.6	81
Kidney donors (American) ^d	1638	873,696			75
Kidney donors and recipients (Spanish) ^e	39	730,000			78
Kidney donors (American) ^f	3	1,236,667			90
Kidney donors (American) ^g	34	631,500			700
Kidney donors (American) ^g	19	641,730	396,554-713,399	1.8	701
Kidney donors (American) ^g	51	605,592			702
Kidney donors, hypertensive (American) ^g	10	359,000	151,000-581,000	3.8	
Kidney donors, normotensive (American) ^g	10	558,000	275,000-847,000	3.1	
Kidney donors (American) ^h	3020	1,170,000	1,160,000-1,190,000	1.0	77
Tumor nephrectomy (American) ^h	1354	990,000	970,000-1,010,000	1.0	

^aFold change from lowest to highest nephron number. Studies used the gold-standard physical disector/fractionator method for estimating nephron number, except for ^bWeibel-Gomez stereology, ^cacid maceration, ^denhanced or unenhanced CT and biopsy (Weibel-Gomez), ^eMRI and biopsy (Weibel-Gomez), ^fMRI with cationic ferritin, ^gwhole-kidney ultrafiltration coefficient (K_f) techniques, and ^himproved MRI or CT and biopsy (less bias—nephron numbers 27% higher than prior calculations). Studies involving infants with potentially incomplete nephrogenesis were excluded.

CT, computed tomography; MCD, minimal change disease; MRI, magnetic resonance imaging.

(Adapted from Puellas VG, Hoy WE, Hughson MD, et al. *Curr Opin Nephrol Hypertens*. 2011;20(1):7–15 and Denic A, Lieske JC, Chakkera HA,

been reported in several other populations (see Table 20.2), with the widest range being a 13-fold variation in the largest sample investigated (159 African American people; 210,332 to 2,702,079 per kidney).⁹⁵

A single study of nephron number in 15 American infants aged younger than 3 months (who died from sudden infant death syndrome or congenital abnormality) showed a range of 246,181 to 1,106,062 nephrons per kidney, confirming that nephron number variability is already evident in early life, likely reflecting an impact of developmental programming, rather than nephron loss,

on nephron number.⁹⁶ A range of 8 to 12 glomerular generations (number of glomeruli alongside a medullary ray, from inner cortex (earliest formed nephron) to outer cortex (most recently formed nephron), a proxy measure of nephron number in cases where a whole kidney is not available for analysis, has also been reported in fetuses and neonates with completed nephrogenesis (Fig. 20.3).^{48,49} Wide variation is also evident in the number of papillae/lobes formed in the human kidneys (a product of variation in early branching morphogenesis and remodeling processes), ranging from 3 to 22^{97,98}; however, whether

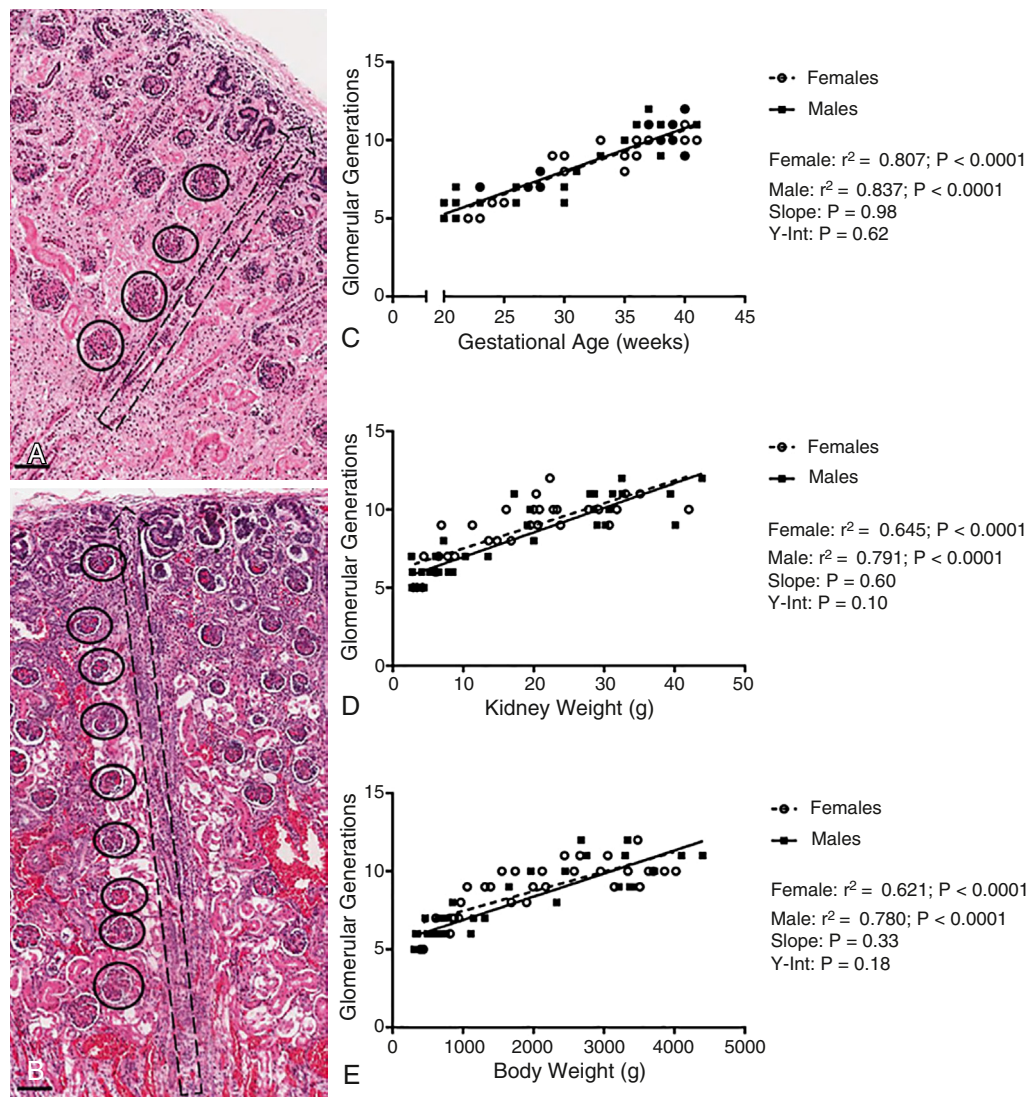


Fig. 20.3 Representative images of medullary rays (dotted lines) and generations of glomeruli (circled) in hematoxylin-eosin-stained sections of renal cortex from fetuses at (A) 22 weeks' gestation (4 glomerular generations formed) and (B) 33 weeks' gestation (9 glomerular generations formed). Scale bar = 100 μ m. Linear regression analyses of the number of glomerular generations formed within the kidney versus gestational age (C), kidney weight (D), and body weight (E) in female (O) and male (■) fetuses and newborn infants. (From Ryan D, Sutherland MR, Flores TJ, et al. Development of the human fetal kidney from mid to late gestation in male and female infants. *eBioMedicine*. 2018;27:275–283.⁴⁸)

an association exists between lobe number and nephron number is currently unknown.

The number of nephrons formed by the completion of nephrogenesis is termed nephron endowment. Three primary processes that are considered to play critical roles in determining nephron endowment are branching of the ureteric tree (branch initiation, elongation, and remodeling); condensation of metanephric mesenchymal cells at the ureteric branch tips; and conversion of these mesenchymal condensates into nephron epithelium (cellular proliferation, migration, metabolism, differentiation, and survival).^{99–102} Even minor alterations in the nephrogenic process may have a major impact on nephron endowment, with a 2% decrease in ureteric tree branching efficiency estimated to result in a 50% reduction in final nephron complement after 20 generations of branching.¹⁰³ Kidney development is a complex process involving the tightly controlled expression of many genes and constant remodeling.^{45,46,99–102,104,105} The molecular regulation of kidney development is exhaustively reviewed elsewhere (see [Chapter 1](#)).^{99–102} The specific molecular mechanisms whereby nephron

numbers may be affected and/or function altered, however, are not yet completely understood. Gene expression can be influenced by genotype, inherited (epigenetically programmed) changes in DNA methylation and histone modification, as well as their interaction with factors in the intrauterine and early postnatal environment. Environmental perturbations that result in reduced nephron endowment are discussed later.

Regardless of the cause, the previously underappreciated variability in total nephron number may influence susceptibility to hypertension and kidney disease.^{72,95,106} Additionally, age-related nephron loss may compound this susceptibility. In general, numbers of viable glomeruli are reduced in kidneys from older people, owing to age-related glomerulosclerosis and obsolescence (see [Chapter 21](#)).^{72,94} Rates of loss of glomeruli per year have been variously reported as approximately 6750 glomeruli per year after the age of 18 years,⁹⁴ 4500 glomeruli per year,¹⁰⁷ and 6200 glomeruli per year.⁷⁵ Denic and colleagues⁷⁵ estimated nonsclerotic and sclerotic glomerular number in 1638 living donor kidneys using a CT/biopsy method. A striking decrease of almost 50% in the numbers of

nonsclerotic glomeruli was observed between 18- and 29-year-olds (an average of 990,661 glomeruli) and donors aged 70–75 years (520,410 glomeruli). Moreover, the total number of sclerotic plus nonsclerotic glomeruli was much lower in the older donors, suggesting that many glomeruli had disappeared without trace following global glomerulosclerosis (and/or older donors were born with fewer nephrons).

PLAUSIBILITY OF THE NEPHRON NUMBER HYPOTHESIS

In support of the nephron number hypothesis, it is known that persons born with severe nephron deficits (e.g., unilateral renal agenesis, bilateral renal hypoplasia, and oligomeganephronia) develop progressive proteinuria, glomerulosclerosis, and renal dysfunction with time.^{3,108} Analogously therefore people born with nephron numbers at or below the median level may be more susceptible to superimposed postnatal factors that act as subsequent “hits” (see Fig. 20.2). Indeed, with a refinement of their technique of *in vivo* nephron counting, Denic and colleagues found that a low nephron number (defined as below the 10th percentile adjusted for age and sex) predicted future CKD risk (hazard ratio [HR] 3.15, 95% confidence interval [CI] 2.10–4.74) and kidney failure (HR 7.12, 95% CI 3.47–14.6) in patients who underwent tumor nephrectomy (Fig. 20.4).⁷⁷ This is the first study to directly prospectively correlate nephron number with kidney function over time and supports the hypothesis that in the setting of a second hit (nephrectomy, comorbid disease), kidneys with fewer nephrons may lose function more rapidly with age.

The counterargument to the nephron number hypothesis, however, is that in experimental animals and in humans, removal of one kidney (presumed reduction in nephron number of 50%) under varying circumstances may be associated with higher blood pressures or low-grade proteinuria but does not always lead to hypertension and kidney disease.^{73,109,110} Of interest, uninephrectomy on postnatal day 1 in rats or fetal uninephrectomy in sheep, at a time when nephrogenesis is not yet completed, leads to adult hypertension before evidence of renal injury.^{111–113} Other experimental studies have shown greater compensatory glomerular hypertrophy and increased susceptibility to hypertension and injury, in response to neonatal nephrectomy compared with nephrectomy in later life.^{114–116} Unilateral *in utero* nephrectomy in sheep was associated with significant renal hypertrophy and a 45% increase in nephron number in the remaining kidney, suggesting that a congenital solitary kidney may have a higher than normal nephron endowment and therefore be relatively protected compared with an acquired single kidney.^{117,118} The timing of kidney loss may impact adaptive changes in the contralateral kidney.

In humans, a study of 97 people with a radiologically normal single kidney, aged 2.5–25 years, found that renal function declined faster over time in those with acquired single kidneys (surgical removal of the other kidney) versus congenital single kidneys, although blood pressures and proteinuria were not different.¹¹⁹ These findings may be confounded by indication for nephrectomy, however, as approximately 25% of the nephrectomies were performed for obstruction, which may impact nephron development in the contralateral kidney, as discussed later.^{120,121} In contrast, among 944 children with a solitary functioning kidney, at

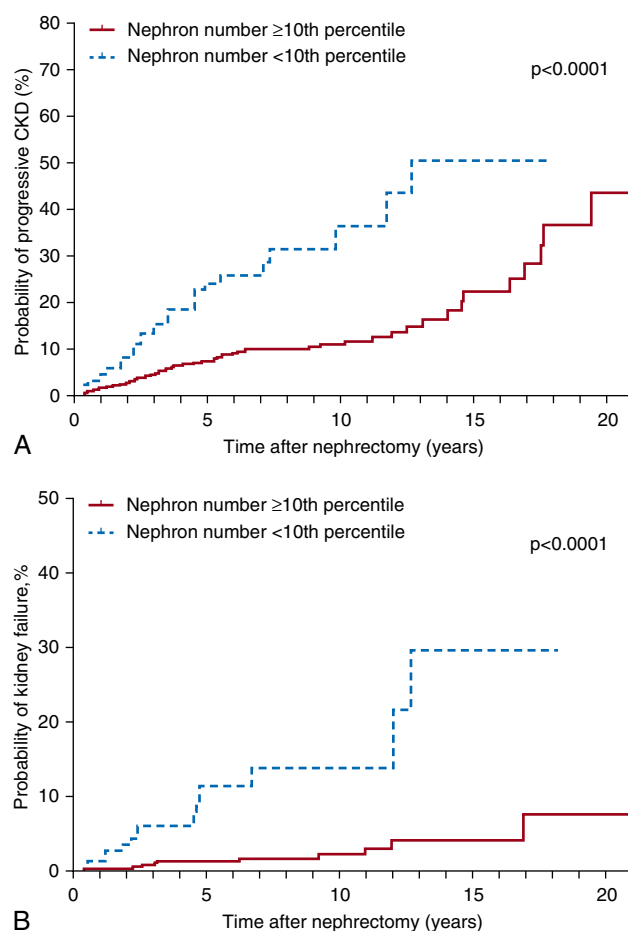


Fig. 20.4 Risk of adverse outcomes after nephrectomy among kidney tumor patients with nephron number above or below the 10th percentile for age. Low nephron number of retained kidney (< 10 th percentile for age and sex in healthy subset) predicted progressive chronic kidney disease (CKD) (A) and kidney failure (B) (Log-rank $P < 0.0001$ for both). (From Denic A, Mullan AF, Alexander MP, et al. An improved method for estimating nephron number and the association of resulting nephron number estimates with chronic kidney disease outcomes. *J Am Soc Nephrol*. 2023;34:1264–1278.⁷⁷)

a median of 12.8 years of follow-up from diagnosis (age 18 years), major risk factors associated with reduced estimated glomerular filtration rate (eGFR)/CKD included congenital (vs. acquired) single kidney, anomalies in the single kidney, and an elevated body mass index (BMI). Birth weights and gestational ages tend to be lower in children with congenital kidney anomalies, likely further impacting their remaining nephron number.^{122,123} Importantly, the proportion of children with kidney dysfunction increased with age in both the congenital and acquired solitary functioning kidney subgroups, suggesting accumulation of kidney injury through hyperfiltration and second hits over time in both groups.¹²⁴

PROGRAMMING OF KIDNEY STRUCTURE

PROGRAMMING OF NEPHRON NUMBER IN HUMANS

As noted earlier, total nephron number varies widely in the normal human population (see Table 20.2).^{95,125} A

significant proportion of the interindividual variability in nephron number appears to be already present perinatally, demonstrating a strong developmental effect.^{49,126} Overall, the data support a direct relationship between nephron number and birth weight and an inverse relationship between nephron number and glomerular volume.^{96,127-129} This relationship suggests that larger glomeruli may reflect compensatory hyperfiltration and hypertrophy in people with fewer nephrons and may therefore be a surrogate marker for a reduced nephron number.^{94,128}

Hinchliffe and associates^{130,131} studied nephron number in preterm or term stillborn infants or infants who died at 1 year of age and who were born AGA or SGA. At both time points, growth-restricted infants had significantly fewer nephrons than controls (Fig. 20.5A). Manalich and colleagues¹²⁸ examined the kidney structure of neonates who died within 2 weeks of birth, in relation to their birth weights (Fig. 20.5B). A significant direct correlation was found between glomerular number and birth weight, as well as a strong inverse correlation between glomerular volume

and glomerular number, independent of sex and race. LBW infants had on average a 12% reduction in nephron number compared with non-LBW infants. Similarly, a study utilizing the acid-maceration technique to count nephrons also showed a linear correlation between nephron number and birth weight percentile; an average of 30% fewer nephrons were counted in the kidneys of SGA infants compared with those that were AGA.¹³² Although only a small number of studies have evaluated nephron numbers in infants, each with a low sample size, they all reinforce that an adverse intrauterine environment, which may manifest as LBW, SGA and/or IUGR, and preterm birth, is associated with a congenital reduction in nephron endowment and an early, compensatory increase in glomerular volume.

An association among birth weight, nephron number, and glomerular size has also been observed in later life (Table 20.3). Hughson and colleagues¹²⁷ reported a linear relationship between glomerular number and birth weight and calculated a regression coefficient predicting an increase of 257,426 glomeruli per 1 kg increase in birth weight (Caucasian

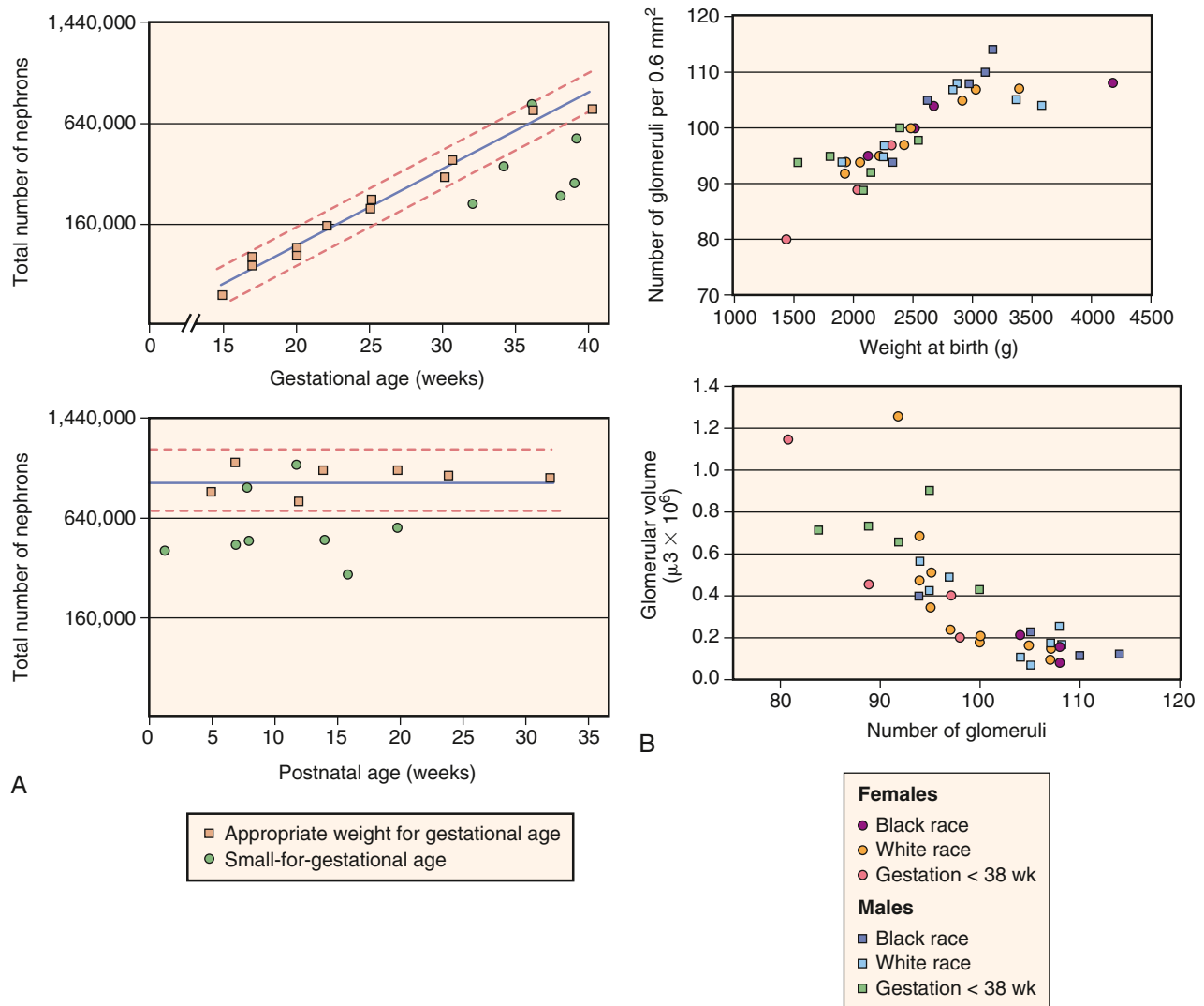


Fig. 20.5 Effect of intrauterine growth restriction on nephron number in humans. (A) Nephron number at birth in relation to gestational age (upper panel) and lack of postnatal catch-up in nephron number (lower panel). (B) Relationship between birth weight and glomerular number (upper panel) and between glomerular number and glomerular volume (lower panel) in neonates. (A from Hinchliffe SA, Lynch MR, Sargent PH, et al. The effect of intrauterine growth retardation on the development of renal nephrons. *Br J Obstet Gynaecol.* 1992;99:296–301¹³¹ and B from Manalich R, Reyes L, Herrera M, et al. *Kidney Int.* 2000;58:770–773.¹²⁸)

Table 20.3 Glomerular Characteristics by Birth Weight in Humans

Birth Weight Mean (Range), kg	N	Number of Glomeruli ^a	Mean Glomerular Tuft Volume ($\mu\text{m}^3 \times 10^6$)	Total Glomerular Tuft Volume (cm^3)
2.65 (1.81–3.12)	29	770,860 (658,757–882,963)	9.2	6.7
3.27 (3.18–3.38)	28	965,729 (885,714–1,075,744)	7.2	6.8
3.93 (3.41–4.94)	30	1,005,356 (900,094–1,110,599)	6.9	6.6

^aAdjusted for age, gender, race, and body surface area.

From Hoy WE, Hughson MD, Bertram JF, et al. *J Am Soc Nephrol*. 2005;16(9):2557–2264.

and African American population across a wide age range; four individuals were born LBW). The association was higher when excluding adults (likely affected by nephron loss), with a calculated increase of 518,038 glomeruli per 1 kg increase in birth weight in infants and children (although only 1 was born LBW). In 140 American adults aged 18 to 65 years who died of various causes, a significant correlation was also observed between birth weight and glomerular number.¹²⁹ Glomerular volume was again inversely correlated with number. Total glomerular number did not differ statistically between African American and white Americans, although the distribution of glomerular numbers among African Americans appeared bimodal. The range of nephron number was also greatest in African Americans. Significantly, however, none of the people in this study had been born LBW.¹²⁹

In humans, approximately two-thirds of the nephrons develop during the last trimester, making this the window of greatest susceptibility to adverse effects, although earlier insults can also impact nephrogenesis.^{130,133} In infants born preterm with developmentally immature kidneys, nephrogenesis can continue postnatally and is therefore susceptible to insults after birth.^{49–51} Rodriguez and colleagues⁵⁰ studied kidneys at autopsy from 56 extremely preterm infants compared with 10 term-born infants as controls. Evidence of active glomerulogenesis was seen in preterm infants who died before postnatal day 40, but it was absent in older infants. The radial glomerular counts (glomerular generations; see Fig. 20.3) correlated with gestational age and were lower in the kidneys of preterm versus term infants, suggestive of a nephron deficit; however, it is to be noted that SGA/IUGR infants were included in this study, which may have confounded these results. Carpenter and colleagues⁵¹ studied 7 preterm infants who survived beyond 40 days and found evidence of ongoing nephrogenesis in 4 infants, including at day 62 in 1 (examined at 34 weeks post conception), suggesting that the duration of ongoing postnatal nephrogenesis is likely dependent on gestational age at birth. In this study, however, nephrogenesis appeared to cease 2 weeks earlier in kidneys from preterm infants compared with controls who had died at the matched postmenstrual age in utero or within 4 days of birth. This observation suggests additional postnatal nephron stresses in the surviving preterm infant kidneys.⁵¹ Another autopsy study by Sutherland and colleagues⁴⁹ of 28 kidneys from preterm neonates (with 2–68 days postnatal survival) compared with 32 stillborn gestational age controls also confirmed that nephrogenesis is ongoing postnatally in infants born preterm. However, there was evidence of accelerated postnatal maturation with an increased number

of glomerular generations and a concurrent decreased width of the nephrogenic zone in the kidneys of preterm neonates compared with stillborn infants at the comparative age post conception, consistent with the observed earlier cessation of nephrogenesis by Carpenter and colleagues.⁵¹

In their earlier study, Rodriguez and colleagues⁵⁰ also stratified the preterm infants by the presence or absence of kidney failure. Among infants surviving longer than 40 days, those with kidney failure (serum creatinine >2.0 mg/dL) had significantly fewer glomerular generations than those without. This cross-sectional observation may suggest that kidney failure inhibited glomerulogenesis or, conversely, that fewer glomeruli lowered the threshold to develop kidney failure in these infants. Preterm infants surviving longer than 40 days without kidney failure exhibited glomerulomegaly, which may reflect, at least in the short term, a compensatory renoprotective response.⁵⁰

The appearance of renal corpuscles with shrunken glomerular tufts and enlarged Bowman spaces (resembling atubular glomeruli) has also been noted in the outer renal cortex of kidneys from preterm infants, suggestive of a developmental defect and/or injury to those nephrons newly formed in the extrauterine environment; 0% to 13% of glomeruli were affected per kidney.⁴⁹ It is probable that the early loss of these affected glomeruli, along with a hastened cessation of nephrogenesis after birth, contributes to a reduced nephron endowment in infants born preterm; however, this is yet to be confirmed. In support of this possibility, reduced glomerular density and higher glomerular volume were associated with lower gestational age and birth weights in kidney biopsies of Japanese children with proteinuric kidney diseases, suggesting reduced nephron numbers and compensatory glomerular hypertrophy.¹³⁴

KIDNEY SIZE AS A CORRELATE FOR NEPHRON NUMBER

An analysis of the association between kidney weight and nephron endowment in infants younger than 3 months of age (a time at which compensatory hypertrophy has likely not yet occurred) has revealed a direct relationship (Fig. 20.6).⁹⁶ The regression analysis predicted an increase of 23,459 nephrons per gram of kidney weight.⁹⁶ Kidney mass is therefore proportional to nephron number in the early postnatal period, and kidney volume is proportional to kidney mass. Kidney volume has therefore been used as a surrogate for nephron endowment in infants in vivo.⁷²

Fetal ultrasound studies have consistently shown that fetuses exposed to IUGR or infants born SGA have

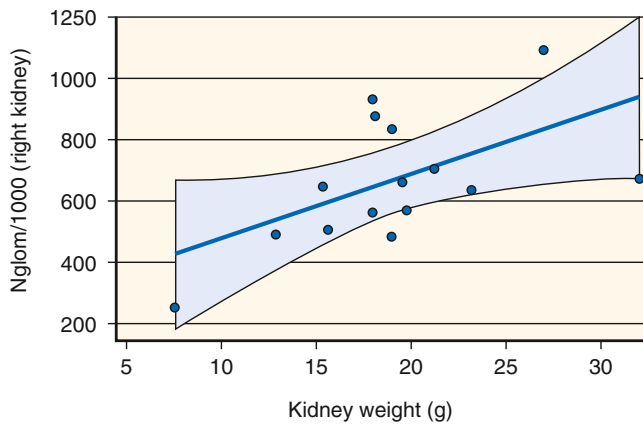


Fig. 20.6 Relationship between nephron number and mass of the right kidney in Caucasian infants aged ≤ 3 months who died within the first 3 months of life. *Ngglom/1000*, Number of glomeruli per kidney in 1000s. (From Zhang Z, Quinlan J, Hoy W, et al. A common *RET* variant is associated with reduced newborn kidney size and function. *J Am Soc Nephrol*. 2008;19:2027–2034.⁹⁶)

significantly smaller kidney volumes than AGA infants.^{135–138} A strong correlation between fetal body size and kidney size is also reported,¹³⁷ indicating that growth of the developing kidneys is highly vulnerable to suboptimal blood flow. Besides a reduction in kidney volume, ultrasound studies have also revealed a reduction in hourly urine volume, a higher prevalence of oligohydramnios, and reduced kidney perfusion in growth-restricted fetuses.^{135,137,139,140} These findings may represent reduced fetal perfusion in situations of intrauterine compromise, however, and do not necessarily reflect altered kidney development. Most (but not all¹⁴¹) studies among preterm infants also show a reduction in kidney size compared with term-born infants at equivalent ages, with reports of reduced kidney volume^{142,143} and slowed postnatal growth.¹⁴⁴

Smaller kidney size can persist into postnatal life. An ultrasound analysis of 178 Danish children born preterm or SGA compared with 717 term AGA children at 0, 3, and 18 months found that weight for gestational age was positively associated with kidney volume at all three time points.¹⁴⁵ Slight catch-up in kidney growth was observed in growth-restricted infants but not in preterm infants. Similarly, in a South Indian population, kidney volumes were lower in LBW and SGA neonates compared with those born AGA.¹⁴⁶ Kidneys in the LBW and SGA infants remained small and grew more slowly than kidneys of AGA infants during the first 2 years of life. In a study of multiethnic children in the Netherlands, lower fetal weight gain and lower early infant weight gain led to smaller kidneys at 6 years of age.¹⁴⁷ However, only lower fetal weight gain was associated with lower eGFR. These findings show that suboptimal early growth affects kidney structure and function in later life.

Given the strong association between body and kidney size, however, it is important to consider kidney volume relative to body size, rather than absolute kidney volumes alone. In Australian Aboriginal children, LBW was found to be associated with significantly lower kidney volumes corrected for body size compared with children born AGA.¹⁴⁸ Comparison of kidney volume between Swedish children aged 9 to 12 years born preterm and either SGA

or AGA, as well as term controls, also found that kidneys were smallest in those who had been preterm and SGA, but when adjusted for body surface area (BSA), there were no significant differences between the groups.¹⁴⁹ Similarly, recent studies in 5-year-old Colombian children also found that the significant reduction in kidney volume in children born preterm compared with children born at term was not evident when corrected for BSA.¹⁵⁰ While a smaller kidney size may be a surrogate marker for a reduced nephron endowment (particularly in fetuses), it is important to recognize that an assessment of kidney size by ultrasound cannot distinguish between normal growth with age (e.g., layering of generations of nephrons in the renal cortex) and kidney hypertrophy in response to the body's metabolic demands. In adults in particular, kidney size is not a reliable marker of nephron number. Findings from a meta-analysis of six studies in adults, in which both kidney weight and nephron number (estimated using the disector/fractionator technique) were analyzed, showed that only 5% of the variation in nephron number was explained by differences in kidney weight.⁹³

GENETIC VARIANTS ASSOCIATED WITH KIDNEY SIZE AND NEPHRON NUMBER IN HUMANS

Rare genetic and congenital anomalies of the kidney and urinary tract (CAKUT) contribute to about 40% of all childhood ESKF (discussed in more detail in [Chapters 44 and 71](#)).^{151,152} Full or partial deletion of more than 25 genes has been shown to result in kidney hypoplasia, and deletion of several of these genes results in low nephron endowment; not all have been studied in humans.^{46,102,153} Of the mutant genes associated with monogenic forms of CAKUT, most encode transcription factors (e.g., *PAX2* and *GATA3*) and growth factor receptors (e.g., *RET*) expressed in ureteric bud/tree cells during branching morphogenesis or in intermediate mesoderm (e.g., *SIX2*, *EYA1*, and *ROBO*) where they set the fate of renal progenitor cells and regulate interactions with the ureteric bud.^{151,153}

There is some evidence that mild mutations of these same genes exert subtler effects on kidney mesenchyme/ureteric bud interactions and may be commonplace in the normal population ([Table 20.4](#) for clinical studies; experimental studies shown in [eTable 20.1](#)). *PAX2* is highly expressed in the ureteric bud, where it suppresses apoptosis and optimizes the extent of branching. An intronic *PAX2* polymorphism, which reduces *PAX* transcript levels from the mutant allele by only 50%, is found in 18.5% of Canadians and is associated with a subtle (10%) reduction in newborn kidney size.¹⁵⁴ Ureteric branching morphogenesis is also highly dependent on *GDNF* signaling from the metanephric mesenchyme to the ureteric bud via the *RET* tyrosine kinase receptor on ureteric tip cells. A polymorphic variant of the *GDNF* receptor, *RET(1476A)*, was associated with a 10% reduction in kidney volume at birth compared with the wild-type (*RET 1476G*) allele.⁹⁶ In this study, newborn kidney volume was shown to be proportional to nephron number, suggesting that the modest kidney hypoplasia seen with the *PAX2* and *RET* polymorphisms represents a reduction in nephron endowment.⁹⁶ Newborn kidney size was reduced by 13% among Polish babies with a common variant of the *BMPRIA* gene, encoding a bone morphogenetic protein

eTable 20.1 Gestational Exposures and Factors Associated With Nephron Number, Kidney Size, Blood Pressure and Kidney Function**Experimental Evidence****Reduction in Nephron Number**

Experimental Model	Animal	Glomerular Number (% change)	Birth Weight	Blood Pressure	Kidney Function/ Pathology
Calorie-restricted diet (CRD) ^{198,390,709,710}	Rat, Mouse	↓ 20-50	↓	↑, NA	↓ GFR, Proteinuria
Uterine artery ligation ^{166,209-211}	Rat	↓ 20-30	↓	↑ (Males)	Proteinuria
Low-protein diet (LPD) ^{163,169,181,711-713}	Rat, Mouse	↓ 16-30	↓/↔	↑	↓ GFR, Proteinuria ↓ Podocyte endowment ↓ Longevity
Postnatal nutrient restriction ²¹¹	Rat	↓ 27	Normal	↑	NA
Iron deficiency ²¹⁸	Rat	↓ 22	↓	↑	NA
Vitamin A deficiency ^{169,220}	Rat	↓ 20	↔	NA	NA
Zinc deficiency ²¹⁹	Rat	↓ 25	NA	↑	↓ GFR, Proteinuria
Ethanol ^{297,299}	Sheep	↓ 11	↔	NA	NA
	Rat		↓	↑	↓ GFR (females), ↑ GFR (males)
Hypoxia ^{265,267,714}	Rat, Mouse	↓ 26-52	↓	NA, ↑	↑ Albuminuria ↓ Podocyte endowment (males)
Cigarette smoke ^{271,302,715}	Mouse, Rat	NA	↓	NA	↓ Kidney mass
	Mouse		↔	NA	↑ Albuminuria
Ureteral obstruction, neonatal ¹²¹	Rat	↓ 50	NA	↑	↓ GFR, ↓ Renal growth post relief of obstruction
Prematurity ^{167,253,716}	Mouse	↓ 17-24	↓	↑	Early differentiation of nephron progenitor cells
	Baboon	↔ (nephrogenesis incomplete)	↓	NA	↓ GFR, ↑ Albuminuria
					Outer cortical glomerular abnormalities
Postnatal hyperoxia ²⁷⁸⁻²⁸⁰	Rat	↔ (↓ 25 with aging)	NA	↑	↓ Creatinine clearance ↑ Crescentic glomeruli (males)
Glucocorticoids ^{158,162,247,717}	Rat	↓ 20	↓/↔	↑	Glomerulosclerosis
	Sheep	↓ 38	↔	↑	↑ Collagen deposition
Maternal diabetes ^{231,370}	Rat	↓ 10-35	↔	↑	Salt sensitivity
Gentamicin ^{285,287}	Rat	↓ 10-20	↓	NA	NA
β-Lactams ²⁸⁹	Rat	↓ 5-10	↔	NA	Tubular dilatation
					Interstitial inflammation
Cyclosporine ^{291,292}	Rabbit	↓ 25-33	↓/↔	↑	↓ GFR, ↑ RVR, proteinuria
Lipopolysaccharide ^{718,719}	Sheep (acute high dose)	↓ 23 (Singleton), ↓ 18 (Twin)	↔	NA	NA
	Sheep (chronic low dose)	↔	↔	NA	NA
Bisphenol A ³¹⁰	Mice	↓ (Female)	NA	NA	Crescentic glomeruli (females)

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eTable 20.1 Gestational Exposures and Factors Associated With Nephron Number, Kidney Size, Blood Pressure and Kidney Function (Continued)**Experimental Evidence****Reduction in Nephron Number**

Experimental Model	Animal	Glomerular Number (% change)	Birth Weight	Blood Pressure	Kidney Function/ Pathology
Di-(2-ethylhexyl)phthalate ³¹²	Rat	↓	↓	NA	↓ Creatinine clearance, Proteinuria, Podocyte injury
Anti-ouabain antibodies ⁵⁷¹	Rat	↓ (12.5, glomerular density)	↓	NA	↓ Kidney weight
Dahl salt-sensitive strain ^{66,720,721}	Rat	↓ 15 / ↔	NA	↑ with Na intake	Proteinuria, Accelerated FSGS
Munich-Wistar-Frömter strain ^{66,722}	Rat	↓ 40	NA	↑ with age	↑ SNGFR, FSGS, Proteinuria
Milan hypertensive strain ^{66,723,724}	Rat	↓ 17	NA	↑	Resistant to FSGS
Oligosyndactylism strain ^{725,726}	Mouse	↓ 50	NA	NA	Glomerular hypertrophy
PAX2 mutations ^{154,330,727}	Mouse Human	↓ 22	NA	NA	Renal coloboma syndrome in humans, small kidneys
GDNF heterozygote ^{728,729}	Mouse	↓ 30	↔	↑	Normal GFR Glomerular hypertrophy
hIGFBP-1 overexpression ²³⁰	Mouse	↓ 18-25	↓	NA	Glomerulosclerosis
Bcl-2 deficiency ³³¹	Mouse	↓	NA	NA	↑ BUN and creatinine
p53 Transgenic ⁷³⁰	Mouse	↓ 50	NA	NA	Glomerular hypertrophy
Fibroblast growth factor (FGF) 7 null mutant ¹⁶⁹	Mouse	↓ 30	NA	NA	Proteinuria, ESKF ↓ Ureteric branching

Augmentation of Nephron Number

Vitamin A supplementation (± maternal LPD) ⁵⁶³	Rat	Normalized	NA	NA	NA
Vitamin D deficiency ^{224,226}	Rat	↑	↔ / ↓	NA	↓ Creatinine clearance and sodium retention (males)
Amino acid (glycine, alanine) or urea supplementation (± maternal LPD) ⁷³¹	Rat	Normalized	NA	Normalized with glycine only	NA
L-citrulline supplementation (± maternal CRD) ⁵⁶⁸	Rat	↑	NA	↑	NA
Methionine supplementation (± maternal CRD) ¹⁹⁸	Mouse	Partial rescue	↔	NA	NA
Tsc1 (hamartin) heterozygote (± maternal CRD/mutant) ^{198,569}	Mouse	Normalized	NA	NA	Prolonged nephrogenesis
Restoration of postnatal nutrition post-intrauterine growth restriction ¹⁶⁶	Rat	Normalized	↓	Normalized	NA
Iron supplementation of iron-deficient pups postnatally ²¹⁷	Rat	Partial rescue (glomerular density)	NA	NA	NA
Folic acid supplementation (± maternal CRD) ⁵⁶⁴	Rat	Partial rescue	NA	NA	NA

eTable 20.1 Gestational Exposures and Factors Associated With Nephron Number, Kidney Size, Blood Pressure and Kidney Function (Continued)**Experimental Evidence****Reduction in Nephron Number**

Experimental Model	Animal	Glomerular Number (% change)	Birth Weight	Blood Pressure	Kidney Function/ Pathology
Ouabain administration (+ maternal LPD) ^{570,572}	Rat	Normalized Partial rescue	NA ↔	NA Normalized	NA Prevented ↑ proteinuria
MitoQ supplementation (+ maternal cigarette smoke) ⁵⁷⁵	Mouse	Normalized (glomerular density)	↑	NA	NA
Grape skin AH09 extract supplement (± maternal LPD) ⁵⁷⁴	Mouse	↑ (glomerular density) Partial rescue (glomerular density)	↔ Partial rescue	NA	Normalized glomerular maturation
Endorepellin treatment (+ maternal LPD) ⁵⁷⁸	Mouse	Partial rescue	NA	NA	NA
Nicotinamide riboside supplementation (+ maternal LPD) ⁵⁷⁶	Mouse	Partial rescue	NA	NA	Normalized podocyte density
Maternal uninephrectomy prior to gestation ⁷³²	Rat	↑	NA	NA	NA
Maternal water restriction (with dehydration-anorexia) ⁷³³	Rat	↑ (17, glomerular density)	↓	↑	NA
Postnatal overfeeding, normal birth weight ³⁶²	Rat	↑ 20	↔	↑	Glomerulosclerosis
Maternal high-fat diet before and during pregnancy and lactation ²³⁹	Mouse	↑ 20–25	↔	NA	NA
Six2 heterozygote ⁷³⁴	Mouse	↑ 18	NA	NA	↑ Progenitor cell proliferation
TGF-β heterozygotes ^{735,736}	Mouse	↑ 23–60	NA	NA	↑ Ureteric branching
PVG/c strain ^{66,737}	Rat	↑ 22	NA	Resistant	Resistant to FSGS

Human Evidence

Clinical Circumstance	Population/Age	Glomerular Number/Kidney Volume (% change)	Birth Weight	Blood Pressure	Kidney Function/Pathology
Low birth weight ¹²⁸	Neonates	↓ 12	↓	NA	NA
IUGR ^{131,132,135,137,138}	Fetuses	↓ 30–35 ↓ Kidney volume	↓	NA	NA
Preterm birth ^{49–51,141–143}	Neonates/infants	↓ Correlated with gestational/ postnatal age ↓ Kidney volume	↓	NA	Outer cortical glomerular abnormalities Accelerated kidney maturation
Gestational diabetes ^{137,703–705}	Fetuses	NA ↔ / ↑ kidney volume	↔ / ↑	NA	NA
Maternal smoking ²⁷⁰	Fetuses	NA ↓ / ↑ kidney volume (dose-dependent)	↓	NA	NA
Maternal vitamin A deficiency ⁷⁰⁶	Indian vs. Canadian newborns	NA ↓ kidney volume	NA	NA	NA
Females vs. males ⁵⁴⁰	Adult	↓ 12	NA	Variable	Variable
Hypertensive vs. normotensive living donors ⁷⁷	44.1 ± 12.1 years 92% Caucasian	↓ (100,428 fewer glomeruli)	NA	↑	↓ eGFR with ↓ nephrons
Hypertensive vs. normotensive Caucasian ^{106,129}	35–59 years	↓ 19–50	NA	↑	No ↑ Glomerulosclerosis
Hypertensive vs. normotensive African American ¹²⁹	35–59 years	NS ↓	NA	↑	No ↑ Glomerulosclerosis

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eTable 20.1 Gestational Exposures and Factors Associated With Nephron Number, Kidney Size, Blood Pressure and Kidney Function (Continued)

Clinical Circumstance	Population/Age	Glomerular Number/Kidney Volume (% change)	Birth Weight	Blood Pressure	Kidney Function/Pathology
Hypertensive vs. normotensive Japanese ⁴⁴³	Adult males	↓ 39 (nonsclerotic) ↔ (sclerotic) ↔ kidney volume	NA	NA	↑ SNGFR ↓ eGFR with ↓ nephrons
Hypertensive vs. normotensive Aboriginal Australian ⁴³⁹	Adult	↓ 30	NA	NA	Glomerular hypertrophy
Aboriginal Australians vs. Caucasian Australians ⁴³⁹	Adult	↓ 30	↓	NA	Glomerular hypertrophy
Senegalese Africans ^{699,707}	5-70 years	NA	NA	NA	↑ Variability of glomerular size with ↓ glomerular numbers
Genetic polymorphisms:					
<i>RET</i> (1476A) polymorphism ⁹⁶	Newborns	↓ 10	NA	NA	NA
<i>PAX2</i> AAA haplotype ¹⁵⁴	Newborns	↓ 10	NA	NA	NA
Combined <i>RET</i> (1476A) polymorphism and <i>PAX2</i> AAA haplotype ⁹⁶	Newborns	↓ 23	NA	NA	NA
<i>I/D ACE</i> polymorphism ⁷⁰⁸	Newborns	↓ 8	NA	NA	NA
<i>BMPRI1</i> ^{rs7922846} polymorphism ¹⁵⁵	Newborns	↓ 13	NA	NA	NA
<i>OSR1</i> ^{rs12329305(T)} polymorphism ¹⁵⁷	Newborns	↓ 12	NA	NA	NA
Combined <i>OSR1</i> and <i>RET</i> polymorphisms ¹⁵⁷	Newborns	↓ 22	NA	NA	NA
Combined <i>OSR1</i> and <i>PAX2</i> polymorphisms ¹⁵⁷	Newborns	↓ 27	NA	NA	NA
<i>ALDH1A2</i> ^{rs7169289(G)} polymorphism ¹⁵⁶	Newborns	↑ 22	NA	NA	NA

Table 20.4 Gestational Exposures and Factors Associated With Nephron Number, Kidney Size, Blood Pressure and Kidney Function**Human Evidence**

Clinical Circumstance	Population/Age	Glomerular Number/Kidney Volume (% change)	Birth Weight	Blood Pressure	Kidney Function/Pathology
Low birth weight ¹²⁸	Neonates	↓ 12	↓	NA	NA
IUGR ^{131,132,135,137,138}	Fetuses	↓ 30-35 ↓ Kidney volume	↓	NA	NA
Preterm birth ^{49-51,141-143}	Neonates/infants	↓ Correlated with gestational/postnatal age ↓/↔ kidney volume	↓	NA	Outer cortical glomerular abnormalities Accelerated kidney maturation
Gestational diabetes ^{137,703-705}	Fetuses	NA ↔ / ↑ kidney volume	↔ / ↑	NA	NA
Maternal smoking ²⁷⁰	Fetuses	NA ↓ / ↑ Kidney volume (dose-dependent)	↓	NA	NA
Maternal vitamin A deficiency ⁷⁰⁶	Indian vs. Canadian newborns	NA ↓ kidney volume	NA	NA	NA
Females vs. males ⁵⁴⁰	Adult	↓ 12	NA	Variable	Variable
Hypertensive vs. normotensive living donors ⁷⁷	44.1 ± 12.1 years 92% Caucasian	↓ (100,428 fewer glomeruli)	NA	↑	↓ eGFR with ↓ nephrons
Hypertensive vs. normotensive Caucasian ^{106,129}	35-59 years	↓ 19-50	NA	↑	No ↑ Glomerulosclerosis
Hypertensive vs. normotensive African American ¹²⁹	35-59 years	NS ↓	NA	↑	No ↑ Glomerulosclerosis
Hypertensive vs normotensive Japanese ⁴⁴³	Adult males	↓ 39 (nonsclerotic) ↔ (sclerotic) ↔ kidney volume	NA	NA	↑ SNeGFR ↓ eGFR with ↓ nephrons
Hypertensive vs normotensive Aboriginal Australian ⁴³⁹	Adult	↓ 30	NA	NA	Glomerular hypertrophy
Aboriginal Australians vs. Caucasian Australians ⁴³⁹	Adult	↓ 30	↓	NA	Glomerular hypertrophy
Senegalese Africans ^{699,707}	5-70 years	NA	NA	NA	↑ Variability of glomerular size with ↓ glomerular numbers

Genetic polymorphisms:

<i>RET</i> (1476A) polymorphism ⁹⁶	Newborns	↓ 10	NA	NA	NA
<i>PAX2</i> AAA haplotype ¹⁵⁴	Newborns	↓ 10	NA	NA	NA
Combined <i>RET</i> (1476A) polymorphism and <i>PAX2</i> AAA haplotype ⁹⁶	Newborns	↓ 23	NA	NA	NA
<i>I/D ACE</i> polymorphism ⁷⁰⁸	Newborns	↓ 8	NA	NA	NA
<i>BMP1A</i> ^{rs7922846} polymorphism ¹⁵⁵	Newborns	↓ 13	NA	NA	NA
<i>OSR1</i> ^{rs12329305(T)} polymorphism ¹⁵⁷	Newborns	↓ 12	NA	NA	NA
Combined <i>OSR1</i> and <i>RET</i> polymorphisms ¹⁵⁷	Newborns	↓ 22	NA	NA	NA
Combined <i>OSR1</i> and <i>PAX2</i> polymorphisms ¹⁵⁷	Newborns	↓ 27	NA	NA	NA
<i>ALDH1A2</i> ^{rs7169289(G)} polymorphism ¹⁵⁶	Newborns	↑ 22	NA	NA	NA

eGFR, Estimated glomerular filtration rate; IUGR, intrauterine growth restriction; NA, not assessed; NS, nonsignificant; SNeGFR, single-nephron eGFR.

receptor on ureteric epithelial cells.¹⁵⁵ Conversely, 22% of Canadian newborns inherit a variant of the *ALH1A2* gene (*rs7169289*) associated with increased production of all-*trans* retinoic acid metabolism in fetal tissues; newborns with the G allele were shown to have a 22% increase in newborn kidney size compared with the wild-type allele.¹⁵⁶ One of the earliest transcription factors marking the nephron progenitor cells of intermediate mesoderm is *OSR1*; a variant of the human *OSR1* gene that interferes with mRNA splicing was identified in about 6% of Caucasians.¹⁵⁷ This *OSR1rs12329305(T)* variant was associated with a 12% reduction in newborn kidney size.

Taken together, these observations suggest that final nephron endowment may represent a complex polygenic trait determined by the additive effects of multiple genes regulating either the extent of ureteric branching or the renal progenitor cell pool during fetal life. The impact of genetic variation on nephron endowment alone, as well as in the context of developmental stresses and the risk of later-life hypertension and kidney disease, however, requires further study.

PROGRAMMED KIDNEY STRUCTURE AND FUNCTION

EXPERIMENTAL EVIDENCE FOR PROGRAMMING OF NEPHRON ENDOWMENT, BLOOD PRESSURE, AND KIDNEY FUNCTION

Developmental programming of nephron number has been the most rigorously studied link to later-life hypertension and kidney disease so far. Numerous animal models have demonstrated the association between in utero growth restriction (e.g., induced by gestational exposure to low-protein or low-calorie diets, uterine ischemia, dexamethasone, vitamin A deprivation, and alcohol) with subsequent hypertension.¹⁵⁸⁻¹⁶⁵ The link between adult hypertension and growth restriction in these animal models appears to be mediated, at least in part, by an associated congenital nephron deficit.^{158,162,163} Corresponding blood pressures and nephron numbers in various programming models are outlined in [eTable 20.1](#). The association among birth weight, nephron numbers, and blood pressure varies between models, as discussed in detail later, underscoring the complexity of developmental programming and the need for better markers than birth weight.

Vehaskari and colleagues¹⁶³ demonstrated an almost 30% lower glomerular number in offspring of pregnant rats fed a low-protein diet, along with a reduction in their birth weight, compared with offspring with a normal-protein diet during gestation ([Fig. 20.7A](#)). As shown in [Fig. 20.7B](#), the low-protein offspring had tail-cuff systolic blood pressures that were around 40 mm Hg higher by 8 weeks of age, and these rats died prematurely.¹⁶³ Similarly, prenatal administration of dexamethasone was associated with reduced offspring body weight and fewer glomeruli than controls.¹⁵⁸ In these nephron-deficient rats, GFR and sodium excretion were reduced and albuminuria was increased.¹⁵⁸ Uteroplacental insufficiency, induced by maternal uterine artery ligation late in gestation, also results in low nephron number and was associated with increased

profibrotic renal gene expression with age, although hypertension developed only in males.^{104,166} Conversely, adequate postnatal nutrition, by cross-fostering onto normal lactating females at birth, restored nephron number and prevented subsequent hypertension in growth-restricted male rats.¹⁶⁶ Mice delivered preterm (i.e., 1–2 days early [normal mouse gestation, 21 days]) had lower nephron numbers, lower GFR, higher blood pressures, and more albuminuria than mice born at term.¹⁶⁷ Nephron numbers were lower in mice delivered 2 days compared with 1 day early, suggesting the degree of preterm birth is important in determining final nephron endowment, even though nephrogenesis normally continues for about 5 days after birth in mice. Not surprisingly, the timing of the gestational insult has been found to be crucial in renal programming, with the greatest impact on nephron number occurring during periods of most active nephrogenesis.¹⁶⁸

The complexity of the effects of perturbations to the fetomaternal environment and genetic variants on kidney development has been highlighted in two studies. Sampogna and colleagues¹⁶⁹ compared the effects of vitamin A deficiency, protein deficiency, and fibroblast growth factor 7 (FGF7) deletion on kidney development and nephron endowment in mice ([Fig. 20.8](#)). Perturbations to kidney development, which varied according to condition, included developmental delay, defects in nephron induction, changes in the growth axis, and alterations in ureteric branching morphogenesis. These produced up to a threefold decrease in glomerular number and a twofold decrease in total branching events. Boubred and colleagues¹⁷⁰ carefully compared the effects of two different perturbations to the fetomaternal environment on nephron number and adult renal physiology in rats. Maternal gestational low-protein diet and betamethasone both led to a similar level of fetal growth restriction, but nephron numbers were lower in betamethasone-exposed than low-protein diet offspring. Betamethasone-exposed offspring had impaired GFR, increased blood pressure, and glomerulosclerosis, whereas kidney function and structure were unaltered in low-protein diet offspring at 22 months of age. These findings suggest that the degree of reduction in nephron number is a risk factor for cardiovascular and renal disease.

The definitive pathophysiologic impact of a reduction in nephron number on the development of kidney dysfunction is difficult to elucidate from the existing literature comprising varied experimental conditions. Compensatory mechanisms, such as glomerular hypertrophy and increased single-nephron GFR (SNGFR), may maintain whole-kidney GFR when nephron mass is low.¹⁷¹ Often there are findings of normalized total kidney filtration surface area through glomerular hypertrophy at the time of assessment (see [Table 20.3](#)), meaning that the impact of nephron number per se on kidney function cannot be confirmed or refuted.¹⁷² A secondary insult is often required for dysfunction to emerge. Overall, the available evidence suggests that the effect of a suboptimal fetomaternal environment on offspring nephron endowment and adult health may be influenced by species; the nature, timing, duration and severity of the perturbation; sex and age of the offspring; the postnatal nutritional environment (including lactation, infant nutrition, and subsequent growth); and superimposed kidney stresses.

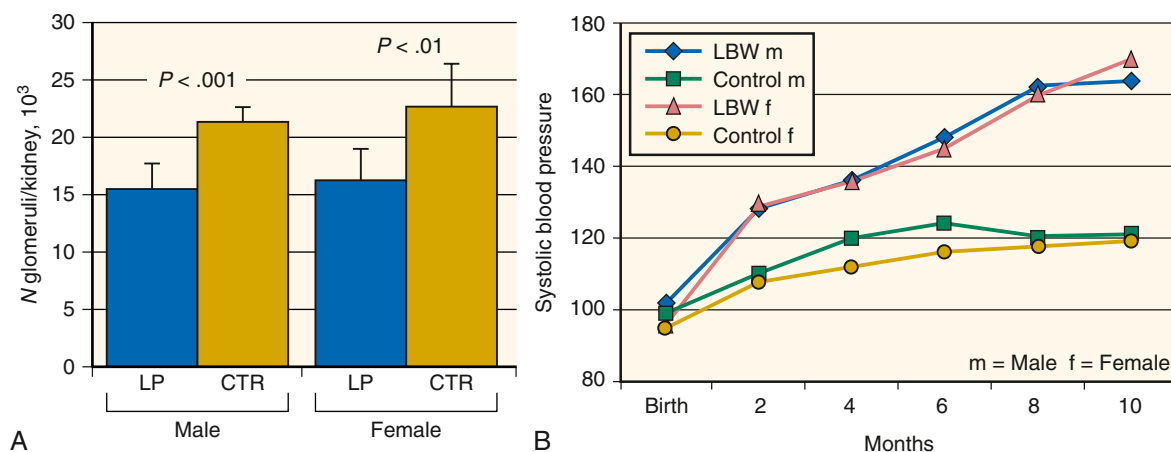


Fig. 20.7 Fetal programming of hypertension in growth-restricted rats (induced by maternal low-protein diet). (A) Total number of glomeruli per kidney at 8 weeks of age and (B) systolic blood pressure throughout life, in male ($n = 7$) and female ($n = 6$) offspring from low-protein pregnancies and in male ($n = 7$) and female ($n = 7$) control rats. CTR, Control diet; LBW, low birth weight (growth restricted); LP, low-protein diet. (Adapted from Vehaskari VM, Aviles DH, Manning J, et al. Prenatal programming of adult hypertension in the rat. *Kidney Int.* 2001;59[1]:238–245.¹⁶³)



Fig. 20.8 Architecture of wild type (WT), fibroblast growth factor 7 (FGF7) null mutant, vitamin A-deficient, and protein-deficient mouse kidneys. Embryonic day 15.5 ureteric tree tracings for each condition (to scale). N is the maximum branching generation number, and $Glom$ represents the mean glomerular count (minimum of three kidneys studied per condition). In each case, glomerular count is reduced compared with the wild type. The number of branching generations is specific to each condition. (From Sampogna RV, Schneider L, Al-Awqati Q. Developmental programming of branching morphogenesis in the kidney. *J Am Soc Nephrol.* 2015;26[10]:2414–2422.¹⁶⁹)

PROGRAMMING OF PODOCYTE NUMBER AND STRUCTURE

The podocyte depletion hypothesis has emerged as a potentially unifying concept in glomerular pathology in recent years, with podocyte depletion defined as a loss of podocytes, decreased podocyte density due to glomerular hypertrophy, and/or a change in podocyte phenotype.¹⁷³ In each situation, the ability of podocytes to fully envelop the glomerular basement membrane to form an effective filtration barrier is impaired, resulting in proteinuria and glomerulosclerosis.^{173,174} In rats, a reduction of 20%, or more, in podocyte number is required to initiate these pathologic changes.^{173,175} Podocyte loss has been associated with kidney disease, hypertension, diabetes, and aging,^{176–178} as well as preterm birth.^{179,180} Offspring podocyte deficits have been observed to be programmed by maternal hypoxia and low-protein diet.^{181,182} Humans and animals born with both a low nephron and low podocyte endowment could be expected to be at additional risk of developing CKD; however, whether an early life podocyte deficit

increases susceptibility to glomerular injury is unclear. Experimental studies examining the impact of postnatal stressors on kidney function and podocyte structure have yielded conflicting results, with either the growth-restricted offspring or normally grown offspring found to be at greater risk.^{181,183} Furthermore, podocyte number has the potential to increase in postnatal life, limiting the initial deficit.¹⁸¹

MECHANISMS OF STRUCTURAL PROGRAMMING IN THE KIDNEYS

In humans, nephrogenesis continues until around 36 weeks in normotrophic fetuses but may continue for several weeks postnatally in preterm infants, as discussed earlier.^{130,133} In rodents, nephrogenesis continues for up to 10 days after birth but is most active mid to late gestation when studies show the most impact from manipulation of environmental factors.¹³³ It is important to note that many differences in the timing of gestation, timing of nephrogenesis, gene expression profiles, patterning of kidney development, and

structure of the kidneys exist between different species,¹⁸⁴ which must be taken into account when interpreting the findings from experimental studies.

Many experimental models of programming, as outlined in eTable 20.1, have been shown to result in a reduced nephron number in affected offspring, and these have been used to gain insight into potential mechanisms contributing to disease outcomes. Clinical associations with nephron number and kidney size in humans are also shown in Table 20.4. Maternal and environmental factors that affect birth weight and gestation length in humans can directly affect nephrogenesis, as illustrated in eTable 20.2, and some of these factors may compound the effects of LBW or preterm birth.^{107,185} Given the wide variety of factors that can influence fetal/neonatal kidney development, occurring alone or in combination, however, the specific mechanisms underlying nephron deficits are not fully understood (eTable 20.3). In addition, other more subtle structural changes within the affected kidneys may occur, which also have the potential to adversely impact later-life kidney function and disease susceptibility, even in the setting of a normal nephron endowment.

EVOLUTIONARY BIOLOGY, CELLULAR METABOLISM, AND NEPHRON NUMBER

Principles of evolutionary theory and bioenergetics may explain the process by which a fetal nephron deficit can occur in the situation of a suboptimal fetomaternal environment, as detailed in recent papers by Chevalier.^{186,187} From an evolutionary biology perspective, the processes of structural and functional development of an organism prioritize what is required for survival into the years of reproductive fitness.¹⁸⁶⁻¹⁸⁸ When the growth of a developing fetus is constrained by the poor availability of necessary energy and substrates, adaptations occur in resource allocation, which favor vital organs to the detriment of other body systems. With respect to the kidneys, this potentially means that only the minimum number of nephrons necessary for adequate renal filtration to support body growth, maintenance, and reproduction into early adulthood may be formed; nephron hypertrophy compensates for the nephron deficit where needed, and CKD may follow especially in the setting of further stresses.^{186,187} The adjustment of nephron endowment to match available energy during development appears to be a highly conserved adaptive evolutionary trait, given that it is evident across disparate branches of the animal kingdom (e.g., birds,¹⁸⁹ rodents, and primates) (see eTable 20.1).¹⁸⁶

The process of resource (energy) allocation is certainly evident in IUGR pregnancies, where fetal hypoxia triggers peripheral vasoconstriction resulting in a redistribution of blood flow toward the fetal brain, heart, and adrenal glands, with a consequent reduction in blood flow to the kidneys.^{137,190} IUGR pregnancies are also associated with changes in fetoplacental endocrine and metabolic function, placental structure and growth, reduced protein accretion, and slowed somatic growth of the fetus.^{190,191}

The reduction in nephron formation within the developing kidneys of affected fetuses likely results from alterations in cellular metabolism. Metabolic plasticity of nephron progenitor cells enables them to coordinate the

high metabolic demand of balancing cell survival and self-renewal (to maintain the mesenchymal population) and differentiation (aggregation and epithelialization) during nephrogenesis, as well as to support tissue homeostasis in a dynamic fetomaternal environment.¹⁹² Activity of the 5' adenosine monophosphate-activated protein kinase (AMPK), mechanistic target of rapamycin (mTOR), and hypoxia-inducible factor (HIF)-related metabolic pathways all directly respond to changes in substrate availability (such as variations in oxygen, glucose, fat, and amino acid levels, and enzyme cofactors).^{186,192-194} A stressed environment can induce metabolic reprogramming, whereby the balance between the modes of energy production, oxidative phosphorylation (promotes cell differentiation and maturation), and glycolysis (promotes cell proliferation), is disturbed.^{186,192} Experimental and cell culture studies suggest that prolonged glycolytic metabolism induced in nephron progenitor cells is associated with significant impairments in nephron formation, whereas inhibiting glycolysis increases the number of induced nephrons (without affecting ureteric branching).^{195,196} Fetal baboons, exposed to a global nutrient-restricted maternal diet, exhibited a significant downregulation in genes involved in mitochondrial oxidative phosphorylation pathways.^{193,197} In a mouse model, activating the mTOR complex 1 (mTORC1) pathway was shown to partially abrogate the adverse effect of a maternal calorie-restricted diet on offspring nephrogenesis.¹⁹⁸ Although cellular metabolism-related mechanisms of nephron deficit have not yet been explored in detail, it is expected that there is a multitude of circumstances (with or without evident SGA, LBW, or IUGR) under which fetal bioenergetics may be affected, including maternal nutrient deficiencies, maternal/placental health conditions, maternal smoking/hypoxia, and postnatal hyperoxia and nutrient restrictions in preterm infants.

MATERNAL NUTRITION

UNDERNUTRITION

Supplementation of maternal balanced energy intake reduces the risk of small and vulnerable neonate births (Fig. 20.9).¹⁹⁹ Suboptimal maternal nutrition during pregnancy has long been thought to be the basis of programmed nephron deficits in offspring.²⁰⁰ Postnatal nutrition of the preterm infant may also impact ongoing nephrogenesis, given that it may continue for weeks after birth. Consistent with this, both intrauterine and extrauterine undernutrition have been associated with significant reductions in GFR in early childhood.²⁰¹

Experimental alterations in maternal dietary composition at different stages of gestation have been shown to program kidney gene expression early in the course of gestation, which later affects nephron number (see eTable 20.3).²⁰² Maternal protein and calorie restrictions during all or the later stages of pregnancy have been the most widely studied models of LBW and reduced nephron number since as early as 1968.²⁰³ In a rat model of maternal low-protein diet, besides a marked 30% nephron deficit in offspring, a significant reduction in the number of podocytes formed per glomerulus was also reported.¹⁸¹ Not all diets, however, have the same programming outcomes. It has been proposed that the source of carbohydrate and the relative

eTable 20.2 Potential Renal Programming Effects of Risk Factors for Preterm Delivery

Maternal Risk Factor for Preterm Birth	Animal		Human		
	BP	Nephron Number/Kidney Size	BP	Nephron Number/Kidney Size ^a	Kidney Function
Vitamin D deficiency	↑	↑ (↑ immature glomeruli)	ND	ND	ND
Anemia/iron deficiency	↑	↓	ND	ND	ND
Smoking	ND	↓	↑	↓	↓
Alcohol intake	ND	↓	ND	ND	ND
Antibiotic use (UTI)	ND	↓	ND	ND	ND
Chorioamnionitis	ND	↓	ND	ND	ND
Steroids	↑	↓	Normal	Not different	Unknown
Cyclosporine/tacrolimus (maternal transplant)	↑	↓	ND	ND	Possibly normal
Maternal dialysis	ND	ND	ND	ND	Some ↓
Maternal diabetes	↑	↓	↑	ND	↓
Preeclampsia	ND	ND	↑	ND	ND

^aKidney size generally determined by ultrasound.

BP, Blood pressure; ND, not determined; UTI, urinary tract infection; ↑, increased; ↓, decreased.

Compiled from references.^{218,225,247,255,257,289,292,297,715,738-745}

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eTable 20.3 Proposed Mechanisms of Developmental Programming in Kidneys

Experimental Model	Possible Mechanism of Nephron Number Reduction	References
Maternal macronutrient deficiency (e.g., low-protein diet)	↑ Apoptosis in metanephros and postnatal kidneys - Altered gene expression in developing kidneys - Altered gene methylation ↓ Placental 11-β HSD2 expression → increased fetal exposure to glucocorticoids - Metabolic reprogramming of progenitor cells - Impaired angiogenesis - RAAS alterations	186, 213, 337, 379
Maternal vitamin A restriction	↓ Branching of ureteric bud ? Maintenance of spatial orientation of vascular development ↓ c-ret expression	220
Maternal iron restriction	- Altered mesenchyme to epithelial transition - Reduced oxygen delivery - Altered glucocorticoid responsiveness - Altered micronutrient availability ↑ Inflammation	217, 218, 746
Maternal zinc deficiency	↑ Apoptosis ↓ Antioxidant activity	219
Gestational glucocorticoid exposure	- Enhanced tissue maturation ↑ Glucocorticoid receptor expression ↑ Na-K-ATPase, α_1 and β_2 subunit expression ↓ Renal and adrenal 11-β HSD2 expression ↓ Glucose delivery to fetus / altered metabolism - RAAS alterations	247, 251, 245, 253, 747, 748
Uterine artery ligation/embolization	↑ Proapoptotic gene expression in developing kidneys: caspase-3, Bax, p53 ↓ Antiapoptotic gene expression: PAX2, bcl-2 - Altered gene methylation - Altered RAAS gene expression - Metabolic reprogramming of progenitor cells - Oxidative stress	186, 210, 274, 749
Maternal diabetes/hyperglycemia	↓ IGF-2/mannose-6-phosphate receptor expression - Altered IGF-2 activity/bioavailability - Activation of NF-κB - Altered ureteric branching morphogenesis - Metabolic reprogramming of progenitor cells	186, 229, 232, 233, 750
Gestational/postnatal drug exposure	↓ Branching morphogenesis ↑ Mesenchymal apoptosis - Arrest of nephron formation ? Via reduced vitamin A levels - Altered gene expression - Vasoconstriction - Inflammation - Tubular injury, necrosis, and obstruction	287, 289, 292, 294, 297, 298, 751
• Gentamicin • β-lactams • Cyclosporine • Ethanol • Nicotine • NSAIDs • Acyclovir • Alcohol		
Perinatal hypoxia	- Altered RAAS gene expression - Increased expression of glucocorticoid receptors - Suppression of β-catenin signaling - Accelerated aging - Metabolic reprogramming of progenitor cells - Oxidative stress - Altered tubule development - Proximal tubule injury - Impaired podocyte development	182, 186, 266, 274, 714, 752
Postnatal hyperoxia/ventilation	↓ Nephrogenic zone width ↓ Glomerular diameter ↓ Glomerular capillary length and surface area - Downregulation of HIF, catalase, and MAPK/ERK signaling	278, 281, 282, 753

eTable 20.3 Proposed Mechanisms of Developmental Programming in Kidneys (Continued)

Experimental Model	Possible Mechanism of Nephron Number Reduction	References
Ureteral obstruction: postnatal	↓ Cell proliferation ↑ Apoptosis of tubular cells Delayed maturation of interstitial fibroblasts → interstitial fibrosis ? Alteration of postinductive processes	121
Preterm birth (and associated postnatal factors)	Accelerated maturation of glomeruli Early differentiation of progenitor cells Altered HIF signaling Impaired angiogenesis Oxidative stress ? Factors associated with shift from intrauterine to extrauterine environment Kidney injury and repair Reduced telomere length	49, 167, 253, 278, 282, 626, 716

ERK, Extracellular signal-regulated kinase; *HIF*, hypoxia-inducible factor; *11-β HSD2*, 11β-hydroxysteroid dehydrogenase; *IGF-2*, insulin-like growth factor 2; *MAPK*, mitogen-activated protein kinase; *NF-κB*, nuclear factor kappa B; *RAAS*, renin–angiotensin–aldosterone system; *VEGF*, vascular endothelial growth factor.

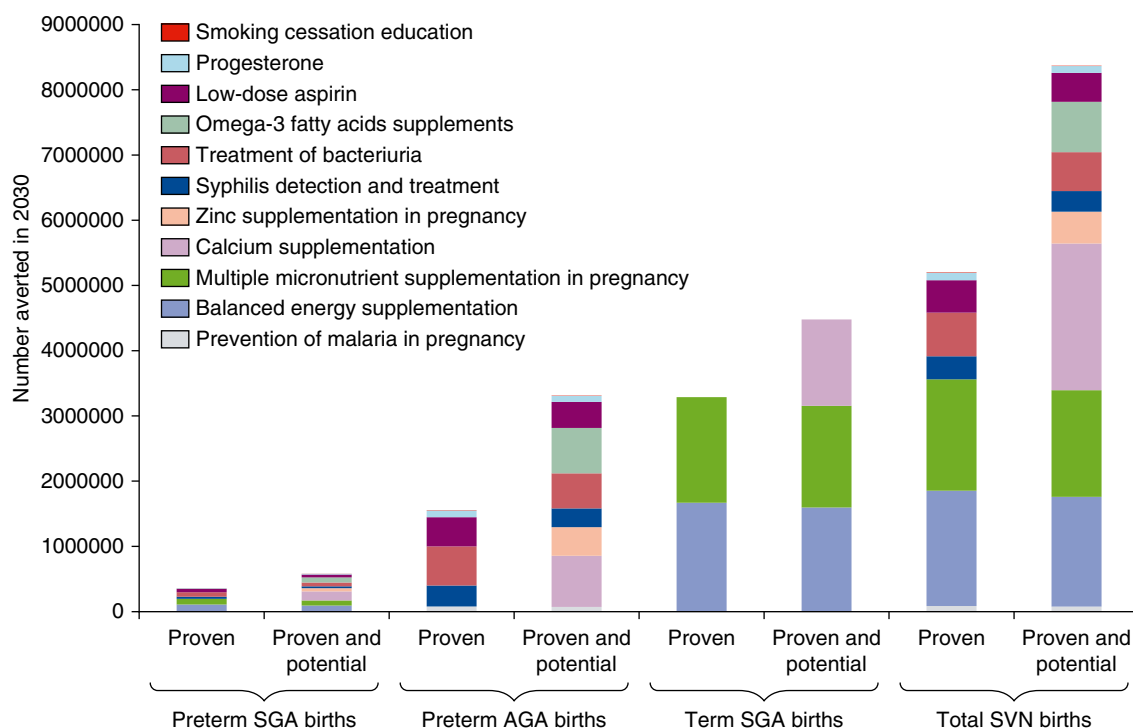


Fig. 20.9 Impact of maternal interventions on averting the risk of birth of small and vulnerable neonates from 81 low- and middle-income countries. AGA, Appropriate weight for gestational age; SGA, small-for-gestational age; SVN, small and vulnerable neonate. (Reprinted with permission from Elsevier, from Hofmeyr et al. Evidence-based antenatal interventions to reduce the incidence of small vulnerable newborns and their associated poor outcomes. *Lancet*. 2023; 401(10389):1733–1744.)

deficiencies of specific amino acids may have a greater impact on organ development than protein restriction per se, potentially through epigenetic modulation of gene expression.^{204,205} Effects are also dependent on the degree of protein restriction and the sex of the fetus, with a more severe restriction required to impair kidney development and program hypertension in female rats than males.^{206,207}

Fetal nutrient supply is also affected by alterations in placental development that affect uteroplacental blood flow and transfer of nutrients to the fetus (see eTable 20.3). Placental insufficiency is the most common cause of fetal growth restriction in the Western world, and the placental phenotype most likely underpins the developmental programming of chronic disease in the offspring.²⁰⁸ Rat models of bilateral uterine artery ligation in late gestation, which reduces placental blood supply and induces fetal growth restriction, have shown offspring nephron deficits, glomerular hypertrophy, and increased blood pressure in males.^{166,209-211}

Increased fetal exposure to glucocorticoids and alterations to the renin–angiotensin–aldosterone system (RAAS) have been proposed as mechanisms whereby fetal nutritional deficits reduce offspring nephron number, both of which are discussed further later.²¹² Other potential mechanisms include reduced vascular endothelial growth factor (VEGF) expression and angiogenesis,²¹³ global downregulation of gene expression,²¹⁴ altered DNA methylation and microRNA activity,^{215,216} and the metabolic reprogramming of nephron progenitor cells.^{186,192}

Supplementation of *maternal micronutrients* calcium and zinc reduces the risk of small and vulnerable neonatal births (see Fig. 20.9).¹⁹⁹ In terms of micronutrients impacting

the developing kidneys, maternal *iron* restriction during pregnancy also leads to a reduction in birth weight, reduced radial glomerular counts and glomerular density, and the development of hypertension and tubulointerstitial fibrosis in rat offspring (see eTable 20.3).^{217,218} Conceivably, fetal anemia may result in reduced tissue oxygen delivery, altered fetal kidney glucocorticoid sensitivity, or altered availability of other micronutrients that may affect nephrogenesis.²¹⁸ Prewaning or postweaning zinc deficiency was also associated with decreased nephron number, reduced GFR, and higher blood pressures in rats, potentially mediated by reduction in the antioxidant and antiapoptotic effects of zinc.²¹⁹

Maternal *vitamin A* restriction has also been associated with a reduction in nephron number in rat offspring (see eTable 20.3).^{161,220} Severe vitamin A deficiency during pregnancy is associated with congenital malformations and renal defects. Vitamin A and all-*trans* retinoic acid have been shown to modulate branching capacity in ureteric epithelial cell culture and the spatial organization of blood vessel development in cultured renal cortical explants.²²⁰ An analysis of fetal rats (just before birth) revealed a direct correlation between plasma retinol concentration and nephron number, with a 50% reduction in circulating vitamin A levels resulting in a 25% reduction in nephron endowment.²²⁰ Maternal smoking and alcohol intake may be associated with reduced levels of circulating vitamin A, and both are associated with LBW and programming of disease (described later). Subtle differences in vitamin A level during pregnancy therefore may contribute to the wide distribution of nephron number observed in the general population.¹⁰⁷

Maternal 25-hydroxy vitamin D (25OHD) deficiency is common²²¹ and is associated with an increased risk of preterm birth, LBW, and SGA births in affected populations.^{222,223} In animal models, offspring from dams that were vitamin D-deficient throughout pregnancy and lactation exhibit augmented nephron number (without affecting kidney size); however, there is also evidence of delayed glomerular development and altered kidney function.²²⁴⁻²²⁶ The hormonally active form of vitamin D is a ligand for a nuclear transcription factor, which regulates gene expression in a number of pathways including cell differentiation; these changes in developmental trajectory may also be associated with the observed upregulation of renal renin expression.^{224,225,227} In a study of 4212 pregnant women and their children at 6 years of age, maternal vitamin D levels were inversely associated with eGFR in childhood, but there was no clear association with combined kidney volume.²²³ Whether human nephron endowment is influenced by vitamin D levels (supplementation or deficiency) is unknown.

OVERNUTRITION, MATERNAL DIABETES, AND OVERWEIGHT/OBESITY

In some populations, *HBW* is associated with an increased susceptibility to proteinuria and kidney disease.^{13,55,68} Therefore focus on long-term programming effects of maternal diet has moved to include the effects of overnutrition, as well as the consequences for offspring born to women with diabetes and/or obesity.

Human offspring of diabetic pregnancies have a higher incidence of congenital malformations, resulting from defects in early organogenesis, and have an increased risk of CKD.^{228,229} The expression and bioavailability of insulin-like growth factors (IGFs), which are important regulators of fetal development, are altered in diabetic pregnancies.²²⁹ In offspring of rat dams with streptozotocin-induced diabetes during pregnancy, there was a significantly increased expression of the IGF-2/mannose-6-phosphate receptor, which leads to reduced IGF-2 bioavailability.²²⁹ The critical role of IGF-2 in renal development is highlighted in cell culture studies showing the addition of IGF-2 increases nephron number by 25% to 40% in a concentration-dependent manner.²³⁰ In another rat model, offspring nephron numbers were reduced by 10% to 35%, correlating with the degree of maternal hyperglycemia (Fig. 20.10).²³¹ Culture of metanephroi in varying glucose concentrations demonstrates that tight glucose control is necessary for optimal metanephric growth and differentiation.²³¹ In mice, offspring of diabetic mothers had fewer nephrons and increased apoptosis in tubules and podocytes, potentially mediated via increased renal angiotensinogen and renin mRNA expression and NF- κ B activation.²³² Other authors suggest altered branching morphogenesis, increased asymmetric dimethylarginine, and reduced nitric oxide levels as potential mediators of reduced nephron numbers in offspring of mothers with diabetes.^{233,234} As discussed earlier, the increased fetal glucose availability likely also affects the metabolic programming of nephron progenitor cells.^{186,192} Multiple pathways are therefore likely to be implicated in hyperglycemia-induced renal programming.

The programming of offspring by *maternal obesity and/or overnutrition* is multifactorial, with the fetomaternal

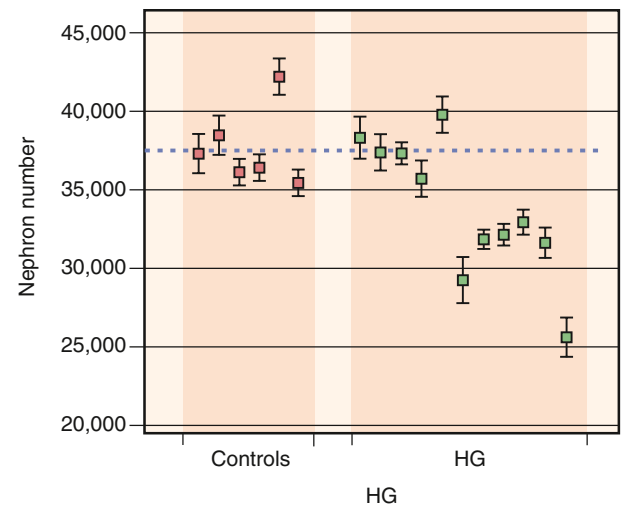


Fig. 20.10 Effects of maternal hyperglycemia on nephron number in rat offspring. Dotted line represents mean value in control group. HG, Hyperglycemia. (From Amri K, Freund N, Vilar J, et al. Adverse effects of hyperglycemia on kidney development in rats: in vivo and in vitro studies. *Diabetes*. 1999;48:2240–2245.²³¹)

environment and offspring outcomes differing according to maternal health and diet; indeed, maternal obesity may be associated with LBW or HBW pregnancies.²³⁵ Exposure to a maternal high-fat diet for 10 days before pregnancy until weaning resulted in female rat offspring with hypertension but no change in nephron endowment, although renal renin and Na/K-ATPase activity were altered.²³⁶ Rat dams fed an obesogenic high-fat/high-fructose diet for a more sustained period before mating resulted in offspring with albuminuria, which was further exacerbated by a postnatal high-fat diet.²³⁷ A similar model also caused increased renal norepinephrine and renin expression in offspring.²³⁸ Conversely, in mice, a maternal high-fat diet caused an increase in offspring nephron number, with no consequences for kidney function.²³⁹ Dams in that study were glucose intolerant before mating but had normal glucose tolerance during pregnancy, suggesting complex interactions between maternal obesity and diabetes. Data suggest that it may not be the high-fat diet per se causing changes in kidney development but rather the resultant maternal hyperglycemia. Of note, many calorie-dense foods are also deficient in essential micronutrients, which may exacerbate programming risk given the “hidden malnutrition” evident in obesity.^{240,241}

MATERNAL AND FETAL EXPOSURE TO GLUCOCORTICOIDS

Elevated maternal glucocorticoids due to stress or glucocorticoid therapy may impact fetal growth and kidney development (see eTable 20.3). Under normal circumstances, the fetus is protected, at least in part, from exposure to excess endogenous corticosteroids by the placental enzyme 11 β -hydroxysteroid dehydrogenase type 2 (11 β -HSD2), which metabolizes corticosterone to the inert 11-dehydrocorticosterone.²⁸ Rats and humans with mutations in the 11 β -HSD2 gene have low levels of placental 11 β -HSD2 and give birth to LBW offspring in whom hypertension develops prematurely.^{242,243} The

administration of high concentrations of cortisol (in sheep) or corticosterone (in rats and mice) may also overwhelm this enzyme and affect the fetus, as evidenced by offspring with low nephron number, and later hypertension and impaired kidney function.²⁴⁴⁻²⁴⁶ These outcomes occurred even after short-term exposure that did not affect birth weight.^{246,247} Prenatal administration of dexamethasone, a steroid not metabolized by 11 β -HSD2, led to fetal growth restriction, a 20% to 60% reduction in nephron number, glomerulomegaly, and subsequent hypertension in rats and sheep.^{158,162,247,248} Metanephric organ culture experiments have demonstrated that dexamethasone can cause a dose-dependent inhibition of branching morphogenesis, in part through regulation of *GDNF* gene expression.²⁴⁹ Excessive fetal steroid exposure may drive inappropriate gene expression and affect growth and nephrogenesis, potentially through more rapid maturation of tissues.²⁰⁴

Studies also implicate glucocorticoids as modulators of nephrogenesis in the setting of maternal low-protein diet, with increased nephron number in offspring of dams treated with an inhibitor of steroid synthesis.^{28,204,250} Expression of steroid-responsive receptors including renal Na/K-ATPase α_1 - and β_1 -subunits were also found to be significantly increased in offspring of rats fed a low-protein diet during gestation.²⁵¹

In humans, antenatal glucocorticoid administration for women at risk of preterm delivery has markedly improved neonatal survival and outcomes, likely through mechanisms that accelerate infant lung maturation.²⁵² Findings from sheep and baboon models suggest that this also increases the structural and functional maturation of the kidneys.^{253,254} Follow-up at age 30 years, of people whose mothers participated in a randomized placebo-controlled study of antenatal betamethasone, did not find any effect on blood pressure or other cardiovascular indicators.²⁵⁵ No difference in blood pressure or kidney function was noted among preterm infants who had received antenatal steroids within the first 2 years of life compared with those who did not.^{256,257} A Cochrane review found that there are, to date, insufficient follow-up data to determine unequivocally whether or not there are any long-term adverse effects of antenatal glucocorticoid treatment, but the substantial benefits for neonates support its use.²⁵² However, a study of 23,363 singletons in Taiwan followed for 10 years, in which CKD was identified using ICD-9/10/CM codes (including CAKUT), found that maternal steroid use during pregnancy was associated with increased risk of childhood CKD (HR 1.69; 95% CI 1.01–2.84).²⁵⁸ The association was stronger in children who were born preterm, male, had exposure during the second trimester, and were exposed to a cumulative dose equivalent above 24 mg hydrocortisone.²⁵⁸ Longitudinal examination of the consequences of sustained glucocorticoid therapy (e.g., throughout pregnancy) has not been reported.

HYPOXIA, HYPEROXIA, AND MATERNAL SMOKING

In addition to maternal iron deficiency (resulting in anemia) and placental insufficiency discussed earlier, a range of conditions, including pulmonary and cardiovascular disease, living at high altitude, maternal smoking, preeclampsia, and

sleep apnea, can result in fetal *hypoxia* (see [eTable 20.3](#)). These conditions have been associated with LBW, SGA, IUGR, and/or preterm births in humans.²⁵⁹⁻²⁶²

Offspring of pregnant sheep maintained in a high-altitude environment from 30 to 140 days' gestation exhibited significantly reduced relative kidney weight, tubule injury, and increased serum creatinine levels.²⁶³ In mice, a short-term severe hypoxic insult (5.5%–7.5% maternal oxygen) from embryonic day (E)9.5 to 10.5 caused a CAKUT phenotype, whereas mild hypoxia in midpregnancy (12% from E12.5–14.5) resulted in low nephron endowment, mediated in part through suppression of β -catenin signaling.²⁶⁴ A similar mild hypoxia in late gestation resulted in fetal growth restriction, altered renal tubular development, low nephron number in male but not female offspring, and hypertension in both sexes.^{265,266} Importantly, in the first reported evidence of podocyte programming, podocyte number per glomerulus was found to be significantly reduced by 15% in the male hypoxia-exposed mice offspring (but not in the females).²⁶⁷

Maternal smoking is a major modifiable risk factor for IUGR and preterm births²⁶⁸ and has been associated with infant kidney malformations,²⁶⁹ as well as altered fetal kidney volume in a dose- and time-dependent manner.²⁷⁰ In a mouse model of cigarette smoke exposure, male offspring had fewer nephrons per area of tissue assessed and increased urine albumin/creatinine ratio.²⁷¹ These adverse effects on kidney development relate to the fetoplacental response to reduced oxygen availability and reprogramming of cellular metabolism, altered activity of the HIF signaling pathway, and the associated rise in oxidative stress.^{272,273} The origin of distinct sex differences requires further investigation. Oxidative stress is linked to tissue injury and inflammation, DNA damage, cell cycle arrest, accelerated cellular senescence, and telomere shortening.²⁷⁴ It occurs due to suboptimal intrauterine (hypoxic) environments but may also affect neonates with adverse postnatal exposures.

Hyperoxia can occur in the situation of preterm birth, whereby infants are exposed to a much higher environmental oxygen environment after birth compared with in utero and, in addition, may receive supplemental oxygen therapy (21%–100% O₂). This is known to impair neonatal vascular growth via alterations in HIF signaling, VEGF expression, and increased oxidative stress (with low levels of endogenous antioxidants) and contributes to conditions such as bronchopulmonary dysplasia and retinopathy of prematurity.^{275,276} Circulating endothelial progenitor cells from hyperoxia-exposed preterm infants exhibit impaired VEGF-nitric oxide signaling and reduced proliferative growth.²⁷⁷ In the developing kidneys, a study of rat pups exposed to 80% O₂ in the last days of ongoing postnatal nephrogenesis exhibited reduced nephrogenic zone width and smaller glomerular diameter²⁷⁸; these changes were completely abrogated through treatment with a HIF-1 α stabilizing drug. Nephron number was not affected in pups, but in adulthood, there was a significant reduction in nephron number and evidence of high blood pressure, kidney injury (in males), and reduced creatinine clearance in hyperoxia-exposed rats.^{279,280} In a similar model, where hyperoxia exposure was commenced 2 days earlier, there was proximal tubule dilation and injury,

increased apoptosis, a thinning of the nephrogenic zone, and an associated downregulation of HIF, catalase, and mitogen-activated protein kinase (MAPK)/extracellular signal-regulated kinase (ERK) signaling.²⁸¹ In preterm and term lambs (both with completed nephrogenesis) mechanically ventilated with supplemental oxygen after birth, there were significant reductions in glomerular capillary length, surface area, and total renal filtration surface area compared with unventilated term controls.²⁸² It is not known whether capillary growth may be affected in other regions of the kidneys or what impact this may have in the long term.

DRUG AND ALCOHOL EXPOSURE

Several medications commonly used during pregnancy or in the early postnatal period have been studied for their effects on birth weight and nephrogenesis (see [eTables 20.1 and 20.3](#)). Among 397 pregnant women, antibiotic use was found to be associated with 138 g lower infant birth weight and altered methylation of some growth-related imprinted genes.²⁸³ Of concern, in some regions antibiotic exposure may occur unavoidably through maternal food and drinking water intake, and ciprofloxacin, norfloxacin, and sulfamethoxazole exposures in particular have been associated with decreased fetal growth and/or preterm birth.²⁸⁴

The aminoglycoside antibiotic gentamicin administered to pregnant rats results in a permanent nephron deficit in offspring.²⁸⁵ Significantly lower numbers of ureteric branch points and nephrons were observed in metanephric explants cultured in the presence of gentamicin.^{286,287} In contrast, other investigators did not find a reduction in nephron number in rat pups administered gentamicin intraperitoneally in the latter stages of nephrogenesis (birth to 14 days of age).²⁸⁸ Administration of the β -lactam antibiotic ampicillin to pregnant rats leads to an 11% reduction in offspring nephron endowment, as well as evidence of focal cystic tubule dilatation and interstitial inflammation.²⁸⁹ Administration of ceftriaxone in vivo did not result in a nephron deficit, but histologically, there was evidence of renal interstitial inflammation. The penicillins were also found to inhibit nephrogenesis in cultured metanephroi in a dose-dependent fashion, an effect that was less evident with ceftriaxone. Importantly, nephrogenesis was affected even at therapeutic doses of penicillins in the rat. The mechanism whereby the β -lactams reduce nephron endowment is likely through an increase in apoptosis in the induced mesenchyme.^{161,289} Further research on antibiotics that are frequently used in human pregnancy and preterm infants is warranted, especially given that treatment of bacteriuria during pregnancy reduces the risk of small and vulnerable neonate births (see [Fig. 20.9](#)).¹⁹⁹

The immunosuppressive medication cyclosporine is a known nephrotoxin in humans that crosses the placenta. Women treated with cyclosporine may have successful pregnancies, although infants tend to have birth weights in the low range, and its effect on the fetal kidneys is not well described.²⁹⁰ In a rabbit model, cyclosporine administration in the later, but not the earlier, period of gestation resulted in smaller litters and growth-restricted pups. All pups exposed to cyclosporine in utero had a 25% to 33% lower nephron number than controls.^{291,292} There

were also findings of glomerulomegaly, glomerulosclerosis, hypertension, proteinuria, and decreased GFR.²⁹² Organ culture studies suggest that the presence of cyclosporine suppresses Wnt signaling pathways and cellular proliferation in the nephrogenic zone and inhibits the conversion of metanephric mesenchyme to epithelium, as well as increasing apoptosis.^{161,293}

Nonsteroidal antiinflammatory drugs (NSAIDs) are sometimes used postnatally in children born preterm. Postnatal administration of a cyclooxygenase 2 inhibitor, but not a cyclooxygenase 1 inhibitor, in rat and mice pups resulted in reduced cortical volume, impairment of nephrogenesis, and reduced glomerular diameter.²⁹⁴ Administration of indomethacin or ibuprofen postnatally did not affect nephron number in rats.²⁸⁸ In the preterm baboon kidney, early postnatal administration of five doses of ibuprofen (consistent with recommended dosing in preterm infants with patent ductus arteriosus) was associated with a reduction in width of the nephrogenic zone, suggesting premature termination of nephrogenesis.²⁹⁵ The impact of these medications on human nephrogenesis is not known.

Regarding maternal alcohol consumption, kidney malformations have been reported in children with fetal alcohol syndrome.²⁹⁶ In fetal sheep, repeated ethanol exposure during the second half of pregnancy resulted in an 11% reduction in nephron number and impaired vascular function in late pregnancy.^{297,298} In rats, 2 days of exposure to a high dose of alcohol in mid-late pregnancy resulted in a low nephron number and offspring with hypertension and sex-dependent impairments in GFR.²⁹⁹ Using metanephric organ culture, alcohol was found to cause dose-dependent decreases in branching morphogenesis, which were prevented by retinoic acid supplementation.³⁰⁰ Among Australian individuals at age 30 years whose mothers had consumed alcohol during pregnancy, investigators reported a 7% prevalence of CKD.³⁰¹ The risk of CKD was doubled among those with exposure to moderate-heavy drinking in late pregnancy and was greater among females.

The impact of nicotine exposure on kidney development has been analyzed independently of the other toxins and hypoxia associated with maternal smoking. Nicotine, which crosses the placenta and is secreted in breast milk, was administered during pregnancy (beginning day 4 or day 7) and throughout lactation in rat dams. With the earlier commencement of nicotine exposure, pups had a significant reduction in body weight and kidney weight at postnatal day 14;³⁰² the later-onset exposure, however, did not affect kidney or body weights at postnatal day 7.³⁰³ The nicotine-exposed offspring exhibited decreased glomerular diameter, tubular injury, and increased collagen deposition, with a significant reduction in angiotensin II receptor type 2 (AT2R) expression and an upregulation in connective tissue growth factor (CTGF); nephron number has not been assessed.^{302,303} Exposure to marijuana during pregnancy has also been associated with greater risks of SGA, LBW, and preterm birth, but the impact on fetal kidney development is not known.³⁰⁴

ENVIRONMENTAL EXPOSURES

The developmental programming research field is also alert to the potential impact of environmental toxins,

with more than 60 different chemicals assessed to date.³⁰⁵ Few studies, however, have examined kidney outcomes (see eTable 20.1). High concentrations of bisphenol A (BPA), a common endocrine-disrupting chemical used in the manufacture of plastics and epoxy resins, have been associated with hypertension and albuminuria in adults.^{306,307} In pregnant women, BPA is associated with a significant reduction in fetal growth rate.³⁰⁸ Experimental studies of BPA administration to pregnant mice showed evidence of disrupted glucose metabolism, oxidative stress, and decreased nephron number in female offspring.³⁰⁹⁻³¹¹ Similarly, another endocrine-disrupting class of chemicals, phthalates (commonly found in shampoo and cosmetics) have been shown to significantly reduce birth weight and nephron number in male and female offspring of exposed rat dams, with no effect on kidney weight; these findings were associated with a dose-dependent reduction in renin, *Wnt11* and *GDNF* expression in the kidneys.³¹² Many other potentially harmful chemicals have been identified, in air pollution, soil, food, and items in everyday use; the specific effects these may have on kidney development and their mechanisms of action, however, are not fully understood.³¹³

CLIMATE CHANGE

Pregnant women are highly susceptible to the effects of climate change given that they are less able to thermoregulate, are more sensitive to dehydration, and are at greater risk of infections.^{22,314} Climate change is leading to more frequent and more widespread exposure of pregnant women to extreme weather events including drought, floods, and hurricanes, as well as heat and wildfire smoke. Climate change is also leading to an increase in vector-borne and water-borne infections.³¹⁵ These varied exposures have been associated with increased risks of LBW and preterm births, preeclampsia, and gestational diabetes, which in turn may impact fetal kidney development and long-term risks for hypertension and CKD (Fig. 20.11).²²

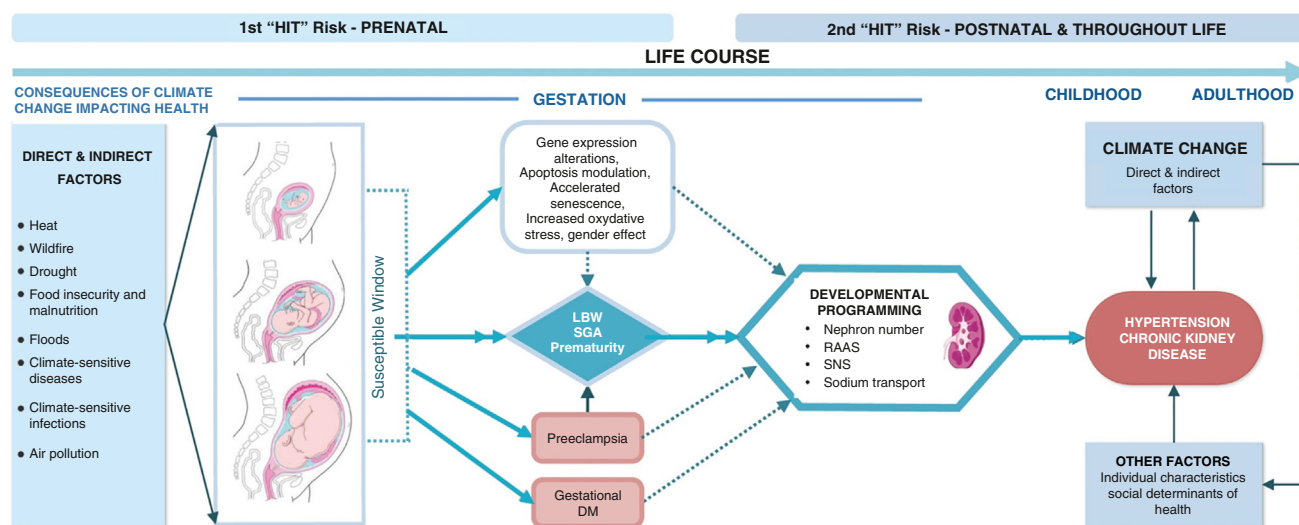
In children, exposure to air pollution has been associated with higher blood pressure and with variable effects on eGFR.³¹⁶⁻³¹⁸

OBSTRUCTION IN THE DEVELOPING KIDNEY

Many developmental abnormalities of the urinary tract are associated with impaired nephrogenesis, which, in turn, is compounded by obstructive injury.^{319,320} From animal studies, it has become clear that perinatal urinary obstruction may lead to reduced nephron numbers, which may exacerbate the impact of other programming factors.^{116,321} In a rhesus monkey model, histologic changes evident in the obstructed kidney include cystic enlargement of the tubules and Bowman spaces, abnormalities in glomerular tuft development, podocyte apoptosis, and a significant reduction in glomerular density; the dysplasia worsened when the obstruction was initiated earlier in gestation.³²² In rats, unilateral ureteral obstruction on postnatal day 1 and relieved on day 5, or on postnatal day 14 and relieved on day 19, reduced nephron number by 50% in both groups.¹²¹ Additionally, renin expression was decreased and glomerulotubular maturation was delayed.³²³ Importantly, temporary neonatal urinary obstruction was also associated with histologic scarring and loss of function of the contralateral kidney in rats.³²⁴ Developing and neonatal kidneys therefore appear to be highly susceptible to obstructive injury, suggesting that early relief of urinary tract obstruction may be important to preserve nephron number.

MOLECULAR PATHWAYS AFFECTED IN FETAL PROGRAMMING OF THE KIDNEYS

The molecular regulation of kidney development, particularly in the mouse, has been comprehensively described elsewhere.⁹⁹ In animal models of developmental programming, changes in expression



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CURRENT OPINION IN NEPHROLOGY AND HYPERTENSION

Fig. 20.11 Schematic illustrating the potential far-reaching impact of climate change on maternal health and kidney development during pregnancy and superimposed kidney injury (second hits) in later life. DM, Diabetes mellitus; LBW, low birth weight; RAAS, renin-angiotensin-aldosterone system; SGA, small for gestational age; SNS, sympathetic nervous system. (From Alvarez-Elias AC, Brenner BM, Luyckx VA, et al. Climate change and its influence in nephron mass. *Curr Opin Nephrol Hypertens*. 2024;33[1]:102–109.²²)

levels of genes regulating branching morphogenesis, apoptosis, and kidney growth have been described (for a review, see Wang and Garrett³²⁵), with some studies determining the effects of maternal perturbation on the whole genome.^{100,202,326}

Affected molecular pathways may be common across different fetomaternal environments or, in other cases, may be specific to individual insults, as described earlier (see eTable 20.3). GDNF signaling through its receptor-tyrosine kinase RET is a key ligand–receptor interaction driving ureteric budding and branching, and this has been shown to be disrupted in models of maternal vitamin A deficiency (see Fig. 20.8),^{161,220} as well as maternal glucocorticoid,²⁴⁹ phthalate,³¹³ and alcohol exposure.²⁹⁹ Ureteric branching is also impaired in maternal diabetes models, which is associated with changes in TGF- β 1 signaling due to increased expression of a hedgehog interacting protein.^{233,327} Suppressed Wnt/ β -catenin signaling has been observed in the kidneys of offspring exposed to fetal hypoxia^{264,265} and phthalates,³¹² as well as in vitro exposure to cyclosporin.^{161,293}

Increased apoptosis in the developing kidneys is a common finding in programming studies and has been observed in experimental models of maternal low-protein diet,²⁰² diabetes,^{232,328} iron deficiency,³²⁶ antibiotic and cyclosporin exposure,^{289,293} ureteral obstruction,³²² and postnatal hyperoxia.²⁸¹ Several studies have suggested that altered regulation of apoptosis in the developing kidneys may be due to downregulation of antiapoptotic factors (e.g., Pax-2 or Bcl-2) and/or upregulation of proapoptotic factors in response to environmental or other stimuli (e.g., Bax, p53, Fas receptor, caspase 3 and 9).^{210,329–333} Inappropriate apoptosis may occur in offspring affected by an adverse fetomaternal environment in successive waves throughout nephrogenesis, even after birth.^{202,210,329,334}

More than 40 microRNAs, involved in posttranscriptional control of mRNA expression, were shown to be deregulated in the kidneys of rat fetuses exposed to maternal low-protein diet, influencing multiple developmental pathways including cellular proliferation, differentiation, and apoptosis.²¹⁶ Epigenetics is thought to play an important role in the programming of health and disease across generations, and there is emerging evidence of its influence on kidney development and disease.^{41,335–337}

IMPACT OF SEX

A meta-analysis demonstrated that adult human kidneys show sex-specific expression of more than 200 genes.³³⁸ In utero, placentation processes and fetal growth patterns are affected by sex,^{339,340} and sex differences in fetal glomerular growth through mid to late gestation have also been observed.⁴⁸ Importantly, fetal sex-specific placental responses to maternal perturbations also occur (reviewed in Kalisch-Smith and colleagues³⁴¹). The placentas of female fetuses are known to have higher levels of the 11 β -HSD2 enzyme compared with males; therefore males are likely to be exposed to more glucocorticoids (endogenous and administered).^{342,343} As described earlier, sexually dimorphic outcomes in kidney structure have been observed in offspring of experimental models of maternal hypoxia^{267,344} and BPA exposure.^{310,311,345} Epigenetic

reprogramming, such as the differential DNA methylation of genes such as *IGF2*, has been proposed as a modulator of these sex differences.^{345,346}

In some clinical studies, programming effects appear more pronounced in males, and in others, the differential effects of gender are modified by age, ethnicity, and BMI.^{347–351} In female rats with similar programmed reductions in nephron numbers, blood pressures are often not as high or increase much later than in males.^{163,352} Often a secondary challenge such as pregnancy, however, will “unmask” a disease phenotype.³⁵³ Sex hormones may in part explain these differences,³⁵² as well as sexual dimorphism in the relative expression of components of the RAAS, potentially differentially altering the balance between vasoconstriction, vasodilation, and sodium handling.^{352,354} Sex differences in the programming of blood pressure, kidney function, kidney disease, and their outcomes are reviewed in detail elsewhere.^{354–358}

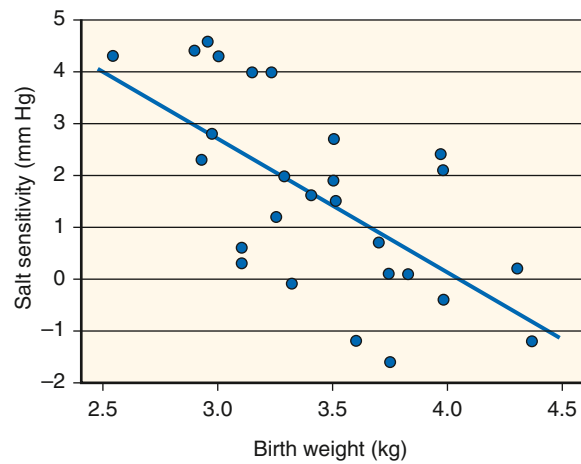
EVIDENCE OF ADDITIONAL PROGRAMMING EFFECTS IN THE KIDNEYS

The relationship between renal sodium handling, intravascular volume homeostasis, and hypertension is well accepted.^{359,360} That factors intrinsic to the kidney itself affect blood pressure has been demonstrated clinically in kidney transplantation, in which the blood pressure of the recipient after transplantation has been shown to be related to the blood pressure or hypertension risk factors of the donor: that is, hypertension “follows” the kidney.³⁶¹ Nephron endowment is likely an independent factor determining susceptibility to essential hypertension and subsequent renal injury. However, low nephron number alone does not account for all observed programmed hypertension (see eTable 20.1).^{204,265,362} Findings suggest that additional factors also contribute to the developmental programming of hypertension and kidney function, as outlined in Table 20.5, such as alterations in tubular sodium handling and vascular function.^{168,363}

SODIUM HANDLING

Both birth weight and kidney size have been shown to be inversely associated with salt sensitivity in children and adults (eFig. 20.1).^{364,365} In each of these studies, salt-induced changes were independent of GFR, excluding confounding by kidney function. In a study of 1512 adults (mean age 62 years), only those with a birth weight <3.05 kg exhibited a progressive 2.48 mm Hg rise in systolic blood pressure per 1 g increase in daily salt intake. Those with a higher birth weight did not show an association between blood pressure and dietary salt, potentially suggesting protection against salt sensitivity.³⁶⁶ This, along with the findings of other smaller studies, suggests LBW (and likely low nephron number) are associated with increased salt sensitivity.^{364–366} Consistent with this, salt sensitivity has been reported in several animal models of low nephron number including maternal uterine artery ligation and gestational diabetes,^{206,367–370} although timing of dietary intervention and age at study appear to play a role.^{371–373}

The structural development of the kidney tubules likely plays an important role in the programming of sodium handling, but little research has been done in this area. In



eFig 20.1 Correlation between birth weight and salt sensitivity in 27 normotensive adults. Salt sensitivity of blood pressure was defined as the difference in mean arterial pressure on high-salt diet (200 mmol/day) compared with low-salt diet (60 mmol/day). $R = -0.06$; $P = .002$. (From de Boer et al. *Hypertension*. 2008;51[4]:928–932.³⁶⁴)

Table 20.5 Developmentally Programmed Changes Observed in Kidneys and Clinical Correlates of Nephron Number

Programmed Changes in Kidneys	Clinical Correlates for Programmed Risk
• Kidney size • Nephron number • Podocyte number • Glomerular volume • Maturation of glomeruli • Alterations in angiogenesis • Tubular structure • Tubular sodium transporter expression and/or activity • Altered renal vascular reactivity • Alterations in renin–angiotensin system • Alterations in sympathetic nervous system activity • Accelerated senescence, especially after catch-up growth • Predisposition to inflammation and fibrosis	• Low birth weight^{128,134} • Small for gestational age/intrauterine growth restriction^{131,132,136,138} • Preterm birth^{49–51,134,142,143} • Neonatal acute kidney injury^{754,755} • Low kidney mass^{72,96} • Reduced kidney volume^{136,145,148} • Glomerulomegaly^{127,128,131,439} • Female sex⁵⁴⁰ • Ethnicity: Aboriginal Australian^{439,448,540} • Older age^{75,77} • Shorter adult height^{77,540}

prehypertensive offspring of rats fed a protein-restricted diet during gestation, expression of the Na-K-2Cl (NKCC2) and Na-Cl (NCC) transporters was significantly increased compared with controls.³⁷⁴ Furosemide administration reduced blood pressure in offspring of protein-restricted rats, supporting increased NKCC2 activity as a mediator of hypertension in the model.³⁷⁵ Maternal high-fructose diet also increased kidney expression of NKCC2, Na-H exchanger 3 (NHE3), Na-Cl cotransporter (NCC), and epithelial Na channels (ENaC) in rat offspring.³⁷⁶ Other programming models have observed significant increases in tubular Na-K-ATPase expression.^{251,370,377,378} Despite differences among models, the data suggest increased sodium transport in all segments of the renal tubule. Whether glomerulotubular balance in the setting of a reduced nephron number indirectly increases sodium transport or sodium transporter activity is independently programmed has not yet been elucidated.

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

RAAS activity is essential for placental angiogenesis, vascular function, and growth factor production during gestation, and its dysregulation can result in poor outcomes such as IUGR.³⁷⁹ Furthermore, all components of the RAAS are expressed in the developing fetal kidneys.³⁸⁰ Alterations in the intrarenal RAAS have been studied in various programming models (eTable 20.4), and a common, though not universal, finding has been inhibition of the system during the period of active nephrogenesis, with an upregulation in adulthood.³⁵⁴ Programmed suppression of the intrarenal RAAS during nephrogenesis is likely to contribute to low nephron

number under adverse fetomaternal conditions, and postnatal upregulation of the angiotensin II subtype 1 receptor (AT1R) expression, possibly mediated by an increase in glucocorticoid activity or sensitivity, may contribute to the subsequent development of hypertension and glomerular hyperfiltration, as reviewed in detail elsewhere.^{118,354} Overall, outcomes are variable and likely reflect species differences, differences in timing of intervention, and timing of study, as summarized in eTable 20.4.¹¹⁸ The mechanisms of RAAS suppression/upregulation are largely unknown, but there is some evidence in rodent models of epigenetic control of gene expression that can be intergenerational.^{381,382}

Underscoring the relevance of the RAAS in developmental programming of blood pressure, angiotensin-converting enzyme (ACE) activity was significantly elevated in children born with LBW compared with normal birth weight. There was a greater frequency of the ACE gene DD genotype among LBW children with the highest blood pressure, suggesting that the programming effect of blood pressure may be in part modulated by ACE gene polymorphisms.³⁸³ The relationship between programming of the RAAS and blood pressure and kidney disease is complex and likely modified by sex, prenatal steroid exposure, and BMI, among other factors.^{384,385}

VASCULAR DEVELOPMENT AND FUNCTION

Hemodynamic adaptations made during a period of suboptimal fetal growth may result in permanent systemic alterations to the structure and function of the vascular tree. In children or adults born LBW (preterm and/or SGA), findings of increased arterial stiffness, reduced arterial diameter, increased arterial intima-media thickness, reduced capillary density (eyes and skin), endothelial dysfunction (impaired vasodilation), elevated antiangiogenic markers, and impaired circulating endothelial progenitor cell function have all been reported.^{386–389} Whether the intrarenal vasculature is affected by developmental insults, however, remains underexplored.

Within the kidneys, the sympathetic nervous system modulates activity of the RAAS, sodium transport, and vascular function and thereby contributes to blood pressure through regulation of vascular tone and volume status.¹¹⁸ Development of the renal sympathetic nervous system and how this may be programmed during nephrogenesis and modulated by the RAAS is expertly reviewed by Kett and Denton.¹¹⁸ An increase in baseline intrarenal vascular resistance and altered contractile responsiveness of the renal artery have been described in several programming models.^{292,390–393} Renal denervation has been shown to abrogate the development of adult hypertension and alter sodium transporter expression in prenatal dexamethasone and uterine ischemia programming models, as well as the age-associated hypertension that develops in growth-restricted rats.^{394–397} Interestingly, in a cohort of White and Black Americans, the effect of birth weight on subsequent blood pressure was significantly modified by β -adrenergic receptor genotype, further underscoring a relationship between birth weight, sympathetic activity, and blood pressure.³⁹⁸

eTable 20.4 Programming Effects on the Renin–Angiotensin System

Model	Species	Timing of Insult	Age and Sex of Offspring at Study	mRNA or Protein Expression	Physiologic Response	Reference(s)
Glucocorticoids	Sheep	Early gestation	40 mo ♀	↔ Plasma renin, ANG II, or Aogen	↑ Basal MAP, females only	170
	Rat		6-7 mo ♂♀	↔ Renal Aogen ↑ PRA, plasma Aogen in females only	↑ Basal TBP, females only	163
	Rat	Mid- to late gestation	6 mo	↑ Renal ACE and renin in males and females	ND	354
	Rat	Mid- to late gestation	4 and 8 wk ♂	↔ PRA, plasma ANG II, and renal ANG II levels ↑ Urine ANG II at 4 and 8 wk	↑ Basal TBP at 8 but not 4 wk of age	32

Continued on following page

eTable 20.4 Programming Effects on the Renin–Angiotensin System (Continued)

PROGRAMMING OF BLOOD PRESSURE AND KIDNEY FUNCTION

Inextricably linked to the programming of nephron endowment and other aspects of the development of kidney structure and function described earlier are the physiologic consequences for blood pressure and kidney function in postnatal life. Unlike nephron number, these can be readily measured *in vivo*, making them valuable markers of individual and population health. The confounding effects of other lifestyle and health factors (both dependent and independent of programming) on blood pressure and kidney function, however, mean that they cannot be directly connected consequentially to programmed kidney structure in the clinical setting.³⁹⁹

Most human data rely on the surrogates of birth weight, preterm birth, and kidney size to reflect risk of developmental programming, although, as discussed earlier, these categories may not capture all infants who may have experienced adverse events during gestation (see Fig. 20.1). Besides nephron endowment, other programming effects in the kidneys may be independent of birth weight or gestational age. Therefore in humans, if birth weight or gestational age are the only surrogate markers used, the impact of programming on any outcome in the kidneys is likely to be underestimated.⁴⁰⁰ Additional clinical surrogates for an adverse intrauterine environment and low nephron numbers are outlined in Table 20.5. Despite the challenges of measuring nephron number *in vivo*, which have hampered the understanding of a direct association with kidney disease risk, the consistency of the data regarding associations with birth parameters at a population level, kidney size, and recent biopsy-based nephron counts in living donors is strongly suggestive of a programming effect (Table 20.6).

BIRTH WEIGHT, PRETERM BIRTH, AND BLOOD PRESSURE

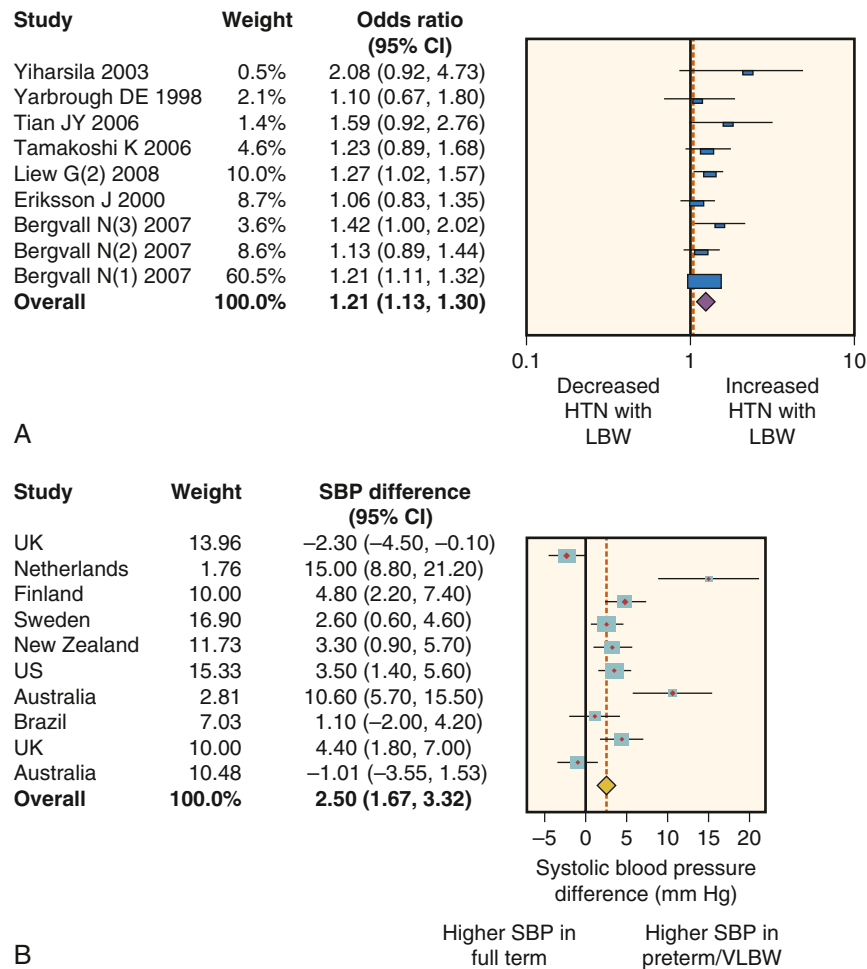
Two meta-analyses and systematic reviews published in 2012 showed consistent associations among lower birth weight, preterm birth, and higher blood pressures in later life (Table 20.7).^{401,402} A meta-analysis of 27 studies found that systolic blood pressures were 2.28 mm Hg (95% CI 1.24–3.33 mm Hg) higher in people born with LBW, compared with those born >2.5 kg (eFig. 20.2A).⁴⁰² Many studies do not discriminate between LBW occurring as a result of SGA/IUGR or as a result of preterm birth with an appropriate (low) weight for gestational age (AGA). Therefore their relative impact is not always clear.⁴⁰³ To investigate this, a study of 50-year-old adults born at term reported an odds ratio (OR) of 1.9 (95% CI: 1.1–3.3) for hypertension among those who were SGA compared with AGA.⁴⁰⁴ Growth restriction before birth *per se* therefore was associated with subsequent higher blood pressure. Regarding the risk among those born preterm, a systematic review of 10 studies found that people born preterm, with a mean gestational age of 30.2 weeks and a mean birth weight of 1280 g, had 2.5 mm Hg higher (95% CI: 1.7–3.3 mm Hg) systolic blood pressures in later life compared with those born at term (eFig. 20.2B).⁴⁰¹ Similar findings have been reported in a

subsequent meta-analysis.⁴⁰⁵ Preterm birth therefore is also independently associated with higher blood pressure, which in some studies is manifest in neonates and already meets the definition of hypertension by 1 to 2 years of age.^{256,406–409}

One systematic review and meta-analysis evaluated blood pressures in individuals aged 2 to 40 years who had been born preterm and either SGA or AGA.⁴¹⁰ Among 25 studies, blood pressures were not different between the two groups, suggesting that preterm birth *per se* may be the overriding risk factor in these individuals. Additional factors, such as maternal preeclampsia, female sex, and current BMI, may impact risk.^{399,411} In contrast, however, a study from Brazil followed 3585 individuals from birth to 22 years of age and found lower blood pressures among females who had been born preterm.⁴¹² The authors suggest that in lower-resource settings, individuals born preterm may remain small, protecting against blood pressure elevations.⁴¹² More data are required from diverse lower-resource settings.

Aging also has an important role in hypertension risk. Blood pressures of people born with LBW and normal birth weight may both still be within the normal range in childhood, but differences become amplified with age; adults who had been born with LBW often develop overt hypertension, with risk increasing with age.^{413,414} Most studies have been conducted in Caucasian populations, but consistent data are also accumulating in other regions.³⁴⁸ Results of a systematic review of African studies, however, suggest that additional factors may contribute to the greater severity of blood pressure in those of African origin.^{348,415–420} An important effect modifier of the association between LBW or preterm birth and blood pressure, noted in diverse populations, is BMI. Current BMI may override an effect of birth weight, especially in children at different stages of growth.^{348,421} Furthermore, blood pressures are often highest in those who “catch up” fastest in postnatal weight (i.e., rapid upward crossing of weight centiles), highlighting the importance of early postnatal nutrition in developmental programming.^{350,420,422–429} Among 513,802 Israeli adolescents eligible for military service, blood pressures were higher at age 17 years in those born with ELBW or VLBW, with the highest risk observed among those who developed subsequent overweight or obesity.⁴³⁰

Associations between increased blood pressure and other markers of potential developmental stresses have also been reported, such as HBW in children (but not adults),^{56,430,431} diabetic pregnancy in children and young adults (especially males),^{432,433} and maternal smoking in children (Table 20.7).⁴³³ Another potential risk factor is maternal gestational hypertension or preeclampsia.^{434,435} In a systematic review of 18 studies among children and young adults, systolic blood pressures were 2.39 mm Hg (95% CI: 1.74–3.05) higher in those exposed to preeclampsia compared with those unexposed (see Table 20.7).⁴³⁶ Whether the effect is mediated by the often-accompanying IUGR or preterm birth and/or may be associated with humoral changes with preeclampsia requires further investigation. In a U.S. birth cohort, the association of offspring blood pressure with maternal preeclampsia was attenuated with higher cord-blood vitamin D levels, which may suggest a role for vitamin D in modulation of long-term cardiovascular risk.⁴³⁷ In a study of young adults born preterm, soluble endoglin and soluble fms-like tyrosine kinase-1 (sflt-1)



eFig. 20.2 Relationship among birth weight, preterm birth, and blood pressure. (A) Meta-analysis of odds for hypertension (*HTN*) in individuals with birth weights below 2500 g (low birth weight, *LBW*) compared with birth weights above 2500 g. The pooled odds ratio is shown as a *diamond*. (B) Meta-analysis of difference in systolic blood pressure (*SBP*) between individuals born preterm or very low birth weight (*VLBW*) compared with full term. Pooled *SBP* difference is indicated by the *diamond* and *dashed vertical line*. (A from Mu et al. *Arch Cardiovasc Dis.* 2012;105:99–113; see original paper for full references.⁴⁰² B from de Jong F et al. *Hypertension.* 2012;59:226–234.⁴⁰¹)

Table 20.6 Clinical Associations of Kidney Programming

	LBW	SGA	Preterm	HBW	Gestational DM	Preeclampsia	Maternal Overweight/ Obesity	Maternal Vitamin A Deficiency	Rapid Catch-Up Growth/ Overweight	Low Nephron Number	Smaller Kidney Size/Weight	Increased Glomerular Volume
Increased blood pressure	✓	(✓)	✓	(✓)	✓	✓	✓		✓	✓	✓	✓
Salt sensitivity	✓				a					a	✓	
Reduced GFR	✓	✓	✓		a				✓	a	✓	
Reduced renal functional reserve	no	✓	✓		✓							
Proteinuria	✓		✓	✓	✓				✓	a	✓	
Acute kidney injury (neonates)		✓	✓			Protective				✓	✓	
Chronic kidney disease	✓	✓	✓		✓	✓	✓		✓	a	✓	✓
End-stage kidney failure	✓	✓	✓	✓	✓	✓				✓		
Transplant outcomes	✓										✓	✓
Death	✓											
Increased glomerular volume	✓		✓		a				a	✓		
Smaller kidney size/weight	✓	✓	✓					✓		✓		
Congenital obstruction	✓			NS	✓		✓			a		
Renal dysplasia/aplasia	✓			NS	✓		NS			✓		✓

✓, Evidence from human studies; (✓), evidence suggested; a, evidence from animal studies only; DM, diabetes mellitus; LBW, low birth weight; NS, not significant; SGA, small-for-gestational age.

Table 20.7 Systematic Reviews, Meta-analyses, and Population-Based Studies of Developmentally Programmed Associations With Blood Pressure and Kidney Disease

Reference	Condition	Age	Number (n)	Study Type	Outcome	Risk (95% CI)
Programmed Associations With Blood Pressure						
401	Preterm birth and/or VLBW ^a	6-22 years	10 studies 3080 subjects	Meta-analysis	SBP	Increase by 2.5 mm Hg (1.7-3.3 mm Hg) vs. term birth
759	Preterm birth	3-49 years	28 studies	Meta-analysis	SBP	Increase by 4.26 mm Hg (3.09-5.43 mm Hg) vs. term birth
405	Preterm birth	>2 years	16 studies	Meta-analysis	DBP	Increase by 2.32 mm Hg (1.35-3.29 mm Hg) vs. term birth
430	Preterm birth	Mean 17.3 years	513,802 subjects	Cohort, military recruits, Israel	SBP DBP Hypertensive range, extremely preterm Hypertensive range, very preterm Hypertensive range, moderate/late preterm Hypertensive range, extremely low birth weight	SMD 0.35 (0.22-0.48) SMD 0.33 (0.20, 0.47) OR 1.47 (1.28-1.71) OR 1.55 (1.35-1.75) OR 1.14 (1.07-1.23) OR 1.36 (1.18-1.53)
412	Preterm birth	22 years	3,010 subjects	Cohort, Pelotas Brazil population based (born 1993)	Hypertensive range, very low birth weight Hypertensive range, low birth weight Hypertensive range, high birth weight SBP	OR 1.34 (1.23-1.45) OR 1.07 (1.05-1.1) OR 1.01 (0.93-1.04) Decrease -1.0 mm Hg (-2.7 to 0.7 mm Hg) vs. term birth in females
760	SGA	14-15 years	1076 subjects	Cohort, Pelotas Brazil population based (born 1993)	DBP SBP DBP	No association in males Decrease -1.1 mm Hg (-2.4 to 0.3 mm Hg) vs. term birth in females No association in males Associated with increase of 2.89 ± 1.32 mm Hg Associated with increase of 3.08 ± 1.23 mm Hg

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Table 20.7 Systematic Reviews, Meta-analyses, and Population-Based Studies of Developmentally Programmed Associations With Blood Pressure and Kidney Disease (Continued)

Reference	Condition	Age	Number (n)	Study Type	Outcome	Risk (95% CI)
761	Preterm birth + SGA	34-49 years	1756 subjects	Cohort, Finland population based (Cardiovascular Risk in Young Finns study)	SBP	Increase by 7.2 mm Hg (2.3-12.1 mm Hg) vs. preterm AGA birth
410	Preterm birth, SGA vs. AGA	5-41 years	26 studies	Systematic review and Meta-analysis	SBP	Increase by 7.3 mm Hg (5.2-9.4 mm Hg) vs. term birth
411	VLBW	Mean 20 years	9 studies 1571 VLBW 777 controls	Meta-analysis	DBP	No difference SGA vs. AGA
402	Low birth weight	4-84 years	20 studies	Meta-analysis	DBP	No difference SGA vs. AGA
	High birth weight				DBP	Increase by 2.1 mm Hg (1.3-3.0 mm Hg) vs. term AGA birth
					Hypertension	OR 1.21 (1.13-1.3)
					SBP	Increase by 2.28 mm Hg (1.24-3.33 mm Hg) for BW <2500 g vs. BW >2500 g
762	Low birth weight ^b	7-17 years	15,324 subjects	Cohort, school-based recruitment across 6 provincial capitals in China (in 2010)	SBP	Decrease by 2.08 mm Hg (-2.98 to -1.17 mm Hg) for BW >4000 g vs. BW <4000 g
56	High birth weight	4-83 years	31 studies	Meta-analysis	SBP	Increase vs. BW 2500-2900 g
763	Birth weight	0-84 years	444,000 subjects	Systematic review	Overall	Increase vs. BW 2500-2900 g
	Catch-up growth				SBP	RR 1.18 (95% CI 1.05-1.32)
					SBP	RR 0.97 (95% CI 0.86-0.97)
					SBP	RR 1.0 (95% CI 0.93-1.06)
					SBP	Decrease by 2 mm Hg per kg increase in BW
					SBP	Increase with catch-up growth
					SBP	Highest blood pressure values in LBW after catch-up
427	Catch-up growth	25 years	225	Cohort, extremely preterm birth cohort, Victoria, Australia	SBP	8.9 mm Hg (5.8-12.0) per 0.1 increase z-BMI/year
420	Birth weight	0-19 years	13 studies	Systematic review, Africa	BP	Inconsistent associations
764	Birth weight	17 years	11,729 subjects	Cohort, Jerusalem, Israel population (born 1974-1976)	SBP and DBP	Inverse association between birth weight and blood pressure
436	Pregnancy complicated by preeclampsia	Children, young adult	45,249 subjects	Systematic review	SBP	Increase by 2.39 mm Hg (1.74-3.05 mm Hg)
					DBP	Increase by 1.35 mm Hg (0.9-1.8 mm Hg)
					BMI	Increase by 0.62 kg/m ²
					SBP	

Table 20.7 Systematic Reviews, Meta-analyses, and Population-Based Studies of Developmentally Programmed Associations With Blood Pressure and Kidney Disease (Continued)

Reference	Condition	Age	Number (n)	Study Type	Outcome	Risk (95% CI)
432	Pregnancy complicated by diabetes	2-17 years	61,852 individuals	Systematic review and meta-analysis	SBP	Increase (0.47-3.28 mm Hg) in offspring of diabetic mothers (only statistically significant in males when stratified by sex)
765	Genes vs. environment	Birth years 1926-1958	16,265 subjects	Same-sex twin study, all	Hypertension	OR 1.42 (1.25-1.69) per 500-g decrease
			595 subjects	Dizygotic pairs ^c	Hypertension	OR 1.34 (1.07-1.69) per 500-g decrease
			250 subjects	Monozygotic pairs ^c	Hypertension	OR 1.74 (1.13-2.70) per 500-g decrease
548	Fetal and/or infant exposure to famine	37-43 years	1339 subjects	Cohort, Biafra famine, Nigeria (1967-1970)	SBP Hypertension	Increase in exposure OR 2.87 (1.9-4.34) in exposure
550		59 years	971 subjects	Cohort, Dutch famine (1944-1945)	SBP DBP Hypertension	Increase (0.25-5.30 mm Hg) after ≥10 weeks' exposure Not significant OR 1.44 (1.04-2.00) after ≥10 weeks' exposure
470		48-53 years	724 subjects	Cohort, Dutch famine (1944-1945)	Albuminuria	OR 3.2 (1.4-7.7) for exposure to famine in midgestation
766		52-53 years	549 subjects	Cohort, siege of Leningrad	SBP, DBP SBP Albuminuria	Not significant Not significant Not significant
Programmed Associations With Kidney Disease						
519	Preterm birth	0-43 years	4,186,615 subjects 4305 cases	Cohort, Swedish national singleton population (born 1973-2014)	Chronic kidney disease	HR 1.94 (1.74-2.16) Inverse correlation with gestational age at birth
405	Preterm birth	>2 years	4 studies	Meta-analysis	Urine albumin:creatinine	SMD 0.25 (0.07-0.43)
515	Preterm birth Low birth weight	0-15 years	381 cases 20,619,622 controls	Case-control study, Japanese national population (born	Childhood chronic kidney disease	RR 4.73 (3.91-5.73) RR 4.10 (3.62-5.01)

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Table 20.7 Systematic Reviews, Meta-analyses, and Population-Based Studies of Developmentally Programmed Associations With Blood Pressure and Kidney Disease (Continued)

Reference	Condition	Age	Number (n)	Study Type	Outcome	Risk (95% CI)
37	Low birth weight	12-75 years	46,249 subjects	Systematic review	Chronic kidney disease End-stage kidney failure Albuminuria Reduced glomerular filtration rate	OR 1.73 (1.44-2.08) OR 1.58 (1.33-1.88) OR 1.81 (1.19-2.77) OR 1.79 (1.31-2.45)
228	Low birth weight Maternal GDM Maternal overweight Maternal obesity High birth weight Low birth weight ^e	<21 years	1994 cases 20,032 controls	Case-control study	Childhood chronic kidney disease ^d	OR 2.88 (2.28-3.63) OR 1.54 (1.13-2.09) OR 1.24 (1.05-1.48) OR 1.26 (1.05-1.52)
351	SGA Preterm birth Low birth weight ^e	26 years (maximum 50 years)	2,663,010 subjects 4,495 CKD, 12,818 other kidney diseases	Cohort, Norway population (born 1967-2016)	Chronic kidney disease	Not significant OR 1.72 (1.60-1.90) OR 1.79 (1.65-1.94) OR 1.48 (1.33-1.66) OR 1.44 (1.33-1.56) OR 1.24 (1.14-1.36) OR 1.35 (1.17-1.56) OR 1.15 (1.06-1.25) aOR 1.33 (1.19-1.49) aOR 1.74 (1.55-1.95)
520	Familial vs. environment	Mean 28.4 years	1,847,565 subjects with 1-7 siblings, 3,336 with CKD	Cohort, Norway population (born 1967-2015)	Acute kidney disease Glomerulonephritis Cystic kidney disease Congenital malformations No LBW ^e , sibling LBW LBW ^e , no siblings with LBW LBW ^e and sibling with LBW	aOR 1.77 (1.54-2.04)
521	Low birth weight ^e SGA Preterm birth Birth weight <2.5 kg Maternal preeclampsia 1 Risk factor ^f 2 Risk factors ^f 3 Risk factors ^f	Mean 27.9 years (maximum 50 years)	2,679,967 subjects, 1181 cases	Cohort, Norway population (born 1967-2016)	End-stage kidney failure	HR 1.68 (1.44-1.97) HR 1.49 (1.27-1.75) HR 1.65 (1.31-2.07) HR 2.0 (1.59-2.56) HR 1.53 (1.12-2.08) aHR 1.05 (0.84-1.31) aHR 1.67 (1.40-1.98) aHR 2.96 (1.84-4.76)

levels were significantly elevated compared with controls born at term and proportional to systolic blood pressures, suggesting a programming effect of preterm birth on angiogenesis and blood pressure (see further discussion on preeclampsia in Chapter 58).³⁸⁹ Sft-1 levels were even further elevated in those who had been exposed to a hypertensive pregnancy, suggesting an additional impact of preeclampsia. Interestingly, elevated blood pressures in young adults born preterm have also been correlated with lower urinary levels of the protein α -Klotho, which has antiaging and organ protective effects, along with decreased urine angiotensin-(1-7).⁴³⁸

NEPHRON NUMBER AND BLOOD PRESSURE

Clinical studies showing an association between nephron number and hypertension are shown in Table 20.4. A study of Caucasians aged 35 to 59 years who died in accidents found that in 10 people with a history of essential hypertension, the number of glomeruli per kidney was significantly lower and glomerular volume significantly higher than in 10 matched normotensive controls (Fig. 20.12).¹⁰⁶ Birth weights were not reported in this study, but the authors concluded that a reduced nephron number is associated with susceptibility to essential hypertension. Kidneys from Indigenous Australian people with a history of hypertension contained approximately 30% fewer nephrons than those with no history of hypertension.⁴³⁹ Although the sample size was small and birth weights were not available, this is a population with high rates of socioeconomic disadvantage and LBW.⁴⁴⁰⁻⁴⁴² Kanzaki and colleagues⁴⁴³ also reported that kidneys from Japanese men with a history of hypertension contained approximately 40% fewer nephrons than age-matched normotensive men. Similarly, among living donors in the United States, those with hypertension had significantly fewer nephrons per kidney than those without hypertension (determined using a biopsy/imaging technique).⁷⁷ These studies attempted to exclude loss of nephrons due to hypertension as a potential confounder of the association, in the absence of birth weight data. The data therefore seem consistent across several populations that higher blood pressures are associated with lower nephron numbers.

Hughson and coworkers¹²⁹ reported significant correlations between birth weight, mean arterial pressure, and glomerular number, as well as between mean arterial pressure and birth weight, among Caucasian (but not African American) people. Among African Americans having nephron numbers below the mean, however, twice as many were hypertensive as normotensive, suggesting a possible contribution of lower nephron number in this group as well.¹²⁹ The relationship between LBW and nephron number was also similar in a cohort of Black and White Cuban neonates (Fig. 20.5B).¹²⁸ Glomerular volumes were found to be higher among hypertensive African American people than hypertensive White Americans.¹²⁹ The consistent finding of larger glomeruli among African Americans may suggest a greater prevalence of low nephron number in this population as a result of the known higher prevalence of LBW or may reflect independent or additional programming of glomerular size. This topic warrants further research.

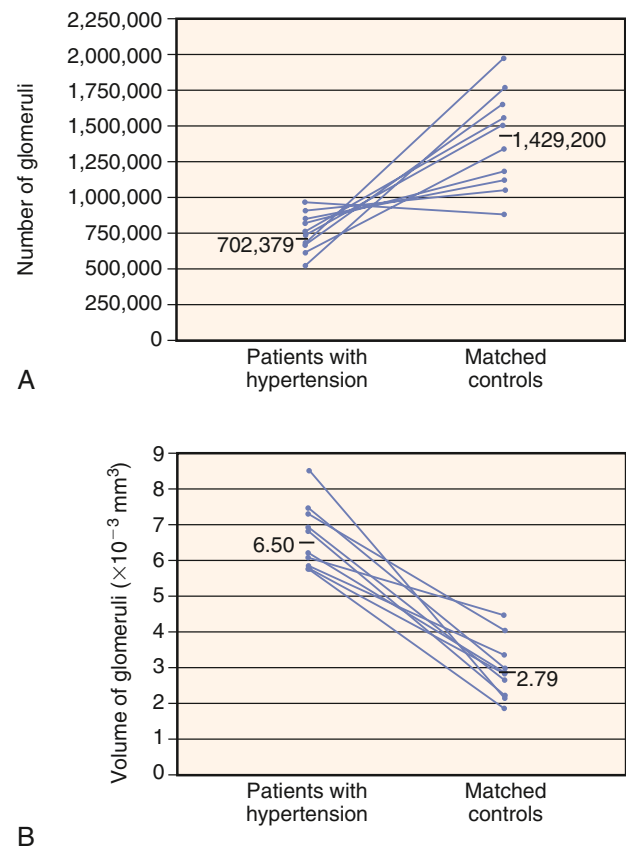


Fig. 20.12 (A) Nephron number, and (B) glomerular volume in Caucasian people with primary hypertension compared with normotensive controls. (From Keller G, Zimmer G, Mall G, et al. Nephron number in patients with primary hypertension. *N Engl J Med*. [2003] 348:101–108.¹⁰⁶)

The pressure-natriuresis curve is shifted to the right in most forms of hypertension. A low total renal filtration surface area associated with a low nephron number is one plausible hypothesis to explain the associated higher blood pressures. Consistent with an association between nephron number and blood pressure, the salt sensitivity of blood pressure has been found to correlate inversely with birth weight and kidney size in adults and children (eFig. 20.1).^{364,365} Similarly, one study found that blood pressure-associated sodium excretion was reduced among preterm compared with term individuals, possibly reflecting a reduced excretory surface area and/or programming of tubular transporters.⁴⁴⁴

BIRTH WEIGHT, PRETERM BIRTH, AND KIDNEY FUNCTION

As with blood pressure, programmed changes in kidney function that occur, at least in the early stages, may not be outside of normal limits. With time or exposure to additional insults, however, these changes may manifest as kidney disease (see Table 20.7).

GLOMERULAR FILTRATION RATE

Amikacin clearance on day 1 of life, measured as a correlate for neonatal GFR, was found to be significantly lower in LBW and preterm neonates than in term controls.⁴⁴⁵ eGFR

correlates with both gestational age and postnatal age in infants born preterm, corresponding with structural and functional maturation of the kidneys over time.⁴⁴⁶⁻⁴⁴⁹ The rate of postnatal renal functional maturation, however, differs according to gestational age at birth.^{448,449} Among preterm neonates, lower eGFR has been associated (independently of age) with SGA, as well as medication exposures, low blood pressure, and poor health outcomes.^{447,448} At 2 years corrected gestational age, 18% of extremely preterm infants were found to have eGFRs <90 mL/min/1.73 m².⁴⁰⁹

In a cohort of children born very preterm (mean age 7.6 years), inulin clearance was significantly lower (although was still within the normal range) among those who had experienced growth restriction compared with no growth restriction perinatally.²⁰¹ Importantly, GFR was lower among children who had been either growth restricted in utero or in the first weeks after birth (in intensive care), highlighting the role of postnatal nutrition as a modulator of renal programming.²⁰¹ Several studies in preterm and LBW children have found similar associations, with an 8% to 25% reduction in eGFR observed in children born with ELBW compared with term infants.^{405,450} Studies using creatinine-based formulas may underestimate the impact of birth weight on GFR, however, suggesting a need to validate measures of renal function in people born LBW and preterm whose body composition may differ over time and/or in comparison to controls.⁴⁵¹⁻⁴⁵⁵

The relationship between birth weight and eGFR appears to track with time, suggesting consistency of the associations. Among 5352 adolescents from the U.S. National Health and Nutrition Examination Survey (NHANES), eGFRs were lower in LBW (OR: 1.49; 95% CI: 1.06–2.10) and VLBW (OR: 2.49; 95% CI: 1.20–5.18) adolescents compared with those with normal birth weight.⁴⁵⁶ This study was the first population-based assessment among adolescents of multiple ethnicities, permitting an estimation of the population-attributable fraction of birth weight to blood pressures and eGFR: 1 in 13 LBW or 1 in 5 VLBW adolescents had systolic blood pressures ≥95th percentile and/or eGFR <90 mL/min/1.73 m².⁴⁵⁶ A subsequent analysis of 6336 participants from NHANES found the prevalence of an eGFR <90 mL/min/1.73 m² to be 30.1% among those with LBW compared with 22.4% among those with a normal birth weight.⁴¹⁴ Similarly, among 57 Norwegian children at age 11 years, eGFRs were lower in those born preterm and AGA compared with term AGA children and lower still in those born preterm and SGA, again suggesting exacerbation of risk with growth restriction.⁴⁵⁷

Overall, a systematic review found an OR of 1.79 (95% CI: 1.31–2.45) for a reduced GFR with LBW (see Table 20.7).³⁷ Linear regression analysis in a cohort of 2192 British adults aged 60 to 64 years revealed that for each 1 kg decrease in birth weight, adult GFR estimated using cystatin C was reduced by 2.25 mL/min per 1.73 m² (95% CI: 0.69–3.58 mL/min per 1.73 m²).⁴⁵⁸ Similarly, a meta-analysis of studies across all ages has shown a significant reduction in eGFR associated with preterm birth (standardized mean difference [SMD] compared with term controls: –0.54, 95% CI –0.85 to –0.22).⁴⁰⁵ Taken together, these findings are consistent with both LBW and preterm birth being risk factors for a reduced kidney function. A study in twins further supported the importance of perinatal health, showing that fetoplacental

factors have a greater impact than genetic factors on adult kidney function.⁴⁵⁹

To investigate the mechanism of the observed reductions in GFR, renal functional reserve, determined by measuring GFR and effective renal plasma flow (ERPF) before and after low-dose dopamine infusion or an oral amino acid load, was studied in young adults who had been born preterm and either SGA or AGA versus term AGA controls.⁴⁶⁰ After renal stimulation, the relative increase in GFR tended to be lower in the preterm SGA group compared with AGA and control groups, and ERPF was lower in both groups of preterm adults; however, statistical significance was not reached, likely because of small numbers. Renal functional reserve was also not significantly different among healthy adults aged 41 to 52 years with birth weights below 2.5 kg compared with those with birth weights from 3 to 4 kg.⁴⁶¹ Reduced renal functional reserve was, however, observed among young adults of mothers with type 1 diabetes, who had been exposed to diabetes during gestation, but not those of fathers with diabetes, suggesting a direct programming rather than genetic effect.¹²⁰ Measurement of renal reserve capacity is cumbersome, and therefore the data are sparse and conflicting. Further work is required to understand whether this may be a useful clinical predictive test (e.g., in people at risk of developmental programming who may wish to become living kidney donors).⁴⁶²

PROTEINURIA

One of the earliest signs of hyperfiltration, which would be expected in the setting of a reduced nephron number and filtration surface area, is microalbuminuria, which may progress to overt proteinuria with ongoing renal injury and worsening hyperfiltration. Indigenous Australians who had LBW evidenced an OR for macroalbuminuria of 2.8 (95% CI: 1.26–6.31) compared with those who had normal birth weights; risk increased with age.⁴⁴⁰ Importantly, proteinuria was also associated with a higher rate of cardiovascular and renal deaths in this population, underscoring its clinical relevance.⁴⁶³ A meta-analysis including eight additional studies reported an OR of 1.81 (95% CI: 1.19–2.77) for albuminuria with LBW (see Table 20.7).³⁷ Hoy and colleagues^{442,464} subsequently reported that LBW, childhood poststreptococcal glomerulonephritis, and current body mass were all independent predictors of albumin to creatinine levels in Indigenous Australian adults. These findings are compatible with the “multihit” model of CKD, of which nephron endowment at birth may be the first “hit” increasing susceptibility to kidney disease throughout the life course (Fig. 20.11).

Preterm birth has also been associated with albuminuria, with a meta-analysis of a small number of studies in children showing increased urine albumin/creatinine levels in those born preterm compared with term controls (SMD 0.25, 95% CI 0.07–0.43).⁴⁰⁵ Similarly, a subsequent study reported a 36% prevalence of albuminuria in extremely low-gestational age neonates evaluated at 2 years corrected gestational age.⁴⁰⁹ Besides albumin, there is additionally some evidence of elevated low-molecular-weight protein excretion (a marker of proximal tubule dysfunction) in infants born preterm, which can also persist into later life.^{448,465-467} Among children aged 4 years who had been born preterm, albuminuria was higher in both boys and

girls who had reached normal height (presumably caught up in growth), and among 19-year-olds who had been born very preterm, albuminuria was higher among those who were SGA, highlighting the interplay among preterm birth, growth restriction, and catch-up growth on later risk of disease.^{468,469} Similarly, among 14-year-old children, current overweight and obesity further increased the association between preterm birth and albuminuria.³⁵⁰ Not all studies have shown these associations, however. For example, in NHANES, despite there being significant differences in blood pressure and eGFR among participants, there was no difference in the urine albumin/creatinine ratio between adolescents born with LBW, VLBW, or normal birth weight.⁴⁵⁶

An analysis of 724 people, aged 48 to 53 years, who had been exposed to malnutrition in midgestation during the Dutch famine (1944–1945) revealed an increased prevalence of microalbuminuria (12%) when compared with those exposed during early gestation (9%), late gestation (7%), or not exposed to famine (4%–8%) (see Table 20.7).⁴⁷⁰ Size at birth was not associated with microalbuminuria, emphasizing the need for surrogate markers in addition to birth weight to better identify individuals at risk for renal programming. In the Aguascalientes region of Mexico, an area with one of the highest global rates of early-onset CKD of unknown cause,⁴⁷¹ persistent proteinuria in adolescents is associated with decreased kidney volume, suggestive of a nephron deficit^{472,473}; any other potential perinatal associations are as yet unknown.

A U-shaped association between birth weight and proteinuria was described among Pima people, with increased risk evident for birth weights below 2.5 kg and above 4.5 kg.¹³ The strongest predictor of proteinuria among HBW individuals in this study was exposure to gestational diabetes.¹³ In a Canadian study, urine albumin/creatinine ratios were lower in infants of mothers with diabetes compared with nondiabetic mothers at 1 year of age but were higher at 3 years, although independent of birth weight.⁴⁷⁴ The effects of gestational diabetes and HBW on renal programming require much more study.

NEONATAL ACUTE KIDNEY INJURY

Preterm birth is an important risk factor for acute kidney injury (AKI) in neonates.^{475–477} A multinational and multicenter study of neonates hospitalized in intensive care reported a U-shaped distribution of AKI incidence according to gestational age, with rates of 47.9% in extremely preterm infants, 18.3% in very and moderately preterm infants, and 36.7% in infants born ≥ 36 weeks' gestation.⁴⁷⁸ Neonates in intensive care are particularly susceptible to kidney dysfunction, not only because of low nephron number (immaturity and/or impaired nephrogenesis) and potentially programmed risk but also because of critical illness (e.g., sepsis), hypotension, malnutrition, and frequent nephrotoxic exposures such as aminoglycosides and NSAIDs.^{477,479,480}

A retrospective analysis of preoperative kidney volume in neonates undergoing congenital heart surgery found higher peak postoperative creatinine values and therefore potentially increased risk of AKI in infants with a kidney volume ≤ 17 cm³.⁴⁸¹ High creatinine values were associated with lower gestational age and lower birth weight z-score.

Among 269 infants, perinatal ibuprofen exposure was associated with an OR of 2.6 (95% CI: 1.2–5.3) for a reduced GFR on day 7 of life, which persisted over the first month.⁴⁸² Other authors found that exposure to aminoglycosides was associated with higher serum creatinine in preterm SGA infants at 2 months.⁴⁸³ A large multicenter study of 8286 preterm infants determined that those with a longer duration of nephrotoxic drug exposure (regardless of specific drug combination) had an increased risk of AKI.⁴⁷⁹ As yet unexplained, maternal preeclampsia appears to be a protective factor for neonatal AKI.⁴⁸⁴

The incidence of AKI in critically ill neonates ranges from around 18% to 70% and is an independent predictor of longer hospital stays and mortality.^{475,477,478,480} Neonatal AKI likely further hinders ongoing kidney development or causes additional nephron loss and is therefore an added risk factor for higher blood pressures and CKD in later life.^{485–489} Importantly, around 1 in 5 neonates develop recurrent AKI during their hospital stays, which further exacerbates long-term risk.^{476,490} Risk factors for this include lower gestational age, lower birth weight, and increased severity of initial AKI episode. Neonatal AKI was also associated with increased diastolic blood pressures in extremely preterm neonates at 2 years corrected age.⁴⁰⁹ Prevention of perinatal AKI is therefore important to reduce the risk of later-life CKD, and long-term follow-up of infants who develop AKI is required to appropriately modify this risk.^{488,491} It is to be noted, however, that there are many constraints regarding the feasibility, methodology, and criteria for AKI diagnosis in neonates of varying maturity, with ongoing efforts to optimize the current diagnostic criteria and also to investigate the utility of other AKI biomarkers.^{477,485,492,493} Current rates of documentation and follow-up of neonatal AKI are suboptimal (9%–15%) and must be improved.^{476,494}

PROGRAMMING OF RISK FOR CHRONIC KIDNEY DISEASE AND END-STAGE KIDNEY FAILURE

The Brenner hypothesis can explain the link between a low nephron endowment and susceptibility to the development of hypertension and CKD.^{66,171,495,496} Briefly, a low total glomerular filtration surface area in the kidneys causes a decreased filtered load, increased sodium reabsorption, and increased systemic fluid volume, with a consequent increase in blood pressure and glomerular capillary pressure. It also triggers structural and functional adaptations by the remaining nephrons to meet filtration requirements, such as glomerular and tubular hypertrophy. The hyperfiltering and hypertrophied glomeruli are susceptible to podocyte injury/loss and glomerulosclerosis, with associated tubulointerstitial fibrosis and vascular sclerosis. As the disease progresses, there is a self-perpetuating cycle of further nephron loss and thus an increasing severity of CKD.^{66,171,495,496} Consistent with evolutionary theory, the majority of people with a low nephron number (excluding CAKUT) will not develop kidney failure until middle-older age, past the peak reproductive years.^{351,497} Besides a nephron deficit, other factors affected by early life exposures also have the potential to contribute to disease initiation and progression, such as podocyte endowment, vascular structure and function, AKI,

inflammation, and oxidative stress.^{173,498,499} Additionally, along with hypertension, people born preterm or SGA may also be at heightened risk of developing comorbidities such as diabetes and obesity, which can further exacerbate CKD risk and progression.^{31,500,501} With such complexity, however, the interplay between early life exposures and later structural and physiologic adaptations, repair processes, cellular senescence, and disease initiation remain to be fully elucidated.

GLOMERULOMEGALY

Given the challenges with measurement of nephron number *in vivo*, glomerulomegaly seen on kidney biopsy may be a useful clinical marker of a reduced nephron number. Human glomeruli have been reported to increase in size up to sevenfold from infancy to adulthood.^{127,502-504} This can be considered normal physiologic growth. Glomerular size also increases in adulthood in people without overt renal disease and is associated with increasing age, increasing body size, and lower birth weight.^{349,505} Mean glomerular volume has also been consistently noted to vary inversely with total glomerular number (Fig. 20.5B), although the correlation appears stronger among Caucasians and Indigenous Australian people than in people of African origin.^{106,107,128,506} This relationship suggests that larger glomeruli may reflect compensatory hyperfiltration and hypertrophy in people with fewer nephrons and may therefore be a surrogate marker for a reduced nephron number.^{94,128} In fact, Hoy and coworkers¹⁰⁷ found that, although mean glomerular volume was increased in association with reduced nephron number, total glomerular tuft volume (a surrogate for total renal filtration surface area) was not different among groups with different nephron numbers (see Table 20.3). This suggests that total renal filtration surface area may initially be maintained in the setting of a low or reduced nephron number but at the expense of glomerular hypertension and hypertrophy, which are maladaptive and predictors of poor outcomes.⁵⁰⁷⁻⁵⁰⁹

Consistent with this possibility, glomerulomegaly is common in renal biopsies from Indigenous Australian people, a population with high rates of LBW births and renal disease, and it has also been associated with faster rate of decline of GFR in Pima people.^{441,463,510} Furthermore, in a study of donor kidneys, maximal planar area of glomeruli was found to be higher in kidneys from African Americans compared with White Americans and was a predictor of poorer posttransplant function.⁵⁰⁸ Among 111 adult males from four ethnic groups, mean glomerular volume and variability were highest among African Americans and Indigenous Australians, likely associated with susceptibility to hypertension and kidney disease.⁵¹¹ In populations at high risk of kidney failure therefore large glomeruli are a common finding at early stages of renal disease and may reflect programmed reductions in nephron number in these populations in which access to prenatal and subsequent health care is often suboptimal.^{473,512-514} An increase in glomerular volume, however, is not always associated with lower glomerular number, and individual glomerular volume varies considerably within kidneys.⁵⁰⁶ Overall, however, glomerulomegaly and greater intrakidney variability in glomerular volume are associated with older

age, fewer nephrons, lower birth weight, hypertension, obesity, and severity of cardiovascular disease and therefore may reflect response to kidney stress.⁵¹¹

LOW BIRTH WEIGHT, PRETERM BIRTH, AND CHRONIC KIDNEY DISEASE

Strong associations have been reported between LBW, gestational age at birth, and the development of CKD in Japanese and North American children attending pediatric CKD clinics.^{228,515} In the U.S. population, LBW, maternal gestational diabetes, maternal overweight, and obesity were all significantly associated with an increased risk of childhood CKD (see Table 20.7).²²⁸ In addition, a significant association was found among LBW, maternal pregestational diabetes, and renal dysplasia/aplasia and among LBW, maternal gestational diabetes, maternal overweight or obesity, and congenital obstruction. In the Japanese population, the associations remained highly significant even after exclusion of children with CAKUT, with an estimated population-attributable fraction of pediatric CKD being 21.1% (95% CI: 16.0%–26.1%) for LBW and 18.2% (95% CI: 16.5–25.6%) for preterm birth.⁵¹⁵

Strong associations between LBW and CKD are also seen in later life. An early meta-analysis of 18 studies reported an OR for CKD of 1.73 (95% CI: 1.44–2.08) with LBW.³⁷ This association has been confirmed in subsequent meta-analyses and further supported by population-wide studies in Nordic countries, the United States, and Japan (see Table 20.7).^{36,58,228,351,515-518} More recent studies have focused on the relative impacts of LBW, preterm birth, and SGA. In a birth registry study of more than 4 million people aged 43 years and under in Sweden, those born preterm were found to develop CKD at almost twice the rate of those born at term; with an inverse association between gestational age at birth and CKD risk, people born extremely preterm were at the highest risk (HR 3.01, 95% CI 1.67–5.45).⁵¹⁹

In a Norwegian birth registry study of more than 2.6 million people using the following definitions, LBW <10th percentile for Norway (<2940 g males, <2850 g females), SGA <10th percentile for gestational age and sex, and preterm birth <37 weeks' gestation, the OR for CKD in people born with LBW was 1.72 (95% CI: 1.60–1.90), with SGA was 1.79 (95% CI: 1.65–1.94), and with preterm birth was 1.48 (95% CI: 1.33–1.66).³⁵¹ Risks were similar for men and women. In this population, risk appeared to be somewhat modulated by familial factors because the risk of CKD was highest among those with LBW and a sibling with LBW (adjusted [a]OR 1.77, 95% CI 1.54–2.04) but also increased in those who were not born with LBW but had a sibling with LBW (aOR 1.33, 95% CI 1.19–1.49).⁵²⁰

LOW BIRTH WEIGHT, PRETERM BIRTH, AND END-STAGE KIDNEY FAILURE

In assessing risk of progression to ESKF, a retrospective analysis of a cohort of more than 2 million Norwegian individuals (mean age 27.9 ± 11.5 years), using the same birth definitions mentioned earlier, found the relative risk of developing ESKF over 50 years of follow-up to be 1.68 (95% CI 1.44–1.97) for LBW, 1.49 (95% CI 1.27–1.75) for SGA, 1.65 (95% CI 1.31–2.07) for preterm birth,

2.0 (95% CI 1.59–2.56) for LBW under 2.5 kg, and 1.53 (95% CI 1.12–2.08) for maternal preeclampsia (see Table 20.7).⁵²¹ No gender differences were observed. As reported previously, ESKF was noted to occur at younger ages among LBW individuals with hereditary disease or urogenital malformations but at similar ages for diabetes, glomerular disease, and interstitial disease.^{58,521} A dose–response relationship was again observed with increasing risk of ESKF with lower birth weights and lower gestational ages.⁵²¹ An earlier analysis had found no evidence of modulation of the association of birth weight and ESKF by familial factors in this population, emphasizing the strong impact of environmental exposures in renal programming.⁵⁸

In a U.S. dialysis-based study, LBW was also associated with an increased risk of ESKF, but the odds for diabetic ESKF were also increased among those having birth weights >4 kg (OR 2.4; 95% CI: 1.3–4.2).⁶⁸ The relevance of HBW was also highlighted by a U-shaped association between kidney disease and birth weight found in two large population-based studies.^{35,36} However, effects differed between males and females, again reflecting potential effect modification by gender under conditions as yet not fully elucidated. Whether HBW in these studies was associated with intrauterine exposure to diabetes is not known.

Among 1850 Pima people younger than 45 years with diabetes, the incidence rate ratio of developing ESKF was 4.12 (95% CI: 1.54–11.02) among those who had been exposed to diabetes in utero compared with those without exposure.⁵²² This effect disappeared after controlling for duration of diabetes in this relatively young cohort, suggesting the ESKF risk may be predominantly mediated by a programmed risk for earlier-onset diabetes among those exposed to diabetes in utero. These findings emphasize that programming of kidney disease risk may be indirect, occurring as a consequence of programming of other disorders such as diabetes, underscoring the need for a holistic and life-course approach to understand and tackle the programming of risk for kidney disease.^{18,26,403,442,522,523}

RELATIVE IMPACT OF BEING BORN SMALL OR PRETERM

As illustrated in Fig. 20.1, LBW may be the result of preterm birth and/or growth restriction. An updated analysis of the Norwegian birth registry, with up to 50 years of follow-up, revealed a “dose–response” relationship was observed with lower birth weights and gestational ages having higher risks, and with greater risk of CKD being observed in those with increasing numbers of risk factors (see Fig. 20.13A).³⁵¹ In a similar analysis by the same group, after adjusted analyses the risk of ESKF (in contrast to CKD) was only observed to increase when two or more risk factors were present (Fig. 13B), with the highest risk being observed in those born preterm, SGA and LBW (HR 2.96, 95% CI 1.84–4.76).⁵²¹ No risk factor alone therefore appeared to predominate, but when occurring together the risk increased. LBW, SGA, and preterm birth often coexist, so many babies are at risk.⁵²⁴ This knowledge may be useful to further stratify the risk, especially regarding intensity of long-term follow-up. Importantly, this study reflects findings in a high-income relatively healthy Caucasian population, possibly less exposed to second “hits” than people living in lower-resource settings. A single birth

risk factor might be enough to initiate CKD if second hits are experienced more commonly or repeatedly. Studies are therefore required from other settings.

PROGRAMMED SUSCEPTIBILITY TO SECOND HITS

As mentioned earlier, it is likely too simplistic to assume that altered kidney development associated with LBW, preterm birth, or other developmental stresses, in itself, is always enough to cause kidney disease, but exposure to additional “hits” (e.g., nephrotoxin exposure, AKI, and glomerulonephritis), or superimposed developmental programming of conditions that are themselves risk factors for kidney disease (i.e., diabetes, cardiovascular disease, metabolic syndrome, obesity) all likely exacerbate kidney disease risk.^{18,26,403,442,522,523}

In the Norwegian population-based studies, LBW and SGA births were associated with increased risk of progression to ESKF in IgA nephropathy.⁵⁸ Subsequently, associations were found between LBW and acute kidney disease (OR 1.44, 95% CI 1.33–1.56), glomerulonephritis (1.24, 95% CI 1.14–1.36), cystic kidney disease (OR 1.35, 95% CI 1.17–1.56), and CAKUT (OR 1.15, 95% CI 1.06–1.25), which are unlikely to be programmed events (see Table 20.7). These findings are therefore consistent with the concept that developmental programming in the kidneys likely represents a “first hit,” making the kidneys more vulnerable to subsequent independent insults. Similarly, among 2240 adults and children in the multicenter Cure Glomerulopathy (CureGN) cohort, patients born preterm had a higher prevalence of FSGS (35% vs. 25%), were 38% less likely to respond to treatment, and were 28% more likely to reach the endpoints of a 40% decline in eGFR or ESKF compared with those born at term.⁵²⁵ Small studies of diverse kidney disease have observed similar associations.^{13,525–532} These findings are also relevant for kidney transplantation, as discussed later.

POTENTIAL IMPACT OF POPULATION DEMOGRAPHICS AND ETHNICITY ON RENAL PROGRAMMING

The nephron number hypothesis was initially put forward as a potential explanation for the disproportionately higher risks of hypertension and kidney disease among disadvantaged populations.⁶⁶ Hypertension, CKD, and ESKF prevalences tend to be highest among Indigenous people in Australia and America, people of African origin, and ethnic minorities in developed countries.^{11,13,68,471,533} The incidence of adverse birth circumstances and maternal risk factors for kidney programming are higher across low- and middle-income countries and in minoritized populations in higher-resource settings, suggesting a potentially significant global population impact in populations with suboptimal access to care.⁶⁵ Most studies regarding programming of blood pressure and kidney disease have thus far been conducted in western Caucasian populations and may not be generalizable to all populations. In the Norwegian population, for example, only 3.3% of newborns have birth weights below 2.5 kg, whereas around 13% to 15% of babies in sub-Saharan Africa are LBW.^{58,240} Hypertensive disorders of pregnancy affect 3% to 10% of pregnancies

worldwide, again more common in lower-resource settings.⁵³⁴ Furthermore, the causes of SGA or preterm birth differ across regions, with maternal nutrition, poverty, and infections being important in lower-income regions and advanced maternal age, multiple gestations, and use of assisted reproduction technology predominating in high-income regions (Table 20.8).⁵³⁵⁻⁵³⁷ How such differences may modulate programming risk across populations is not known.⁵³⁸⁻⁵⁴¹

Among 715 South Asian infants (of Pakistani, Indian and Bangladeshi origin) from the Born in Bradford (United Kingdom) cohort study, kidney size was smaller than among 872 White babies, and the difference persisted after adjustment for potential confounders including birth weight.⁵⁴² In adolescents from the Aguascalientes region of Mexico, kidney dimensions were found to be smaller than in all other comparable studies internationally⁴⁷²; associations with birth weight or gestational age are unknown. Total kidney volume was also found to be lower in LBW and preterm Indigenous Australian neonates at 32 and 38 weeks of gestation and at term compared within non-Indigenous Australian neonates, suggesting a lower nephron number in the Indigenous children.⁵⁴³ However, in this study, calculated GFR (serum cystatin C) was similar in both groups, suggesting SNGFR was increased in Indigenous children. In another larger study of Australian neonates born preterm, however, Indigenous ethnicity was shown to be significantly associated with increased serum creatinine, fractional excretion of sodium, and urine protein excretion compared with non-Indigenous neonates.⁴⁴⁸ Additionally, the Indigenous infants were most susceptible to kidney dysfunction with nephrotoxic drug exposure.⁴⁴⁸ In a study comparing eGFR in 152 African American and White American children at a mean age of 1.5 years, the eGFR was found to be significantly lower among African American children (82 vs. 95 mL/min/1.73 m²).⁵⁴⁴ Birth weight was significantly and positively associated with eGFR in African American but not White children, suggesting that some of the racial disparities in adult CKD may have origins in the prenatal period.

Nephron numbers have thus far been counted using gold-standard stereologic techniques only in small cohorts of Caucasians, African Americans, Indigenous Australians, Senegalese Africans, and Japanese males, many of whom had unavailable birth weights (see Table 20.2). Therefore the relationship between birth weight and nephron number remains unknown in most populations. A study of nephron number and hypertension among older Japanese males found the nephron counts in those who were normotensive were among the lowest reported to date (~640,000 nephrons), suggesting that Japanese people may be especially vulnerable to adult hypertension and CKD, particularly if a second “hit” occurs.⁴⁴³ This hypothesis is consistent with Japan having the second-highest incidence of ESKF in the world.⁵⁴⁵

LBW is strongly associated with reduced nephron numbers and susceptibility to kidney disease among Indigenous Australian people, a population that is also significantly disadvantaged compared with their White counterparts.^{439,441,463} The association between nephron number and birth weight does appear to be consistent among people of African origin, as in Caucasian

populations.^{127,128,546,547} The relationship between low nephron numbers and higher blood pressures seen in Indigenous Australians and Caucasians, however, was not as consistent among African Americans.⁵⁴⁰ Similarly, the association between LBW and blood pressure is also less consistently reported among African American compared with Caucasian children; therefore additional factors likely augment blood pressure risk in the African-origin population.^{415,544}

Evidence for developmental programming of blood pressure in an African population has been shown among Nigerian adults exposed to the Biafran famine (1967–1970) during fetal life and early childhood.⁵⁴⁸ Blood pressures were elevated in adult Nigerians exposed to famine, which is consistent with observations among Caucasians exposed to the Dutch famine (1944–1945).⁵⁴⁸⁻⁵⁵⁰ However, blood pressures in Biafran people were elevated 15 to 20 years earlier than among the Dutch (see Table 20.7), potentially again suggesting aggravating factors present in African-origin populations.^{548,550} In addition, in both populations, exposure to famine was also associated with increased risks of glucose intolerance and obesity, underscoring the multisystem consequences of developmental programming.^{548,549}

Gain of function variants in the *APOL1* gene has been strongly linked to risk of CKD in African American and West African people.^{4,9} It is an evolutionary trait that protects against *Trypanosoma* infection, with strong positive selection.⁴⁹⁷ Glomerular number and size were not affected among African Americans with one or two *APOL1* variant alleles, but accelerated loss of nephrons after the age of 38 years has been observed, which was exacerbated by obesity.⁵⁵¹ Consistent with a potential superimposed programming effect, the *APOL1* high-risk genotype was observed more frequently among individuals who had been born preterm than those born at term (9.4% vs. 4.2%).⁵²⁵ Preterm birth is more common among African American women in the United States and in high-risk populations in Africa; therefore this association with *APOL1* genotype may be a contributor to the greater risk of kidney disease, which is well recognized in populations of African origin.⁵⁵²⁻⁵⁵⁴

Many social determinants of health impact maternal health, pregnancy health, and birth outcomes in high- and low-resource settings (see Table 20.8). Addressing these risk factors at a policy and population level is imperative to reduce the rates of adverse pregnancy and birth outcomes globally.⁵⁵⁵

POTENTIAL FOR RESCUE OF NEPHRON NUMBER

The possibility that interventions could be designed to modulate developmentally programmed changes in the kidneys and reduce long-term disease risk has been explored to some extent. The optimization of maternal health and nutrition before and during pregnancy to attenuate any risk factors for IUGR and preterm birth is the most obvious intervention, as it has been estimated that intrauterine factors determine around 60% of the variation in birth weight.⁵⁵⁶ Prevention of the initiating factors leading to a nephron deficit is likely more realistic than rescue.

Table 20.8 Maternal Factors Associated With Increased Risk of Small and Vulnerable Newborn Births

	Maternal Factor	Global Incidence/Prevalence (%)	SpecificReferences
Developmental	Maternal birth weight <2.5 kg or >4.0 kg	—	
	Maternal gestational age	—	
	Short stature women of reproductive age (<155 cm)	19.7-68.5%	629, 631
Behavioral	Maternal smoking	9-16% HIC 2.6% LMIC	767, 768
	Caffeine consumption	—	
	Alcohol consumption	—	
	Substance and/or drug abuse	—	
Demographic	Age <18 years or >40 years	—	
	Ethnicity	—	
Health related	Undernutrition in women 20-49 years (BMI <18)	12%	535, 629
	Anemia in pregnant women (Hb <110 g/L)	38.2% (95 CI: 34.3-42.0)	769, 629
	Iron deficiency in pregnant women	19.2% (95 CI: 17.1-21.5)	
	Vitamin A deficiency in pregnant women (night blindness)	7.8 % (95 CI: 6.5-9.1)	
	Zinc deficiency (general population)	17.3% (95 CI: 15.9-18.8)	
	Malaria (pregnancy, submicroscopic+microscopic)	21.5%	770
	Hyperglycemia in pregnant women 20-49 years (diabetes/gestational diabetes)	16.9%	771
	Hypertensive disorders of pregnancy	5.2-8.2%	772
	Preeclampsia	4.6% (95 CI: 2.7-8.2)	773
	Eclampsia	1.4% (95 CI: 1.0-2.0)	
	Chronic kidney disease, transplant, dialysis	—	
	Maternal chronic illness	15.8% (Denmark, 2013)	774
	Multiple-gestation pregnancy	—	
	Multiparous (≥3)	—	
	Assisted reproduction technology	2-3% annual births HIC	775, 776
	Infections	—	
	Maternal overweight (BMI 25–29.9 kg/m ²)	30%	629, 777, 778
	Maternal obesity (BMI ≥30 kg/m ²)	11%	629, 634
	Highly active antiretroviral therapy for HIV	—	
Social	Racism, discrimination	—	
	Health literacy	—	
	Recommended 4 antenatal care visits	69%	779
	Unplanned pregnancy, birth spacing	26.5% (range 19.25-61.71)	780
	Teenage pregnancy (<19 years)	21 million per year	781
	Conflict, war, stress	—	
	Environmental conditions	—	
	Education level	—	
	Poverty	—	
Environmental	Seasonal variations in nutrient availability	—	
	Toxin or pollutant exposure	—	
	Consequences of climate change	—	

BMI, Body mass index; CI, confidence interval; HIC, high-income country; HIV, human immunodeficiency virus; LMIC, low- and middle-income countries.

Experimental interventions that focus on the augmentation of nephron number have also been investigated; however, this area of research is still in its early stages and clinical feasibility is largely unknown (eTable 20.1).

Encouragingly, a review of evidence-based antenatal and intrapartum interventions identified multiple things that are feasible to implement and which are associated with a reduction in the rates of small and vulnerable neonate births and/or stillbirths in low- and middle-income countries (see Fig. 20.9).¹⁹⁹ Multiple micronutrient supplementation,

balanced energy and protein supplementation, screening and treatment for syphilis, and use of insecticide-treated bed nets for malaria were associated with reduced risk of preterm birth, SGA, and LBW; screening and treatment for bacteriuria was associated with reduced risk of LBW; preconception food fortification and folic acid supplementation, low-dose aspirin, calcium supplementation, and vaginal progesterone in women at risk of preterm delivery were associated with reduced risk of preterm birth; and zinc supplementation was associated with reduced risks of preterm birth and LBW.

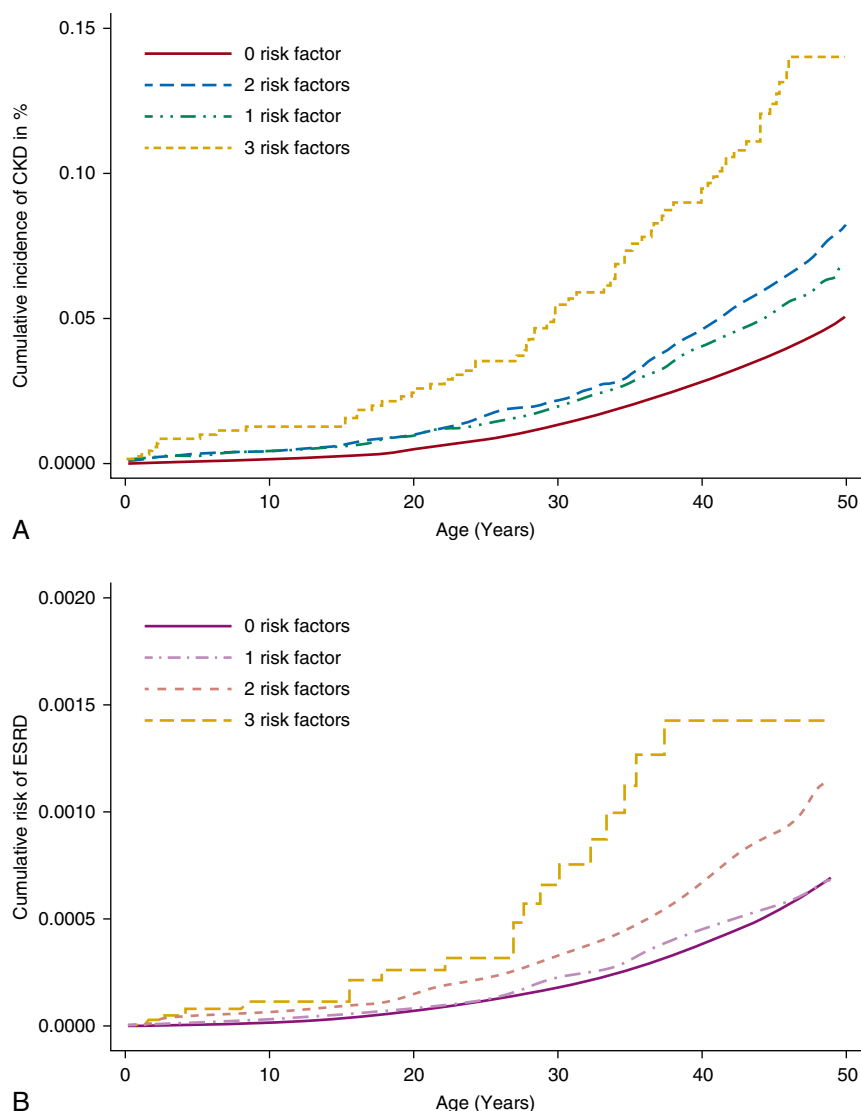


Fig 20.13 (A) Cumulative incidence (%) of chronic kidney disease (CKD) according to number of birth-related risk factors (birth weight <10th percentile, birth weight for gestational age <10th percentile, and preterm birth before 37 weeks' gestation). (B) Cumulative incidence of end-stage kidney failure (ESKF) according to number of birth-related risk factors (listed above). Compared with having none of the exposures, individuals with one exposure had a significantly higher risk of CKD but not ESKF. Individuals with two exposures had a higher risk than individuals with one exposure, and individuals with three exposures had the highest risk for both CKD and ESKF. ESKF, End-stage renal disease/ESKF. (From Gjerde A, Reisæter AV, Skrunes R, et al. Intrauterine growth restriction and risk of diverse forms of kidney disease during the first 50 years of life. *CJASN*. 2020;15:1413–1423 and Gjerde Lillås BS, Marti HP, et al. Intrauterine growth restriction, preterm birth and risk of end-stage renal disease during the first 50 years of life. *Nephrol Dial Transplant*. 2020;35[7]:1157–1163.^{351,521})

Effective intervention approaches may require improvements in public health strategies across at-risk communities, alongside individualized treatments; for example, sildenafil citrate administration improves uteroplacental perfusion and increases fetal weight gain in cases of IUGR.⁵⁵⁷ Health optimization during development would not only improve the long-term health of the individual infant but also positively influence the epigenetic programming of future generations (described later).

MATERNAL DIET

Maternal dietary interventions have been most widely studied with respect to optimizing fetal kidney development

(see Lee and colleagues⁵⁵⁸ for a systematic review of clinical studies). Maternal nutritional deficiencies are common (see Table 20.8), and maternal and neonatal replacement of micronutrients has been implemented as a public health policy in several countries (see Fig. 20.9).⁵⁵⁹ Long-term follow-up of children is sparse, but effects should become clearer as cohorts of children age and more data emerge.^{559,560} A placebo-controlled clinical trial of vitamin A supplementation in Nepalese women before, during, and following pregnancy found no significant difference in blood pressure or albuminuria in children at 9 to 13 years of age; kidney size was not assessed.⁵⁶¹ A significant reduction in risk of microalbuminuria was reported, however, in 6- to 8-year-old Nepalese children whose mothers received folic acid supplementation during pregnancy.⁵⁶² In a study

of 2335 children at 4.5 years of age in rural Bangladesh, those whose mothers received 60 mg versus 30 mg of iron supplementation per day during pregnancy had significantly higher eGFR, but there was no effect on kidney volume or blood pressure.⁵⁶⁰ Whether there was any influence on the children's nephron development is unknown.

Supplementation with retinoic acid (the active metabolite of vitamin A),⁵⁶³ folic acid,⁵⁶⁴ and iron²¹⁷ in experimental programming models have resulted in improved offspring nephron endowment. Iron supplementation of iron-deficient lactating dams from postnatal day 7 delayed the cessation of nephrogenesis in rat offspring,⁵⁶⁵ highlighting the possibility of extending nephrogenesis for the rescue of nephron deficits.

In animals, improvements in macronutrient availability may augment nephron endowment. Offspring of mice fed a high-fat diet before and during pregnancy and lactation exhibited a 20% increase in nephron endowment compared with controls.²³⁹ Although a low-protein diet is commonly used to model nephron deficits, maternal high-protein consumption was shown to have no impact on rat offspring body weight, kidney weight, or nephron number.⁵⁶⁶ High-protein postnatal feeding of growth-restricted rat pups did not affect nephron number but was also detrimental to glomerular function in adulthood.⁵⁶⁷

Individual amino acids may have differential potential to rescue nephron number, as highlighted in [eTable 20.1](#). Supplementation of glycine, alanine, or also urea in a rat maternal low-protein diet model prevented low nephron numbers in offspring; only glycine prevented subsequent hypertension.²⁰⁴ L-citrulline (a modulator of nitric oxide metabolism) has been shown to augment nephron endowment in rat offspring above that of controls, in both normal and calorie-restricted maternal diets; however, high blood pressure was evident in treated offspring from an early age.⁵⁶⁸ A marked 50% reduction in nephron number in mice offspring exposed to a 30% reduction in maternal calorie intake was partially prevented by supplementation with the gluconeogenic amino acid methionine, whereas supplementation with asparagine had no impact.¹⁹⁸ The nephron deficit due to caloric restriction was also attenuated in transgenic mice with nephron progenitor cells heterozygous for *Tsc1*, a component of the primary inhibitor of the metabolic mTORC1 pathway (hamartin).¹⁹⁸ Promisingly, this reduction in hamartin activity in nephron progenitor cells has been shown to result in augmented nephron number due to a delay in the cessation of nephrogenesis.^{198,569}

MODULATION OF NEPHROGENESIS

The steroid hormone ouabain⁵⁷⁰ is highly expressed in the fetal and maternal circulation during pregnancy, with significantly greater levels associated with normal birth weight deliveries compared with SGA births.⁵⁷¹ Reduced maternal ouabain levels lower rat offspring birth weight, kidney weight, and nephron density.⁵⁷¹ Ouabain administration throughout pregnancy partially prevented the reduction in nephron number in rat offspring subjected to maternal low-protein diet and prevented subsequent proteinuria and elevated blood pressure.^{570,572}

Antioxidants (some amino acids and vitamins mentioned earlier can be included in this category) can successfully prevent adult-onset hypertension in animal models.⁵⁷³ The addition of a grape skin extract, a polyphenol-rich antioxidant with vasodilator properties, to the drinking water of pregnant mice on a low-protein diet resulted in the partial rescue of offspring body weight and nephron density.⁵⁷⁴ In offspring of pregnant mice exposed to cigarette smoke, MitoQ (mitochondrial-targeted coenzyme Q10, a naturally occurring antioxidant) treatment during gestation normalized nephron density and reduced oxidative stress.⁵⁷⁵ Mitochondrial function was also targeted by maternal supplementation of pregnant mice fed a low-protein diet with nicotinamide riboside. This treatment resulted in a partial rescue of offspring nephron number and a normalization of podocyte density.⁵⁷⁶

Modulation of kidney remodeling, apoptosis, and the regression of nephron number during development has also been suggested as a potential pathway to augment final nephron number, but effects may differ according to stage of development.⁵⁷⁷ Early postnatal injections of endorepellin (an inhibitor of apoptosis via activation of the antiapoptotic ERK1/2 pathway) could partially rescue nephron endowment in mouse pups exposed to a maternal low-protein diet.⁵⁷⁸

OPTIMIZATION OF PERINATAL MANAGEMENT OF THE HIGH-RISK NEONATE

As discussed earlier, multiple medications commonly used during pregnancy and administered to sick neonates may affect the developing kidneys and thus impact subsequent risk of higher blood pressure and kidney disease (see [eTables 20.1](#) and [20.3](#)). Optimizing treatment strategies in the clinical setting may help to mitigate risk.

Oxytocin receptor (OTR) antagonists such as Atosiban are commonly used tocolytics.^{579,580} Atosiban activates the G_i pathway and has been observed to inhibit growth of human embryonic kidney cells in vitro.^{579,580} Nifedipine is an alternative tocolytic agent. Its impact on nephrogenesis is unknown, but its renal vasodilatory effect could be protective in the fetal and neonatal kidneys.^{579,581}

Advances in neonatal care and therapeutics have dramatically improved survival of very and extremely preterm infants.⁵⁸² The infants are, however, vulnerable to toxicity from various drugs given their reduced kidney function, reduced nephron numbers, and immature drug metabolism.^{583,584} Given that nephrogenesis may continue for a period after preterm birth, optimization of management in high-risk neonates may permit nephrogenesis to continue or reduce additional avoidable postnatal “hits,” which may arrest nephrogenesis.^{51,52,201} The Baby NINJA (Nephrotoxic Injury Negated by Just-in-Time Action) study showed that a systematic surveillance program can significantly reduce AKI episodes along with nephrotoxic drug exposures.⁵⁸⁵

The frequently administered NSAIDs,^{294,295,586} antibiotics,^{283-285,289} and supplemental oxygen therapy^{278,281,282} can all adversely impact the developing neonatal kidneys. A Cochrane review suggested that paracetamol may be as effective as NSAIDs in the treatment of patent ductus arteriosus.⁵⁸⁷ Neonates with heart failure

may require treatment with ACE inhibitors (ACEi), which may impact nephrogenesis, alter renal blood flow, and result in AKI.^{588,589} Caution has been advised with the use of ACEi before 44 weeks post-menstrual age.⁵⁹⁰ Caffeine may be used to prevent or treat apnea of prematurity and may increase kidney tissue oxygenation and reduced AKI risk.⁵⁹¹⁻⁵⁹³ Given the complexity of, and interaction between, multiple intrauterine and extrauterine factors contributing to an infant's nephron deficit, it is likely that a multifaceted approach to nephron protection will be most effective.

Reports from the Assessment of Worldwide Acute Kidney Epidemiology in Neonates (AWAKEN) cohort studies have identified risk factors for, and the burden of, neonatal AKI and the risk of longer-term CKD.^{478,585,594} Communication of risk upon discharge from delivery rooms and neonatal intensive care units is poor. Awareness among obstetricians and pediatricians about long-term risks for mothers and babies, as well as referrals for follow-up, are suboptimal.⁵⁹⁵ A survey among community pediatricians in Switzerland revealed low awareness and low rates of screening for blood pressure and kidney function among children born preterm, LBW, or who had experienced neonatal AKI.^{595,596} There are no evidence-based guidelines for follow-up, screening, or management of small and vulnerable newborns at risk for development of CKD. At a minimum, periodic monitoring of blood pressure, urine protein, and BMI should be feasible in all settings. Children who experienced neonatal AKI and were also born preterm or SGA should be followed closely. Families should be counseled regarding the importance of healthy lifestyles to reduce future potential risk of developing noncommunicable diseases. Hypertension or proteinuria should be treated promptly, even in children, with ACEi/angiotensin receptor blockers (ARB) and now potentially sodium-glucose cotransporter 2 (SGLT2) inhibitors.^{1,595,597-599}

CATCH-UP GROWTH

Postnatal nutrition is important for infant growth, especially in the setting of preterm birth or growth restriction, and can impact nephron number and long-term kidney function. Animal data suggest that restoration of normal dietary components after growth restriction may permit some reversal of programmed changes, but overfeeding certain nutrients can be harmful (eTable 20.1).^{166,237,567,600,601} Nutrition in preterm infants is challenging.⁶⁰² Among 7-year-old children who had been born preterm, GFR was reduced in those who had experienced either intrauterine or extrauterine growth restriction, highlighting the importance of adequate postnatal nutrition for kidney development.²⁰¹ Catch-up growth has been advocated in poorer countries to improve child survival from infectious diseases and reduce stunting and malnutrition.⁶⁰³ Cochrane reviews have shown that both high-volume and high-protein feeds are associated with accelerated weight gain in hospitalized LBW and/or preterm infants, but specific impacts on kidney development are unknown.^{604,605}

In assessing later-life risk, a narrative review of 57 studies concluded overall that, although not all studies are consistent, a rapid (and especially excessive) weight gain after LBW is associated with increased blood pressure and risk of CKD in

childhood and later life.⁶⁰¹ The quality of postnatal nutrition is important. A systematic review found that breastfeeding is associated with reduced adult blood pressures, in part likely mediated by lower adult BMI.^{601,606} The timing of catch-up growth in early infancy and childhood additionally appears to impact long-term risk.^{603,607-609} Rapid weight gain within the first 4 months in children born SGA was associated with higher systolic blood pressure at age 5 years.⁶¹⁰ Among British 22-year-olds, systolic blood pressure increased by 1.3 mm Hg (95% CI: 0.3–2.3 mm Hg) for each standard deviation (SD) decrease in birth weight and increased by 1.6 mm Hg (95% CI: 0.6–2.7 mm Hg) for each SD increase in weight gain between the ages of 1 and 10 years (see Table 20.7).⁶¹¹ The consequences of catch-up growth may differ, however, if growth is linear in height or in weight.⁶¹²

Among those born LBW or preterm, the development of overweight or obesity exacerbates the risk of elevated blood pressure.^{426,607-609,613} In the NHANES cohort,⁴²⁸ the prevalence of hypertension was significantly greater among those with obesity compared to those without obesity, across the spectrum of birth weights, with prevalence ratios of 2.06 (95% CI 1.59–2.67) for normal birth weight, 3.85 (95% CI 1.88–7.86) for HBW, 1.94 (95% CI 1.04–3.60) for LBW, and 7.73 (95% CI 2.80–21.33) for VLBW.⁴²⁸ Risk factors for childhood overweight and obesity also include HBW and exposure to gestational diabetes.⁶¹⁴ Obesity, in turn, is a risk factor for kidney disease.^{615,616} Finding the balance where postnatal nutrition is optimized to improve short-term survival and reduce long-term risk of chronic disease requires further study. Avoidance of overweight through diet and exercise are safe principles.^{617,618}

IMPACT OF EARLY GROWTH ON KIDNEY FUNCTION

Grijalva-Eternod and colleagues⁶¹⁹ demonstrated that a “mismatch” between small kidneys and a (relatively) larger body, as would occur with catch-up growth and overweight after being born small, is associated with hypertension. Similarly, excessive weight gain and obesity were associated with more rapid progression of proteinuric kidney disease in children who had been born preterm⁵⁰⁰ and was a predictor of worse renal function at age 7.5 years in a cohort of very LBW preterm infants who had experienced AKI.⁶²⁰

Accelerated senescence has been proposed as a potential mechanism whereby catch-up growth may increase the risk of cardiovascular and kidney disease.⁶²¹ In animals, rapid weight gain in growth-restricted animals was associated with evidence of accelerated senescence in the kidneys and cardiovascular system, as well as premature death, consistent with accelerated aging.⁶²¹⁻⁶²³ Of interest, in an Argentinian study, neonatal telomere length was shorter in babies born with HBW compared with AGA and SGA infants, and shorter telomeres were also associated with maternal hypertension.⁶²⁴ Leukocyte telomere length was found to be significantly shorter among 5-year-old Bangladeshi children born LBW compared with those of normal birth weight and among young adults born preterm, lending support to the hypothesized association between adverse perinatal exposures and accelerated aging.^{625,626} In contrast, however, leukocyte telomeres were not found to be shorter in 68-year-old survivors of the Dutch famine, although this could reflect

a survivor effect.⁶²⁷ Oxidative stress is a driver of senescence, and in SGA children markers of oxidative stress were higher in those who experienced catch-up growth compared with controls.⁶²⁸ A programmed link among low nephron number, catch-up growth, and accelerated senescence as a potential mediator of hypertension and kidney disease in humans has yet to be proven.

MATERNAL HEALTH AND INTERGENERATIONAL EFFECTS OF PROGRAMMING

Maternal health and nutrition are crucial determinants of a healthy pregnancy, offspring birth weight, and fetal kidney development (see [Table 20.8](#)).²⁴⁰ Socioeconomic and structural factors are important determinants of maternal health and thereby the health of future generations.¹

Short maternal stature, maternal underweight, or iron deficiency are risk factors for LBW offspring.^{539,629-631} Maternal prepregnancy underweight is associated with a higher odds of having an SGA (OR: 1.81; 95% CI: 1.76–1.87) or LBW infant (OR: 1.47; 95% CI: 1.27–1.71), as well as a higher risk of preterm birth.^{632,633} Maternal obesity and gestational diabetes are increasing worldwide and are established risk factors for both LBW and HBW.^{596,634,635} The risks of gestational diabetes and preeclampsia double with prepregnancy maternal overweight.⁶³² Maternal chronic disease increases the risk of having a preterm or SGA infant; therefore planning of pregnancies in such women is important to minimize these risks.^{636,637} Specifically, pregnant women with CKD have a higher risk of LBW, SGA, and preterm birth (discussed in detail in [Chapter 58](#)).⁶³⁸⁻⁶⁴² CKD is also a major risk factor for preeclampsia, in addition to maternal preexisting hypertension, diabetes, anemia, heart disease, and urinary tract infections/pyelonephritis.⁶⁴²⁻⁶⁴⁴ Maternal solitary kidney also increases risk for reduced birth weight and preterm birth.⁶⁴² Teenage pregnancies are at higher risk of preeclampsia, LBW, and preterm birth.⁶⁴⁵⁻⁶⁴⁷

Obesity in pregnant women who had themselves been born with LBW was associated with a 3.65-fold increased risk of preterm delivery compared with those who were not obese.⁶⁴⁸ Maternal LBW or HBW is also associated with a higher risk of developing gestational diabetes.⁶⁴⁹ Similarly, in a population-based cohort, the odds of gestational diabetes or gestational hypertension (including preeclampsia or eclampsia) among those who had been born preterm were significantly increased. Odds were higher with lower gestational age, suggesting a “dose–response” relationship with degree of maternal prematurity (OR: 1.95; 95% CI: 1.54–2.47 if mother born at <32 weeks; OR: 1.14; 95% CI: 1.03–1.25 if mother born at 32–36 weeks).⁴⁰⁷ Gestational hypertension and preeclampsia, in turn, are risk factors for LBW and preterm birth, and gestational diabetes is a risk factor for HBW; therefore the programming cycle continues across generations.⁶⁴³ Maternal LBW or preterm birth are also risk factors for having their own LBW or preterm infants.^{1,648,650,651} This effect, however, was not observed among Indigenous Australian mothers for whom the impact of genetics and environmental stressors may predominate.⁶⁵² A direct intergenerational programming effect of maternal preterm birth on subsequent offspring preterm birth is,

however, supported by the observed increasing risk of preterm birth with lower maternal gestational age and the absence of risk if the father was born preterm.⁶⁵³

EPIGENETICS

The programming of the health of future progeny can be multigenerational, whereby direct exposure to an insult during pregnancy affects the germ cells of the developing offspring and may also be directly transferred to the second generation and/or transgenerational/intergenerational, whereby persistent epigenetic modifications can be passed to the second and third generations or beyond.⁴¹ These effects have been proposed to be mediated by changes in DNA methylation, histone modifications, and microRNA activity, resulting in germline and/or somatic epigenetic changes in gene expression, as evidenced in animal programming models.^{41,210,216,335,381,382,654,655} Besides epigenetic mechanisms, a mother who had been subjected to adverse intrauterine events may have a reduced ovarian and uterus size⁶⁵⁶ and/or experience changes in kidney function, blood pressure, or other disrupted metabolic/endocrine adaptations during pregnancy that may, in turn, de novo impact the development of her offspring.⁴¹

RELEVANCE OF NEPHRON ENDOWMENT FOR TRANSPLANTATION

Consideration of nephron endowment before transplantation could impact long-term kidney health in both donors and recipients.

IMPLICATIONS OF NEPHRON ENDOWMENT FOR THE DONOR

Living kidney donation, after appropriate donor screening, has generally been presumed to be safe, though there may be some delayed adverse consequences. Data, predominantly from Caucasian cohorts, suggest that GFR declines while hypertension and proteinuria increase over time in living donors, but kidney function remains generally well preserved over the first decades.^{110,657,658,659} Obesity significantly increases this risk over time.⁶⁶⁰ Some donor groups are at increased risk of loss of kidney function including the young and obese, the older donor, and some ethnic groups.^{661,662} Indeed, among the U.S. donor population the median estimated risk of ESKF at 20 years was 34 cases per 10,000 donors; however, the risk exceeded 256 cases per 10,000 in the 1% of donors at highest risk. Risk factors for developing ESKF were Black race, elevated BMI, donation to a first-degree relative, and older age among non-Black donors.⁶⁶² Studies have also found higher risks of hypertension, proteinuria, kidney dysfunction, and ESKF among Indigenous Australians and Canadians, African Americans, and Hispanic kidney donors ([Table 20.9](#)).⁶⁶³⁻⁶⁶⁶ These minoritized populations are at greater risk of gestational programming, having higher rates of preterm birth, LBW, SGA, and HBW, which may contribute at least in part to the increased risks post donation.⁶⁶⁷ Better understanding of renal risk in these populations is urgently needed.

A study from Germany found that GFRs were lower and proteinuria and hypertension were more frequent at 5 years post donation among donors with birth weights <2.5 kg (see Table 20.9).⁶⁶⁸ A group of potential donors that has not been studied is women who have had preeclampsia, who are known to be at an increased, albeit slight, risk of ESKF.⁶⁶⁹ The risk of gestational hypertension and preeclampsia is also increased among female kidney donors, suggesting that reduced nephron mass may be a risk factor.^{670,671}

The development of CT or MRI/biopsy techniques to count nephrons *in vivo* is opening up the possibility to directly assess the associations between nephron number and kidney function.⁷⁵⁻⁸³ Among 1334 kidney donors, at 4 months post donation, large glomeruli, and lower nephron numbers were found to predict albuminuria.⁸³ Among 673 healthy living donors who were followed up at a median of 10.5 years post nephrectomy by the same research group, having a nephron number below the 10th percentile for age at the time of kidney donation was associated with a progressive increase in risk of developing CKD (Table 20.9 and Fig. 20.4).⁷⁷

IMPLICATIONS OF NEPHRON ENDOWMENT FOR THE RECIPIENT

Allocation of donor kidneys is largely based on immunologic matching. In animal experiments of renal transplantation, however, the impact of transplanted nephron mass, independent of immunologic factors, on the subsequent development of chronic allograft nephropathy has been demonstrated.⁶⁷²⁻⁶⁷⁶ Despite such evidence, prescription of kidneys on the basis of the physiologic capacity of the donor organ to meet the metabolic needs of the recipient has not generally been considered.⁶⁷⁷

Demographic and anthropomorphic factors associated with late renal allograft loss include donor age, sex, and race, as well as recipient BSA.⁶⁷⁸⁻⁶⁸⁰ In general, kidneys from older, female, and African American donors fare worse.^{72,78,509,681} Indirectly, these observations suggest that the intrinsic nephron endowment of the transplanted kidney contributes to chronic allograft loss. As nephron numbers in donated kidneys have not been counted until recently, studies have evaluated relative recipient and donor BSA, kidney weights, or renal volumetric measurements by ultrasound as surrogates for metabolic demand and nephron mass.⁶⁸²⁻⁶⁸⁶ The preponderance of evidence supports the hypothesis that smaller kidneys perform less well when transplanted into larger recipients, as do kidneys from small or older recipients transplanted into large recipients.^{682-686,687,688} Analysis of donor kidney weight to recipient body weight (DKW/RBW) found that allograft function was significantly better in recipients with a DKW/RBW ≥ 4.5 g/kg compared with <3.0 g/kg.⁶⁸⁹ The DKW/RBW ratio was lowest when male recipients received kidneys from female donors. Other investigators found the risk of proteinuria above 0.5 g/24 h was higher, and developed earlier, in those with DKW/RBW <2 g/kg.⁶⁸¹ At 7-year follow-up, GFR had declined most rapidly in the low versus high DKW/RBW groups (eFig. 20.3).⁶⁹⁰ A small kidney transplanted into a large recipient may therefore not have an adequate capacity to meet the metabolic needs of the recipient without imposing glomerular

hyperfiltration, which ultimately leads to further nephron loss and eventual allograft failure.^{691,692}

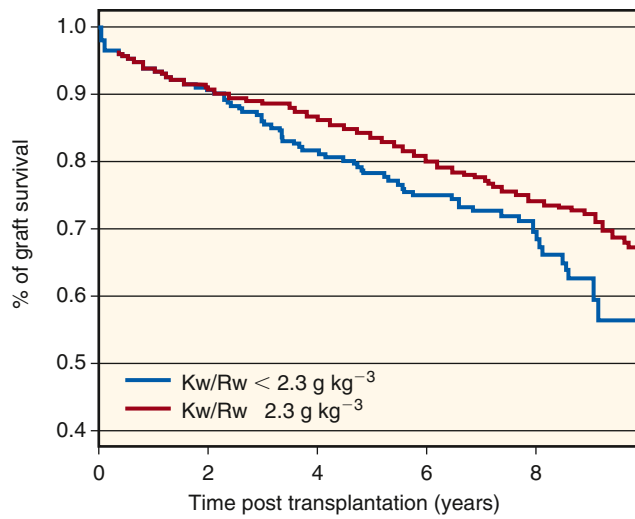
Transplanted nephron mass may therefore be a function of congenital endowment and attrition of nephrons with donor age but is also impacted by peritransplant kidney injury and nephron loss. It may become possible to determine nephron numbers in all living donors pretransplant. Until then, the clinical correlates of donor/recipient size ratios, donor gender, and donor age should be closely considered, in addition to immunologic matching, to enhance selection of the most appropriate recipients in whom the allograft is likely to function well for the longest time.

GLOBAL HEALTH RELEVANCE OF KIDNEY PROGRAMMING

The known perinatal risk factors LBW, SGA, preterm birth, and HBW are prevalent worldwide in both low- and high-income settings, although underlying causes may differ (see Table 20.8).^{23,57} Millions of infants born each year are at risk therefore of future hypertension and kidney disease, as well as other programmed conditions. How much developmental programming contributes to the growing burden of chronic noncommunicable diseases worldwide is unknown but may be significant, especially in regions where access to preventive health care is limited. In the current era of sustainable development goals, it may be possible to reduce the impact of developmental programming on future generations through a multisectoral approach, encompassing not only the health system but also education, infrastructure, and reducing social inequities (see Chapter 73 for discussion of global inequities).^{555,693} Strategies to reduce the impact of developmental programming in individual infants should be simple to implement and, as reviewed in depth elsewhere,^{1,694} include documentation of birth weight and gestational age in each child to identify those at risk; heightened awareness of the risk of neonatal AKI in preterm infants and implementation of strategies to reduce these risks; and education of parents about healthy lifestyle choices to prevent obesity and stunting, LBW, SGA, and preterm birth. Children born to mothers with preeclampsia, diabetes, or obesity should be screened for hypertension and proteinuria, especially if they become overweight; as adults, these individuals require long-term screening, early intervention with renoprotective strategies, and education with a strong emphasis on healthy lifestyles.^{1,63,695}

CONCLUSION

The association between an adverse fetal environment and the programming of susceptibility to develop hypertension and kidney disease in later life is now quite compelling and appears to be mediated, at least in part, by impaired nephrogenesis and suboptimal nephron endowment (Fig. 20.14). The Brenner hypothesis shows how a nephron deficit with concomitant glomerular hypertrophy and altered expression of sodium transporters in the programmed kidneys can contribute to the vicious cycle of glomerular hypertension, glomerular injury, and sclerosis, which subsequently leads to worsening systemic hypertension



eFig. 20.3 Correlation between donor kidney weight (*Kw*)–to–recipient body weight (*Rw*) ratio and long-term graft survival in renal transplantation. Graft survival declines faster with a *Kw/Rw* <2.3 g/kg (*P* = .016). (From Giral et al. *J Am Soc Nephrol*. 2010;21:1022–1029.⁶⁹⁰)

Table 20.9 Hypertension and Kidney Function in Living Kidney Donors at Risk of Adverse Renal Programming

Population	USA			Australia		Canada		Germany		USA	
	Black Donors	White Donors	Indigenous Donors	Indigenous Donors	Non-Indigenous Donors	Indigenous Donors	White Donors	BW ≤2.5 kg	BW >2.5 kg	Nephron number < 10% for age	Nephron number ≥ 10% for age
Donor number	12,387	71,769	22	38	28	38	76	18	73	228	445
Putative programmed risk	LBW, prematurity	ref	LBW	HBW (offspring of diabetic pregnancies)	ref	HBW (offspring of diabetic pregnancies)	ref	LBW	ref	unknown	ref
Hypertension	–	–	50%	42%	6%	42%	19%	39%	15%	↑	–
Proteinuria	–	–	81%	21%	6%	21%	4%	81%	35%	↑	–
↓ GFR	–	–	81%	Not different	38%	Not different		Not different		OR 2.72 (1.69–4.36)	ref
ESKF	74.4 per 10,000 vs. 23.9 per 10,000 in non-donors	22.7 per 10,000 vs. 0.0 per 10,000 in non-donors	19%	1	0%	1	0	0	0	0	0
Follow-up (years)	7.6 (IQR 3.9–11.5)	16.1 (1.27–20.2)	14.6 ± 9.3	13.4 ± 9.5	6.37 (2.54–21.2)			≥5	≥1–3	10.5	
Reference		782	663	666	668	77					

BW, Birth weight; ESKF, end-stage kidney failure; GFR, glomerular filtration rate; HBW, high birth weight; LBW, low birth weight; OR, odds ratio; ref, reference group.

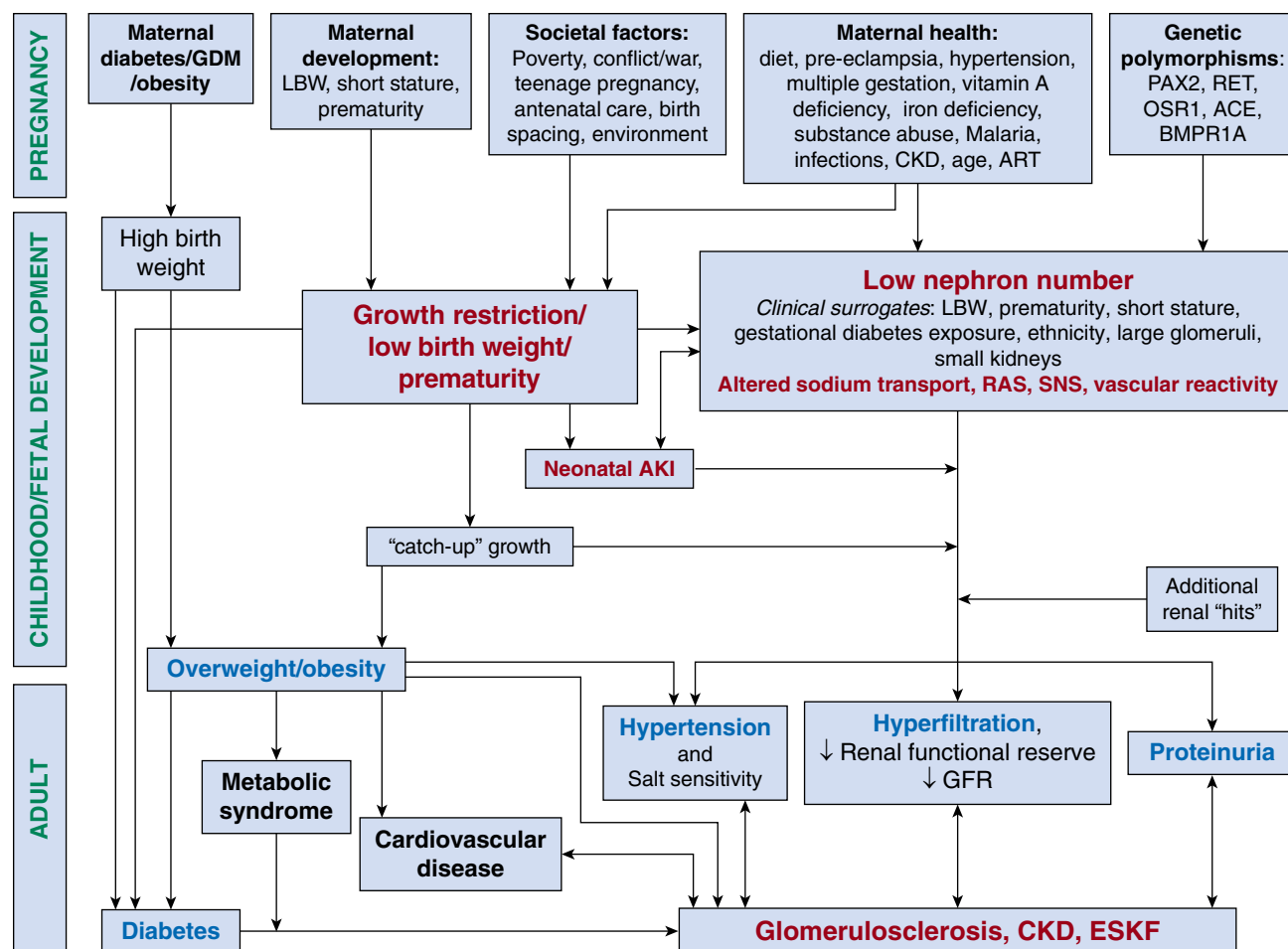


Fig. 20.14 Proposed mechanisms of fetal programming of hypertension and kidney disease. ART, Assisted reproductive therapies; CKD, chronic kidney disease; ESKD, end-stage kidney disease/failure; GDM, gestational diabetes mellitus; GFR, glomerular filtration rate; LBW, low birth weight; RAS, renin-angiotensin system; SNS, sympathetic nervous system. (Adapted from Luyckx VA, Bertram JF, Brenner BM, et al. Effect of fetal and child health on kidney development and long-term risk of hypertension and kidney disease. *Lancet*. 2013;382:273–283.¹⁶)

and ongoing kidney injury. In addition, numerous other factors, such as vascular dysfunction, podocyte deficits, increased oxidative stress, renal inflammation, accelerated senescence, and catch-up body growth, are all likely contributors to ongoing hypertension, nephron loss, and eventual CKD.

A reduced complement of nephrons constitutes a “first hit,” meaning kidneys would have less renal reserve to adapt to dietary excesses and health stressors or to compensate for injury. The number of nephrons in the kidneys of people without kidney disease varies widely, with at least a 2 million nephron difference per kidney between individuals at the highest and lowest ends of the spectrum. Genetic and epigenetic mechanisms, as well as maternal, fetal, and neonatal environmental exposures all contribute to the determination of individual nephron endowment. A significant proportion of the general population with low nephron numbers, particularly in areas where high or low birth weights are prevalent, may be at increased risk of developing later-life hypertension and kidney dysfunction; especially if subjected to superimposed kidney injury (second hits) such as through obesity, diabetes, or AKI. People with multiple exposures to perinatal risk factors begin life with the greatest level of susceptibility to later developing CKD.

Measurement of nephron number *in vivo* is becoming possible but is not yet mainstream. This may become useful in transplantation. In routine clinical care the best-known surrogate markers of kidney risk are LBW, HBW, preterm birth, IUGR, SGA, maternal preeclampsia, gestational diabetes, and obesity. In the absence of other known kidney diseases, a reduced kidney volume on ultrasound (especially in children) and glomerular enlargement on kidney biopsy may also suggest reduced nephron endowment. The mechanisms through which fetal programming exerts its effects on nephrogenesis are varied and likely complementary and intertwined. Although in some experimental studies nephron number and blood pressure can be “rescued” by good postnatal nutrition, supplementation, or therapeutics, applicability of these findings to humans requires further study.

The fact that even seemingly minor influences, such as the specific composition of maternal diet during fetal life, can have such major consequences on kidney development in offspring underscores the critical importance of optimization of perinatal care and early nutrition. In turn, this could have a major positive impact on population health in the future.

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